ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

Doctoral thesis

Experimental studies and thermodynamic analyses of CH_4-CO_2 replacement in natural gas hydrates and hydrate-based desalination

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Department of Urban and Environmental Engineering

(Environmental Science and Engineering)

Ulsan National Institute of Science and Technology

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Junghoon Mok

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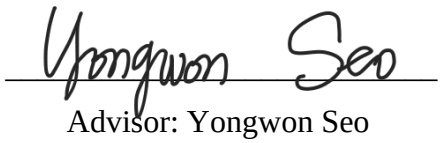
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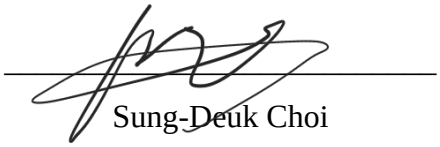
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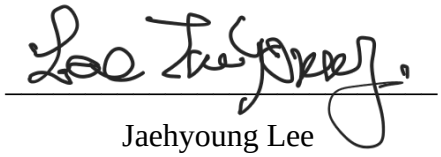
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Abstract

Gas hydrates are crystalline inclusion compounds that consist of well-defined hydrogen-bonded water frameworks (host cages) into which guest molecules are incorporated. As water is the main component required for gas hydrate formation, gas hydrates are considered environmentally benign. Therefore, gas hydrates have been applied to various energy and environmental fields. This study deals with two main topics as hydrate application technology.

In Topic 1, guest replacement and hydrate dissociation/reformation behavior during injection of CO_2 and flue gas into NGH layer were comprehensively observed through an experimental method. we investigated CH_4 - CO_2 replacement behavior according to the experimental conditions or changes in the injected gas. Firstly, the time-dependent behavior during CH_4 - CO_2 replacement was quantitatively examined via a multi-methodological approach, which is a combination of PXRD analysis and in-situ Raman measurement, demonstrated that a significant guest exchange in the large $5^{12}6^2$ cages had a greater effect on the extent of replacement. Secondly, dissociation behavior of CH_4 hydrate after N_2 gas injection was confirmed. Time-dependent Raman spectra confirmed that the N_2 molecules began to be captured in the hydrate cages immediately after the N_2 gas injection at certain conditions, and the extent of N_2 incorporation in the hydrate phase increased at a higher N_2 gas injecting pressure, which indicates that N_2 molecules revealed the dual functional roles of inhibitor for hydrate dissociation and external guest for guest replacement. Next, condition-dependent CH_4 - CO_2 + N_2 replacement behavior was observed in terms of reaction kinetics and extent of replacement. At a higher temperature, the extent of replacement did not change, but CO_2/N_2 ratio in the replaced hydrates decreased slightly. An increase in the pressure led to an accelerated CO_2 inclusion rate in the large ($5^{12}6^2$) cages at the initial stage and an enhanced N_2 inclusion in the small (5^{12}) cages at the final stage. In case of injecting different composition of CO_2 + N_2 mixed gas, the extent of replacement was increased as N_2 composition in the injecting gas increased at same partial pressure of injecting CO_2, but the CO_2 storage was decreased, which was due to the additional participation of N_2 occurred in both large and small cages. In addition, the CO_2 enclathration behavior in various mixed gas conditions are observed for CO_2 sequestration. The presence of NaCl shifted the equilibrium conditions of CO_2 + N_2 hydrates to the higher-pressure or lower-temperature region, whereas it increased thermodynamic CO_2 selectivity at a specified temperature and pressure. *In situ* Raman spectroscopic measurements demonstrated that CO_2 was kinetically selective at the early stage of CO_2 + N_2 hydrate formation and that kinetic CO_2 selectivity was more noticeable in the saline water system. In case of CH_4 + CO_2 + N_2 mixed hydrates, the CO_2/CH_4 ratio in the hydrate phase was not changed in different additional N_2 compositions in same CO_2/CH_4 ratio of injecting gas.

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In Topic 2, we focused on an evaluation of hydrate-based desalination efficiency from gaseous hydrate formers (propane, R134a, R152a, and R22). The thermodynamic stability, crystallographic information, dissociation enthalpy, and kinetic growth behavior of various hydrate formers were experimentally measured. The Hu-Lee-Sum (HLS) correlation was employed to predict the equilibrium shift and hydrate depression temperature. A novel approach to examine the maximum achievable salinity and maximum water yield using the HLS correlation was introduced. The theoretical HBD efficiency increased as the initial salinity decreased, the operating pressure decreased, and the initial subcooling temperature increased. In addition, theoretical HBD efficiency was employed as a quantitative standard for evaluating the kinetic performance of the HBD. At a fixed initial subcooling (2 K), R134a gave faster formation kinetics in the early stage, but R22 eventually offered highest hydrate conversion. At a fixed temperature (272 K), R152a showed fastest formation kinetics and highest HBD efficiency due to its milder hydrate equilibrium conditions. In addition, the estimation of theoretical desalianiton efficiency was also conducted on the liquid phase-hydrate former, cyclopentane (CP). The operating condition-dependent cooling demand for the process was simply examined from a thermodynamic point of view. The study provides a valuable theoretical foundation for the further development of gas hydrate-based desalination technology.

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Chapter 1. Introduction

1.1. Gas hydrates

Gas hydrates are solid inclusion compounds formed when guest molecules, such as methane (CH_4), carbon dioxide (CO_2), and sulfur hexafluoride (SF_6), are incorporated into well-defined cages made by hydrogen-bonded water frameworks. In general, gas hydrates are thermodynamically stable at high pressure and low temperature and exist in three distinct structures depending on the size and shape of the guest molecules: structure I (sI), structure II (sII), and structure H (sH)[1]. Due to the environemtnally benign features of gas hydrates, it has many technological applications in the energy and environmental fields. A representative example is hydrate-based gas separation (HBGS) technology[2]. It emphasizes its cheaper material cost and eco-friendliness compared to conventional separation methods, including liquid absorption, membrane separation, and solid adsorption, because water is the only separation medium for the HBGS process. In the HBGS process, a gas component, which has a lower hydrate equilibrium pressure at a given temperature, is collected predominantly in the hydrate phase, and then the concentrated component can be retrieved after dissociation of the gas hydrates formed from a gas mixture[3]. The separation efficiency of the HBGS process is generally affected by the thermodynamic conditions of the process operation (temperature and pressure), hydrate formation kinetics of target components, and mechanical settings of the hydrate formation/dissociation reactor[4]. There have been extensive efforts to apply HBGS to various combinations of gas mixtures. CO_2 can be captured from post-combustion flue gas (CO_2 + N_2) mixtures using HBGS[5]. As the hydrate equilibrium pressure of CO_2 at a given temperature is significantly lower than that of N_2, the remarkable separation efficiency of the HBGS process was observed for the CO_2 + N_2 gas mixtures[6]. Fluorinated gases (F-gases) like SF_6 and R134a, can also be effectively separated from F-gas + N_2 mixtures using HBGS[7]. In particular, when the HBGS method was combined with the liquefaction method, an equilibrium recovery ratio of 0.90 from the SF_6 (50%) + N_2 (50%) mixture was achieved at much lower pressure compared to the liquefaction method alone[8].

1.2. Guest replacement

Energy demands have increased because of economic development and world population growth, and the continued supply and price stabilization of energy resources will be an indispensable issue in the future. Due to the enormous amount of natural gas stored in global NGH reserves, NGHs are considered promising future energy resources[9]. To exploit natural gas from NGHs, several production methods, such as depressurization, thermal stimulation, and inhibitor injection methods have been devised[10,11]. As depressurization is regarded as the most economical and feasible production method because of its simplicity and lower cost of production, it has been applied successfully in several field test programs

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to produce natural gas from NGHs[12]. However, the drastic temperature decrease caused by hydrate dissociation during production, which is a strong endothermic process, not only lowers the driving force (ΔT) for hydrate dissociation but also hinders mass transfer by inducing hydrate/ice reformation from residual water[13]. Additionally, as the fluid pressure inside the pore void decreases, the effective stress, which is the difference between total stress and pore fluid stress, increases, thereby resulting in a decrease in the permeability of the reservoir and gas production efficiency due to the consolidation of sediments[14-16]. Moreover, there is a risk of pipeline blockage caused by hydrate reformation during the transportation of the produced natural gas and water contents. The inhibitor injection method can be an alternative to depressurization because inhibitors can serve the dual function of dissociating the original NGHs and preventing the blockage problem by hindering hydrate reformation[17]. However, the injection of commonly used liquid hydrate inhibitors or highly concentrated NaCl solutions not only reduces economic feasibility but also causes environmental pollution and mass transfer limitations[18,19]. To supplement these weaknesses of the conventional inhibitor injection method, a novel concept of injecting air or gaseous N_2 into NGHs has recently been attempted[20-24]. When N_2 gas is injected into the NGH layer, the gas hydrates can be destabilized because N_2 has much harsher hydrate equilibrium conditions; thus, the large chemical potential difference between N_2 molecules and guest molecules triggers hydrate dissociation. Furthermore, the hydrate reformation after hydrate dissociation can be prevented due to the additional presence of N_2 in the vapor phase. As N_2 is injected in the gaseous state, it can be readily dispersed in plugged pipelines or hydrate-saturated layers with high permeability. However, as these conventional methods are based on NGH dissociation for CH_4 production, the geo-mechanical instability of the NGH layers in the deep ocean sediment due to gas hydrate dissociation can lead to local sea floor subsidence and the collapse of well casings, thereby causing enormous economic losses and geological hazards[25].

Conversely, extensive research has been conducted to capture and sequester CO_2 as anthropogenic CO_2 emission from fossil fuel burning is blamed for global warming[26]. Carbon capture and storage (CCS) technologies, which refer to comprehensive methodologies of selectively capturing CO_2 from anthropogenic emission sources and isolating CO_2 in geological formations or the ocean, have been recognized as indispensable tools for coping with global warming caused by the increasing greenhouse gas emissions[27,28]. CO_2-enhanced oil recovery, saline aquifer CO_2 storage, and mineral carbonation are well-known examples of long-term geological CO_2 storage[29]. Although, ocean storage and geological storage methods have their own concerns and drawbacks: ocean acidification due to CO_2 dissolution for ocean storage and higher possibility of CO_2 leakage into atmosphere/marine systems, its simple concept and large storage capacity still make it an attractive option[28].

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Recently, an innovative method that produces CH_4 by injecting CO_2 into the NGH layers (i.e., CH_4 – CO_2 replacement) has been suggested as a promising option for CH_4 exploitation and CO_2 sequestration[30,31]. The ocean sediments remain stable after CH_4 production because CH_4 molecules in the NHGs can be swapped with CO_2 molecules while maintaining the gas hydrate lattices. Therefore, the CH_4 – CO_2 replacement is an attractive method that enables the non-destructive production of the NGH layers with a double duty of producing clean-burning CH_4 and sequestrating greenhouse gas CO_2. Several studies revealed that the CH_4 – CO_2 replacement in sI hydrates occurs without the significant decomposition of the initial gas hydrates. Ersland et al. used magnetic resonance imaging to examine a spontaneous exchange of CH_4 and CO_2 in the sI hydrates and demonstrated that no free water was observed during the CH_4 – CO_2 replacement, indicating that the initial sI CH_4 hydrate did not dissociate with liquid water during the exchange process[32]. Lee et al. monitored the heat flow changes during the CH_4 – CO_2 replacement using a micro-differential scanning calorimeter and found no detectable evidence of gas hydrate dissociation[33]. When pure CO_2 gas is injected into the NGH layer for guest exchange, CO_2 is easily liquefied under high-pressure conditions of the target site, which leads to the formation of multiphase fluid flow[34]. However, CO_2 liquefaction makes the injected fluid a more complex and intractable system and can cause blockage issues in the pipelines or sediment. In addition, the applicable temperature range for CH_4 − CO_2 replacement using pure CO_2 is limited because CO_2 and CH_4 + CO_2 hydrates are not thermodynamically stable at temperatures above their corresponding upper quadruple points[35]. To overcome these drawbacks, the concept of CO_2 + N_2 gas injection for guest replacement has been proposed. Park et al. demonstrated that the replacement efficiency could be increased due to the additional participation of N_2 molecules in the guest exchange process[36]. From an engineering point of view, the economic feasibility is improved because the flue gas (CO_2 + N_2) emitted from fossil fuel-fired power plants can be directly used without CO_2 separation or capture[36]. Furthermore, the presence of N_2, which has relatively higher hydrate equilibrium pressure than CO_2 at any given temperature, enables a significant reduction in the risk of plugging and an expansion of the applicable temperature ranges for guest replacement. It is important to analyze and understand the tendency or features of CH_4 recovery and CO_2 sequestration in CH_4 − CO_2 + N_2 replacement under changes in pressure and temperature, because over 230 gas hydrate deposits have been discovered, which have a wide range of thermodynamic conditions[3738]. Therefore, understanding the guest replacement behavior by CO_2 and N_2 molecules under various conditions is essential to increase the efficiency of guest replacement technology.

In addition, a huge volume of CO_2 is also can be stored in the solid form of gas hydrates in such places[39,40]. During the Iġnik Sikumi gas hydrate replacement field experiment in the Alaska North Slope in 2011 and 2012, it was confirmed that in such a free water-bearing reservoir system, the CO_2 storage through “solid-state” guest replacement in native NGHs is difficult to distinguish from that through CO_2

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hydrate formation in free water[41]. In either case (guest replacement or direct CO_2 sequestration), gas hydrates are obviously newly formed by the reaction between the injected gas and free water in marine sediments, and hydrate formation can be affected by the sediment properties and environment around the injection well. Several previous studies have reported that enriched organic matter in the pore water of marine sediments could affect the growth behavior of CO_2 hydrates[42]. In addition, various salts contained in seawater are well known to depress phase equilibrium temperatures or increase the equilibrium pressures of gas hydrates, functioning as thermodynamic hydrate inhibitors [43,44]. During hydrate formation in saline water, salts are excluded from hydrate crystals and concentrated in the residual liquid[45]. Some hydrate-bearing sediments in the Arctic regions showed abnormally high salinity in pore water, possibly due to the salt enrichment during NGH formation in the sediments with low permeability[46]. Therefore, understanding the CO_2 enclathration behavior with various mixed gas are essential for effective hydrate-based CO_2 sequestration.

In this topic, to improve the efficiency of guest replacement technology the behavior of each guest molecules inside the hydrate structure during the replacement was experimentally observed in detail for various thermodynamic conditions. Firstly, the time-dependent guest distributions during the $CH_4 – CO_2$ replacement were closely investigated by adopting various experimental and multi-methodological approaches. Secondly, the injection of N_2 gas as an inhibitor and an external guest into the CH_4 hydrate was closely examined in a closed system to elucidate the dual-functional roles of N_2 and to reveal the guest behaviors in the N_2-induced NGH production. Next, a gas mixture of $CO_2 + N_2$ that simulated post-combustion flue gas was injected into the sI CH_4 hydrate to investigate the thermodynamic and kinetic guest exchange behaviors under various thermodynamic conditions. In addition, the formation behavior of $CO_2 + N_2$ or $CH_4 + CO_2 + N_2$ mixed hydrate was observed to verify the effect of saline solution and N2 on the hydrate growth and CO_2 storage behavior.

1.3. Hydrate-based desalination

While the demand for fresh water continues to rise due to the rapid growth of the world population and industrial development, only about 0.014% of the water resources on Earth are directly available to living organism[47]. Research on desalination, which refers to the removal or separation of salts and minerals from saline water to obtain fresh water, is considered inevitable for the sustainable development of human civilization. Thermal desalination (distillation) and membrane-based separation using reverse osmosis (RO) have been widely used in many actual sites[48-50]. However, these methods still have several technological issues. Thermal desalination requires an excessive amount of heat for water evaporation, which results in lower energy efficiency for the thermal process[51]. The RO process is subject to frequent membrane fouling in treating saline water with high salt concentration and it also requires high operational pressure, which can be a serious disadvantage in terms of operational costs[52].

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As an alternative to these processes, freeze desalination (FD) has been regarded as a fascinating technology because of its lower tendency to foul and fewer corrosion problems[53,54]. Additionally, several researchers have reported that the energy efficiency of FD can be significantly improved by using cold energy from the regasification of liquified natural gas (LNG)[55-57]. The process efficiency of FD is generally determined by a trade-off between the salt separation efficiency of the process and the amount of cooling energy required for freezing and subsequent crystal growth. Therefore, hydrate-based desalination (HBD), which is part of FD, is of strong interest because the crystallization process can be operated at a much higher temperature compared to FD (usually above the freezing point of water) and thus, the energy consumption for crystallization can be drastically reduced[58-62].

In particular, when gas hydrates are formed all other impurities except the guest and host components are excluded from hydrate crystals. Thus, gas hydrate formation from wastewater or brine water leads to the production of purified water and further concentrated solution. HBD as a method to produce fresh water from seawater began to receive significant attention in the 1960s, and there were attempts to operate pilot-scale plants in Hawaii and San Diego to investigate its technical feasibility in the 1990s[63]. One prominent issue that hampered technical progress of HBD was the morphology of synthesized gas hydrates. Because a substantial amount of salts was entrapped in the interstitial pore space of dendritic hydrate crystals, a filtration and an additional wash column were employed as separation methods. The estimated cost of water produced by HBD was reported to be comparable to that by multi-stage flash (MSF) and RO[64]. For efficient solid-liquid separation in the HBD process, several researchers introduced a new apparatus, which can produce solid hydrate pellets in a continuous manner, and evaluated the ion removal efficiency for actual seawater samples using it[65,66]. They produced solid gas hydrate pellets after the formation of gas hydrates using CO_2 and CH_4, without the pretreatment of seawater to show the effects of the hydraulic pressure (to produce pellets) and the types of hydrate-forming gases on the removal efficiency of ions.

However, the studies on the assessment of separation efficiency for the HBD process are still insufficient and an understanding of salt enrichment during gas hydrate formation remains incomplete for applying HBD technology to the actual desalination process. When gas hydrates are formed from saline water, the residual solution is gradually concentrated as the crystal growth progresses, and thus its salinity also gradually increases. A gradual increase in the salt concentration of the residual solution causes the change in the thermodynamic environment and driving force for gas hydrate formation, which implies that it also affects the change in hydrate growth behavior[67]. Furthermore, the maximum achievable separation efficiency of the HBD process might vary depending on the types of hydrate guests, operating conditions, and initial salt concentration of the solution. Therefore, the efficiency evaluation, process design, and optimal operation of the HBD process should be performed by

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considering both the kinetics of the hydrate growth in the saline water and the maximum achievable desalination efficiency.

In this topic, to select a hydrate former with higher HBD efficiency, a novel method of comparing the potential desalination efficiency for each guest condidates according to thermodynamic conditions was presented through experiments and thermodynamic calculations. The HBD efficiency and kinetic performance of various gaseous hydrate formers, such as propane (C_3H_8), 1,1,1,2-tetrafluoroethane (CF_3CH_2F; R134a), chlorodifluoromethane ($CHClF_2$; R22), and 1,1-difluoroethane (CH_3CHF_2; R152a), were closely investigated to determine the energy-efficient hydrate former for the HBD process. The maximum achievable salinity and maximum water yield were calculated under various operating conditions (initial salinity, pressure, and initial subcooling) using the Hu-Lee-Sum (HLS) correlation to evaluate the theoretically achievable HBD efficiency of the gaseous hydrate formers. Furthermore, the hydrate formation behaviors in saline water at both a fixed initial subcooling and a fixed temperature were observed by gas uptake measurements to examine the time-dependent desalination performance of gaseous hydrate formers. In addition, the theoretically achievable HBD efficiency was also estimated for liquid-phase hydrate former, cyclopentane (CP).

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Chapter 2. Experimental investigation

2.1. Materials

CH_4 (99.999%), CO2 (99.99%), N2 (99.99%), Propane (99.5%), and a CO_2 (20%) + N_2 (80%) gas mixture were purchased from MS Gas Co. Ltd. (Republic of Korea). R134a (99.9%), R22a (99.9%), and R152a (99.9%) were supplied by RIGAS Co. (Republic of Korea). NaCl with a purity of 99.5% was purchased from Sigma-Aldrich (USA). Double-distilled, deionized water was used to make NaCl solution. Porous silica gels (SiliCycle Inc., Canada) with a nominal pore diameter of 100 nm and pore volume of 0.88 mL/g were used to increase the contact area between water and gas and between the CH_4 hydrate and gas, as well as to enhance the reaction kinetics for CH_4 hydrate formation and guest replacement. After being dried at 413.2 K for 24 h, porous silica gels were mixed with double-distilled, deionized water and vibrated by sonication for 72 h to obtain water-saturated porous silica gels. All experiments were conducted using water-saturated porous silica gels for a consistent and uniform experimental setup. The materials were used without any additional purification process.

2.2. Apparatus and procedure

2.2.1. Hydrate phase equilibrium measurement

The hydrate phase equilibria were measured using an isochoric pressure search method with stepwise heating/cooling. A high-pressure 316 stainless steel autoclave reactor with an impeller-type stirrer was filled with 50 g of NaCl (3.5 wt%) solution. The reactor was immersed in a water-ethylene glycol bath connected to an external circulator (RW-2025G, JEIO Tech, Republic of Korea) for temperature control. The temperature and pressure in the reactor were monitored with a resistance temperature detector sensor calibrated using an ASTM 63C thermometer (H-B Instrument Company, USA) with an accuracy of ±0.03 K and a pressure transducer (S-10, Wika, Germany) calibrated using a Heise Bourdon tube pressure gauge (CMM-137219, Ashcroft, USA) with a maximum error of ±0.02 MPa, respectively. By pressurizing the reactor to a desired pressure with target gas mixture and lowering the temperature, gas hydrate formation was induced. After sufficient conversion of water into gas hydrates with agitation, the temperature slowly increased to dissociate the hydrates. In the temperature range near the expected equilibrium point, an extensive time interval (10 h per 0.1 K) was given for sufficient equilibration time at each temperature step. The three-phase (H-L_W-V) equilibrium point was determined by the intersection between the thermal expansion and hydrate dissociation lines in the pressure-temperature trace graph.

2.2.2. PXRD measurement

For PXRD measurement, the collected hydrate samples were ground into fine powders with an

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approximate diameter of 100 μm. The PXRD patterns were collected using a synchrotron monochromatic powder X-ray diffractometer attached to the 6D beamline of the Pohang Accelerator Laboratory (PAL, Republic of Korea). The wavelength of the beamline was 1.54056 Å, and the measurements were performed in the θ/2θ scan mode with a step size of 0.0256° in the 2θ range from 2° to 65°. The diffraction patterns were obtained with an exposure time of 120 s at 133 K. The acquired PXRD patterns were refined by profile matching (Le Bail method) within the FullProf program to examine the hydrate crystal structure[68,69]. Rietveld refinement using the FullProf program was conducted to obtain the site occupancy and rotations of the rigid bodies of the guest molecules[70,71].

2.2.3. NMR measurement

A 400 MHz solid-state NMR spectrometer (Avance II$^+$, Bruker, USA) at the Korea Basic Science Institute (KBSI) Seoul Western Centre was used to confirm the enclathration and distributions of $CHClF_2$ and CH_4 molecules in the hydrate cages. The finely powdered hydrate samples were loaded into a 4 mm o.d. Zr-rotor and ^{13}C NMR spectra were obtained at a Larmor frequency of 100.6 MHz with a magic angle spinning (MAS) of 2–4 kHz at 190 K under high-power proton decoupling (HPDEC). A pulse length of 1.8 μs and a pulse repetition delay of 10 s with a radio-frequency field strength of 50 kHz corresponding to 90° pulses of 4.5 μs duration were used. The downfield carbon resonance peak of tetramethylsilane, which was assigned a chemical shift of 0 ppm, was used as an external chemical shift reference.

2.2.4. HP μ-DSC measurement

The dissociation enthalpy of gas hydrates formed was measured using a high-pressure micro-differential scanning calorimeter (HP μ-DSC VII evo, Setaram Inc., France). The equipment consisted of two Hastelloy C276 alloy cells (a sample cell and a reference cell) with an inner volume of 0.5 cm^3. After the sample cell was charged with approximately 12 mg of solution and pressurized with R152a, cooling-heating cycles between 253.2 K and 274.2 K were repeated for complete conversion of water to gas hydrates until no further change in heat-flow peaks was observed. Then, the cell temperature was increased with 0.5 K/min up to 300 K to dissociate gas hydrates. The area of the endothermic heat-flow curve was integrated to determine the dissociation enthalpy of gas hydrates.

2.2.5. Raman spectroscopy measurement

A high-pressure reactor equipped with a fiber-optic Raman probe and a sample loop was used for the *in situ* Raman and gas composition measurements. The reactor with a water jacket was filled with 50 g of solution. After ventilation, the reactor was pressurized with target gas, and the temperature was lowered

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to experimental conditions. In addition, gas compositions in the vapor phase and guest distributions in the hydrate phase were simultaneously observed via a gas chromatograph (GC; 8890 II, Agilent, USA) and a fiber-coupled Raman spectrometer (SP550, Horiba, France) equipped with a multichannel air-cooled CCD detector and a 1800-groove/mm grating, respectively. Each Raman spectrum was obtained with an exposure time of 120 s.

2.2.6. Gas uptake measurement

The amount of gas consumed during the formation of R152a and R134a hydrates in the presence of NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) was measured in isobaric and isothermal conditions. The high-pressure reactor was charged with 10 g of solution and the pressure was maintained at target pressure using a micro-flow syringe pump (500D, Teledyne ISCO, USA). The subcooling (which refers to the difference between the actual operating temperature and hydrate equilibrium temperature) was given for hydrate formation. The volume of gas consumed during hydrate formation was continuously recorded using the LabVIEW software. The accumulated amount of gas consumption ($G_{c,t}$, n_{gas}/n_{water}) at time t was calculated using the following equation:

$$G_{c,t} = \frac{\Delta V_t \rho}{W/m_w},$$

where ΔV_t, ρ, W, and m_w represent the accumulated volume of gas consumed at time t, the molecular density of gas, the weight of initially charged water, and the molecular weight of water, respectively. In addition, the time-dependent hydrate conversion ($H_{c,t}$, $n_{hydrate}/n_{water}$) at time t was observed by multiplying the $G_{c,t}$ values and the hydration number (n):

$$H_{c,t} = \frac{\Delta V_t \rho n}{W/m_w}$$

2.2.7. Calculation of depressed hydrate equilibrium temperature in saline water

To predict the hydrate equilibrium temperature at a given pressure and at specific salinity conditions, the HLS correlation was employed. Hu et al. reported that the hydrate depression temperature ($\Delta T = T_0 - T$) can be derived as follows[72]:

$$\frac{\Delta T}{T_0 T} = -\frac{nR}{\Delta H_d} ln\, a_w$$

in which T_0 and T are hydrate equilibrium temperatures in pure water and saline water, respectively, and ΔH_d, n, and a_w are hydrate dissociation enthalpy, hydration number, and water activity, respectively. According to their results, the left term of the equation was independent of temperature

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and pressure, and strongly dependent on the salt species and concentration. They developed the water activity term to a polynomial form, given by

$$ln\, a_w = B_1 X + B_2 X^2 + B_3 X^3$$

in which B_1, B_2, and B_3 are fitted coefficients from the universal correlation and X is an effective mole fraction, $X = \sum_{i=ions} |z_i| x_i$, in which z_i and x_i are the charge of the ions and the mole fraction of the salts, respectively. The hydrate dissociation temperature can be calculated with the following equation:

$$T = T_0 \left[1 + \left(\frac{\Delta T}{T_0 T}\right) T_0\right]^{-1}$$

For the HLS correlation, the fitted coefficients (B_1 = -1.028743829, B_2 = 2.929237521, and B_3 = -36.5112452) were obtained from the literature[61].

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Chapter 3. N_2-assisted CH_4 - CO_2 replacement

3.1. Cage-specific CH_4 - CO_2 replacement behavior

This chapter includes the published contents:

Mok, J., Choi, W., & Seo, Y. (2020). Time-dependent observation of a cage-specific guest exchange in sI hydrates for CH_4 recovery and CO_2 sequestration. Chemical Engineering Journal, 389, 124434.
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3.1.1. Abstract

CH_4–CO_2 replacement in naturally occurring gas hydrates has been considered a promising method for both energy recovery and CO_2 sequestration. In this study, the time-dependent guest exchange behaviors and guest distributions during CH_4 – CO_2 replacement were closely examined at two different CO_2 injecting pressures (2.2 and 3.5 MPa) using nuclear magnetic resonance (NMR), in-situ Raman spectroscopy, powder X-ray diffraction (PXRD), and gas chromatography. The ^{13}C NMR spectra confirmed that the cage occupancy ratio of the CH_4 molecules in the large $5^{12}6^2$ and small 5^{12} cages ($\theta_L/\theta_{S,CH4}$) after the replacement was significantly smaller than that before the replacement because of the preferential occupation of CO_2 in the large $5^{12}6^2$ cages. The time-dependent Raman spectra revealed that the rate of CO_2 inclusion and the resultant CH_4 depletion in the hydrate phase during the replacement was faster at a higher CO_2 injecting pressure. The Rietveld refinement of the PXRD patterns offered a quantitative cage occupancy of CH_4 and CO_2 molecules before and after the replacement. The time-dependent cage occupancy values of CH_4 and CO_2 during the replacement obtained from a multi-methodological approach, which is a combination of PXRD analysis and in-situ Raman measurement, demonstrated that a significant guest exchange in the large $5^{12}6^2$ cages had a greater effect on the extent of replacement and that the kinetics of the CH_4 – CO_2 replacement was accelerated at a higher CO_2 injecting pressure. The results provide a better understanding of the kinetics and mechanism of the cage-specific CH_4 – CO_2 replacement occurring in the sI hydrates for CH_4 recovery and CO_2 sequestration.

3.1.2. Time-dependent replacement behavior using Raman spectroscopy

Raman spectroscopy is a powerful tool for identifying the guest enclathrating behavior in the hydrate cages[73]. In this study, in-situ Raman spectroscopy was used to examine the exchange behavior of CH_4 and CO_2 in the hydrate cages during the replacement process. A Raman band of symmetric C–H

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stretching vibration of CH_4 molecules varied with the state of CH_4 molecule. As shown in Fig. 3.1.1 (a), a Raman peak of gaseous CH_4 was located at 2,917 cm^{-1}, but that of the CH_4 molecules encapsulated in the small 5^{12} and large $5^{12}6^2$ cages of sI hydrate appeared as two distinct bands at 2915 cm^{-1} and 2905 cm^{-1}, respectively. The guest exchange behavior of CH_4 in the hydrate cages of sI was observed in the range of 2880–2940 cm^{-1}. As shown in Fig. 1 (a), the intensity of the Raman peaks for the CH_4 molecules in the small 5^{12} and large $5^{12}6^2$ cages gradually decreased during the replacement, indicating that the CH_4–CO_2 replacement occurred in both cages of the sI hydrate.

Gaseous CO_2 is known to have major bands (at 1285 cm^{-1} and 1388 cm^{-1}) and minor bands (at 1265 cm^{-1} and 1409 cm^{-1}) because of Fermi resonance[74,75]. When CO_2 molecules were incorporated into the sI hydrate cages, only two major bands appeared at 1276 cm^{-1} and 1380 cm^{-1} [76]. Time-dependent Raman spectra were collected in the range of 1,200–1,500 cm^{-1} and provided in the range of 1,260–1,300 cm-1 in Fig. 3.1.1 (b) to examine the CO_2 enclathration in the sI hydrate during the replacement. The intensity of the Raman peak at 1276 cm^{-1} originating from the enclathrated CO_2 molecules gradually increased as the Raman intensity at 1,285 cm^{-1} originating from the gaseous CO_2 molecules gradually decreased. Fig. 1 clearly indicates that a gradual guest exchange of CH_4 and CO_2 molecules in the sI hydrate occurred without a structural transformation, that is, an iso-structural process.

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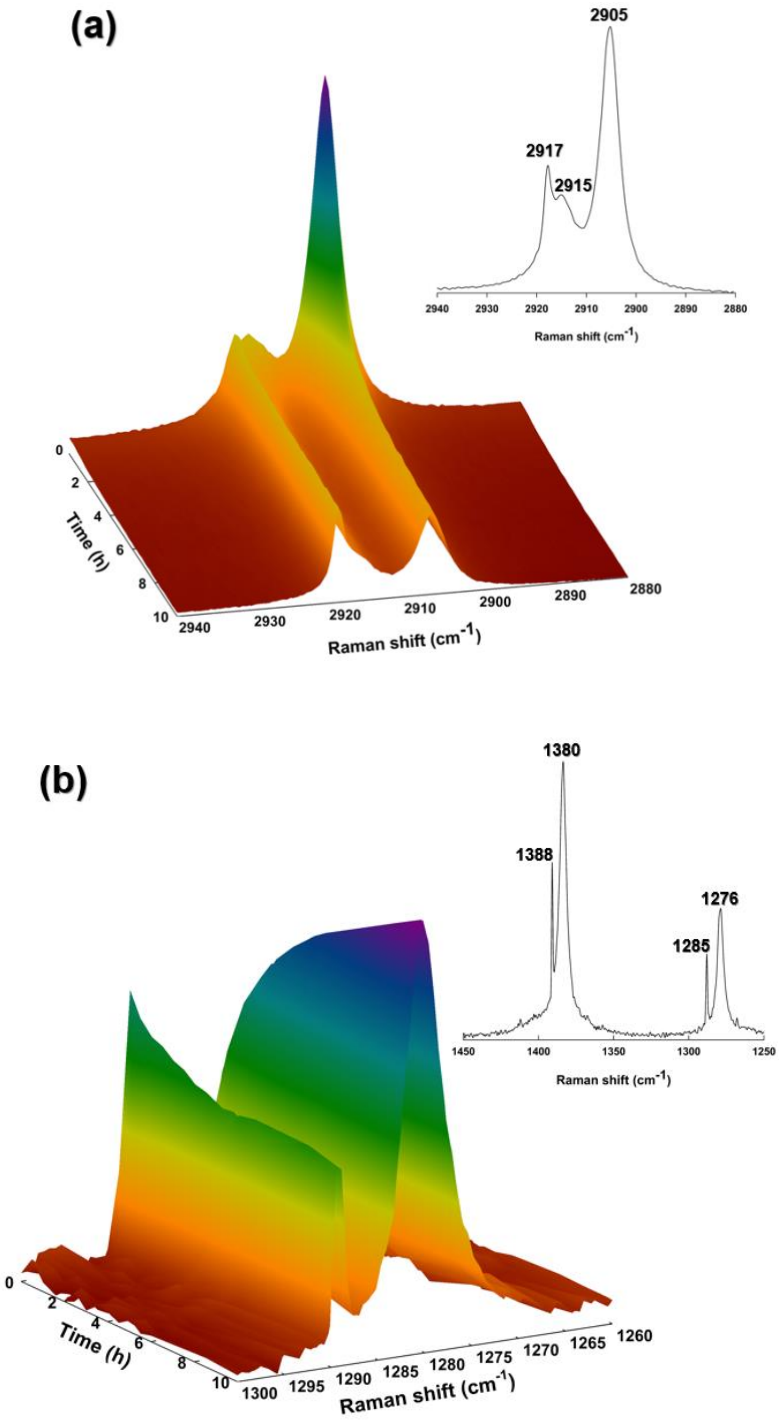


Fig. 3.1.1 Time-dependent Raman spectra of the (a) CH_4 and (b) CO_2 molecules during the initial 10 h of the replacement at PCO_2 = 2.2 MPa. The Raman spectra of the inset figures in (a) and (b) were collected at t = 1 h and 72 h, respectively.

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The Raman peaks corresponding to the enclathrated CH_4 and CO_2 molecules were deconvoluted to examine the guest exchange behaviors during the replacement. In Fig. 3.1.2 (a), the changes in the normalized area of the Raman peaks from the enclathrated CH_4 molecules in the hydrate phase are depicted at two different PCO_2s of 2.2 MPa and 3.5 MPa. The area of the two Raman peaks for the CH_4 molecules in the small 5^{12} and large $5^{12}6^2$ cages was added up and normalized to observe the CH_4-depleting behavior in the hydrate phase. The replacement kinetics at PCO_2 = 3.5 MPa was faster than that at PCO_2 = 2.2 MPa at the early stage of the replacement, even though the difference became insignificant as the replacement proceeded. The changes in the normalized area of the Raman peaks for the enclathrated CO_2 molecules are presented in Fig. 3.1.2 (b). The Raman peak areas of the CO_2 hydrate at 1,276 cm^{-1} and 1,380 cm^{-1} were combined and normalized. As was observed for the enclathrated CH_4 molecules in Fig. 3.1.2 (a), at a higher PCO_2 (3.5 MPa), the growth rate of the CO_2 hydrate was also faster at the early stage of the replacement despite almost the same final rate at two different PCO_2s. The changes in the normalized area for the CH_4 and CO_2 hydrates during the replacement in the sI hydrates demonstrated that the kinetics of the guest exchange between CH_4 and CO_2 molecules at the early stage of the replacement could be affected by the driving force for the replacement (i.e., PCO_2).

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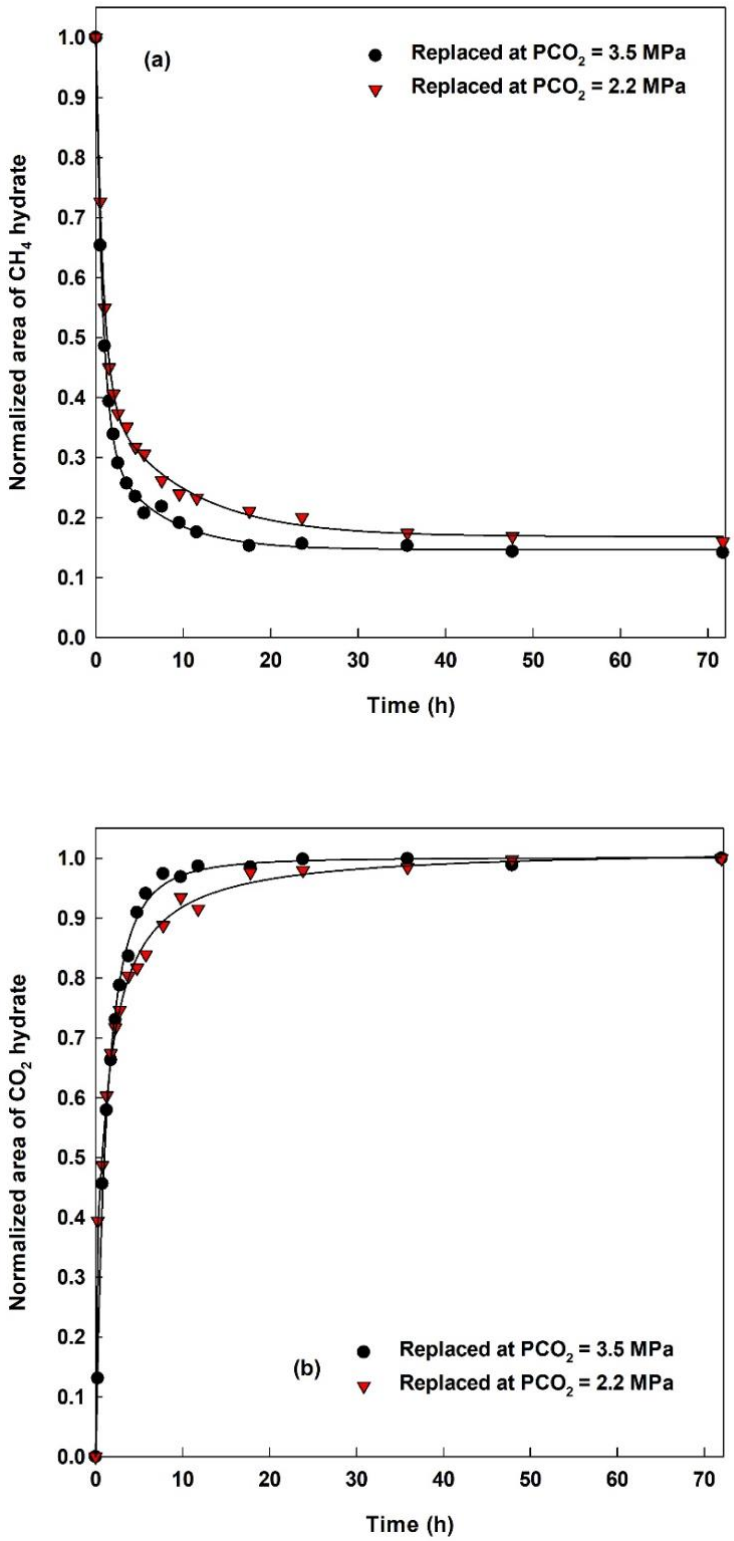


Fig. 3.1.2 Changes in the normalized area of the Raman peaks for the enclathrated (a) CH_4 and (b) CO_2 molecules at PCO_2 = 2.2 and 3.5 MPa.

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To observe the cage-dependent exchange behavior of the CH_4 molecules during the replacement at PCO_2 = 2.2 and 3.5 MPa, the ratio of the peak area for the CH_4 molecules in the large $5^{12}6^2$ to that in the small 5^{12} cages ($A_L/A_{S,\ CH4,\ Raman}$) over time is illustrated in Fig. 3.1.3 along with the cage occupancy ratio ($\theta_L/\theta_{S,\ CH4,\ Raman}$). As shown in Fig. 3.1.3, a rapid decrease in both $A_L/A_{S,\ CH4,\ Raman}$ and $\theta_L/\theta_{S,\ CH4,\ Raman}$ was observed at the early stage of the replacement at both PCO_2 = 2.2 and 3.5 MPa, and it was attributed to the predominant exchange in CH_4 and CO_2 in the large $5^{12}6^2$ cages at the early stage. The molecular size of CH_4 is small enough to be entrapped into both the small 5^{12} and large $5^{12}6^2$ cages of sI hydrate, whereas the inclusion of CO_2 in the small 5^{12} cages is restricted to some extent because the molecular diameter of CO_2 is similar to the diameter of the 5^{12} cage[1]. The cage-dependent guest exchange resulted in the rapid decrease in the $A_L/A_{S,\ CH4,\ Raman}$ and $\theta_L/\theta_{S,\ CH4,\ Raman}$ values at the early stage of the replacement. At a higher PCO_2 (3.5 MPa), the steeper decrease at the early stage and the lower value at the end of the replacement for the $A_L/A_{S,\ CH4,\ Raman}$ and $\theta_L/\theta_{S,\ CH4,\ Raman}$ indicate that the CO_2 injecting pressure can affect not only the guest exchange kinetics but also the equilibrium compositions of the replaced hydrates in the closed system replacement.

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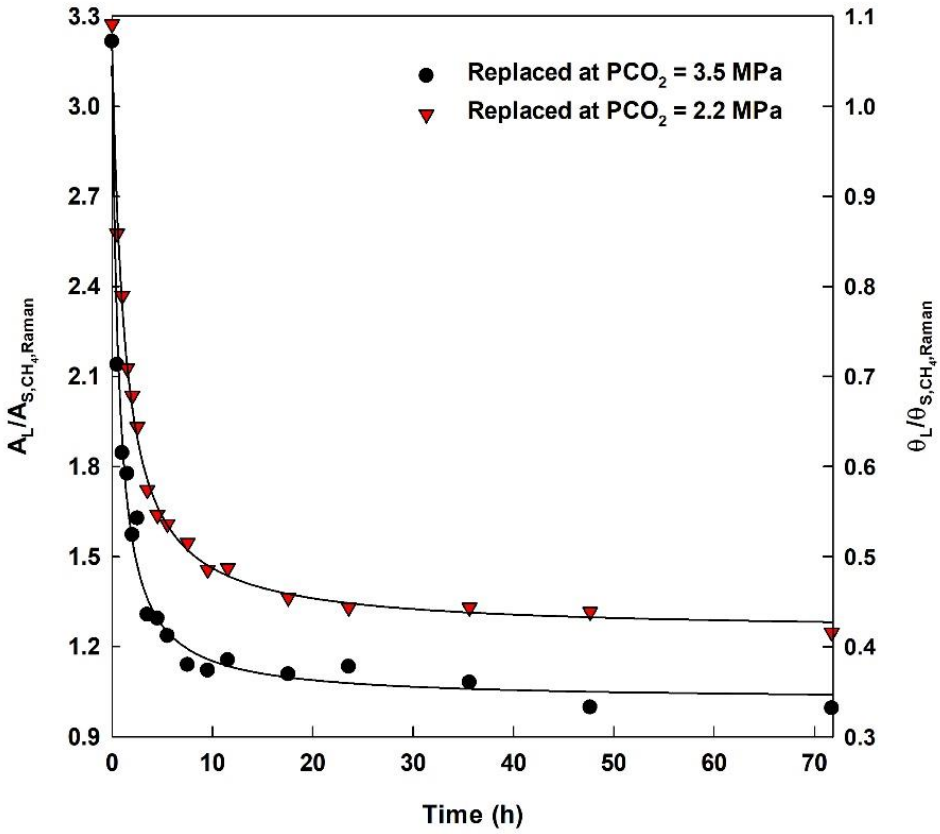


Fig. 3.1.3 Time-dependent area ratio and cage occupancy ratio of the enclathrated CH_4 molecules in the large $5^{12}6^2$ cages and small 5^{12} cages at PCO_2 = 2.2 and 3.5 MPa.

3.1.3. Cage-dependent guest occupancy in the hydrate phase

To verify the value of $\theta_L/\theta_{S,\,CH4,\,Raman}$ before and after the replacement, the cage occupancy ratio of the CH_4 molecules in the large $5^{12}6^2$ and small 5^{12} cages ($\theta_L/\theta_{S,\,CH4,\,NMR}$) at two different PCO_2s of 2.2 MPa and 3.5 MPa was also obtained using NMR spectroscopy. Fig. 3.1.4 shows a stacked plot of the ^{13}C NMR spectra for the hydrate samples before and after the replacement. The ^{13}C NMR spectrum of the initial CH_4 hydrate exhibited two signals at -4.3 and -6.6 ppm that correspond to the CH_4 molecules entrapped in the small 5^{12} and large $5^{12}6^2$ cages of the sI hydrate, respectively. The positions of the resonance peaks from the replaced hydrates were the same as those from the initial CH_4 hydrate, again indicating that the iso-structural replacement occurred in the sI hydrate.

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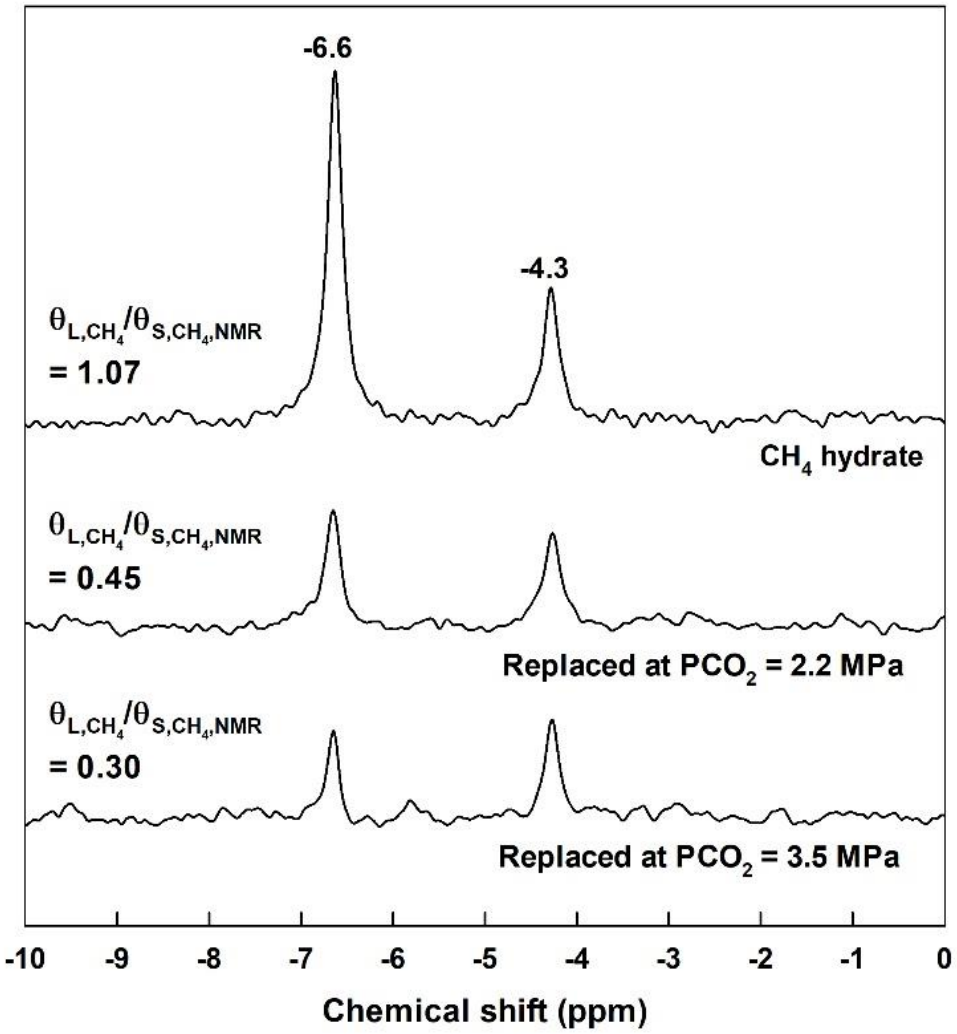


Fig. 3.1.4 ^{13}C NMR spectra of the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa.

The $\theta_L/\theta_{S,\,CH4,\,NMR}$ value of the initial CH_4 hydrate was 1.07, which was comparable with that measured by Raman spectroscopy ($\theta_L/\theta_{S,\,CH4,\,Raman}$ = 1.08). The $\theta_L/\theta_{S,\,CH4,\,NMR}$ values for the replaced hydrates were 0.45 and 0.30 at PCO_2 = 2.2 and 3.5 MPa, respectively, which were in good agreement with those from the Raman spectra ($\theta_L/\theta_{S,\,CH4,\,Raman}$ = 0.41 and 0.35 at PCO_2 = 2.2 and 3.5 MPa, respectively). The NMR result again confirmed that the driving force (PCO_2) for the replacement has a significant effect on the cage-dependent guest exchange during the CH_4–CO_2 replacement.

The crystallographic information on the gas hydrates and the cage occupancy of the guest molecules before and after replacement are important factors to understand the mechanism of the replacement

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process and to quantify the amount of incorporated CO_2 during the replacement. The PXRD patterns of the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa are shown in Fig. 3.1.5. The Rietveld refinement of the PXRD patterns for the hydrate samples before and after the replacement demonstrated the presence of the sI hydrate with a space group $Pm\bar{3}n$ and hexagonal ice (Ih) with a space group P63/mmc and also confirmed the iso-structural replacement in the sI hydrate. The lattice parameters were found to be 11.8822(7) Å (R-weighted pattern (R_{wp}) = 7.2% and goodness-of-fit (χ^2) = 4.6) for the initial CH_4 hydrate, 11.8985(0) Å (R_{wp} = 9.4% and χ^2 = 4.9) for the replaced hydrate at PCO_2 = 2.2 MPa, and 11.9072(5) (R_{wp} = 13.0% and χ^2 = 7.5) for the replaced hydrate at PCO_2 = 3.5 MPa, which were all in good agreement with those for the sI hydrate in the literature[7778].

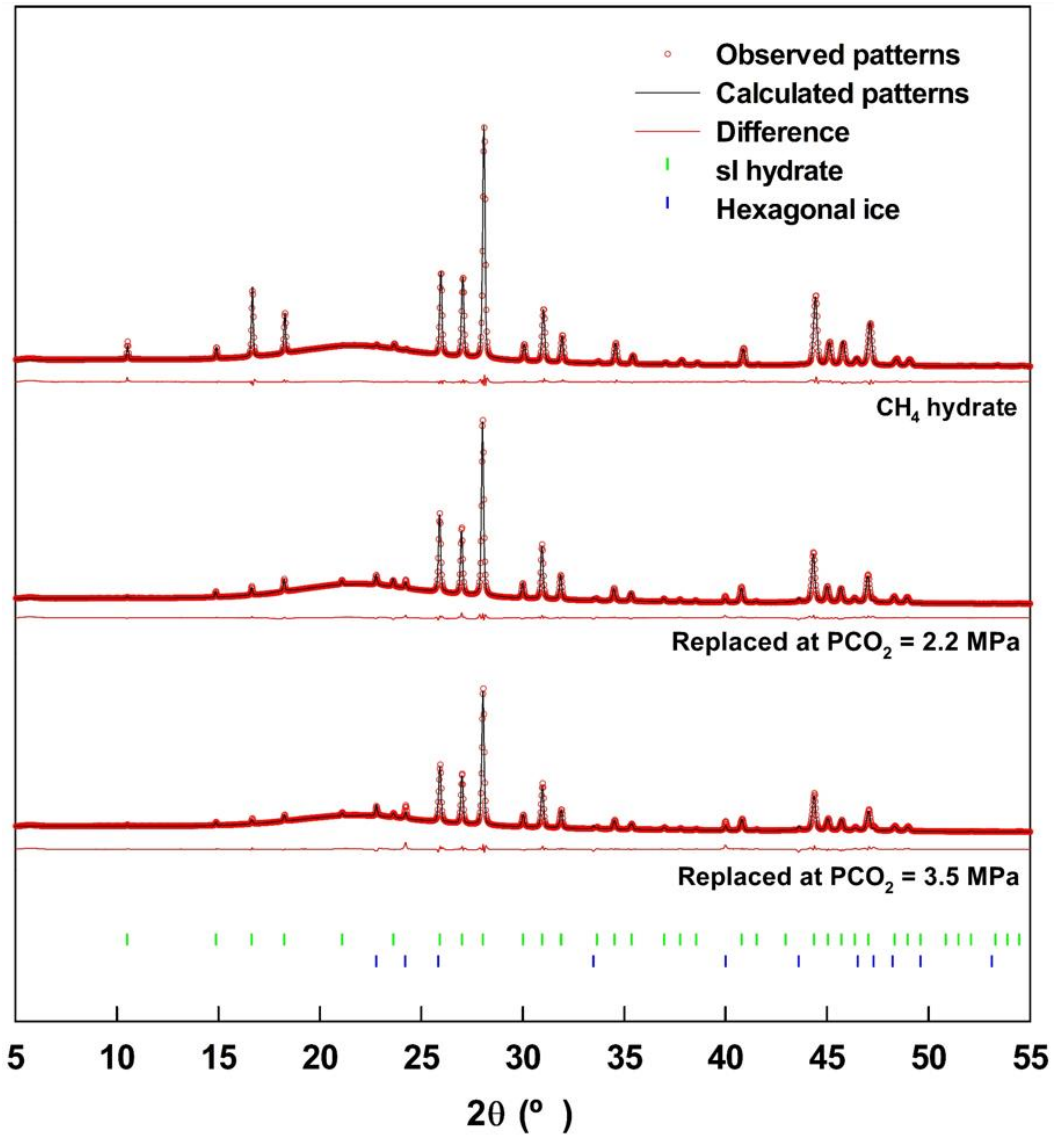


Fig. 3.1.5 PXRD patterns of the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa.

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The cage occupancy of the CH_4 and CO_2 molecules and the weight fraction of ice in the initial CH_4 hydrate and the replaced hydrates were obtained using the Rietveld refinement and are presented in Table 1 and Fig. 6. The low weight fraction of ice for both the initial CH_4 hydrate and the replaced hydrates in Table 1 confirms that no significant hydrate dissociation occurred during the hydrate sampling or during the measurement. For the initial CH_4 hydrate, the cage occupancy of CH_4 in the small 5^{12} cages ($\theta_{S,CH4}$) and large $5^{12}6^2$ cages ($\theta_{L,CH4}$) was found to be $\theta_{S,CH4} = 0.918$ and $\theta_{L,CH4} = 0.998$ with a hydration number of 5.88, which agreed well with the values in the literature[78]. Interestingly, the cage occupancy of the CH_4 molecules before the replacement ($\theta_S = 0.918$ and $\theta_L = 0.998$) was almost the same as that of CH_4 and CO_2 molecules in each hydrate cage of sI after the replacement: $\theta_S = 0.873$, $\theta_L = 0.998$ at $PCO_2 = 2.2$ MPa and $\theta_S = 0.879$, $\theta_L = 0.998$ at $PCO_2 = 3.5$ MPa. Previous studies indicated that the iso-structural replacement in the sI hydrates occurred without a significant formation or dissociation of the hydrate phase[33]. Thus, the one-to-one exchange of CH_4 and CO_2 in the sI hydrates led to the almost constant cage occupancies of the guest molecules in the small 5^{12} and large $5^{12}6^2$ cages of the sI hydrates before and after the replacement.

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Table 3.1.1 Cage occupancy of the guest molecules, occupancy ratio of the CH_4 molecules, weight fraction of ice, and R_{wp} of the PXRD refinement for the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa.

	θ_L in sI hydrate		θ_S in sI hydrate		θ_L / θ_S for CH_4 in sI hydrate	Ice (wt %)	Rwp (%)
	CO_2	CH_4	CO_2	CH_4			
CH_4 hydrate	-	0.992	-	0.918	1.09	0.71	7.17
Replaced at P_{CO2} = 2.5 MPa	0.731	0.264	0.238	0.635	0.42	4.53	9.41
Replaced at P_{CO2} = 3.5 MPa	0.795	0.203	0.253	0.626	0.32	5.93	13.0

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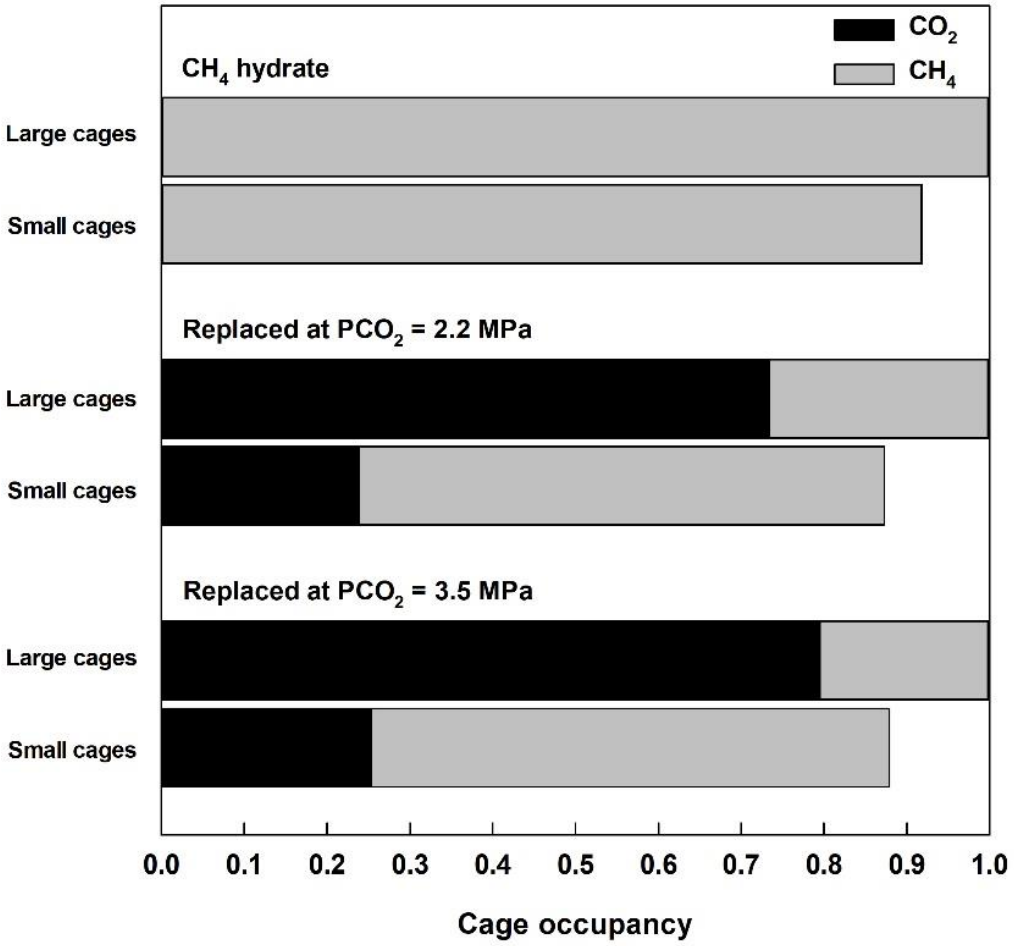


Fig. 3.1.6 Cage occupancy of the CH_4 and CO_2 molecules in the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa.

Fig. 3.1.6 shows guest distributions in the small 5^{12} and large $5^{12}6^2$ cages of the sI hydrates before and after the replacement. 73.5% and 79.7% of the initial CH_4 in the large $5^{12}6^2$ cages were replaced with CO_2, whereas only 30.8% and 31.8% of initial CH_4 in the small 5^{12} cages were produced with CO_2 injection at PCO_2 = 2.2 and 3.5 MPa, respectively. This result corresponded well with the behavior observed by the NMR and Raman spectroscopic measurements and indicated that the continuous decrease in θ_L/θ_S during the replacement was attributed to the predominant exchange of CH_4 and CO_2 in the large $5^{12}6^2$ cages. Additionally, the CO_2 occupancy in the large $5^{12}6^2$ cages ($\theta_{L,CO2}$ = 0.795) at a higher PCO_2 (3.5 MPa) was larger than that ($\theta_{L,CO2}$ = 0.734) at a lower PCO_2 (2.2 MPa), whereas the CO_2 cage occupancy in the small 5^{12} cages was similar at two different PCO_2s ($\theta_{S,CO2}$ = 0.238 at PCO_2 = 2.2 MPa and $\theta_{S,CO2}$ = 0.253 at PCO_2 = 3.5 MPa). The larger increase in θ_{CO2} in the large $5^{12}6^2$ cages

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than that in the small 5^{12} cages at a higher PCO_2 indicated that the CO_2 injecting pressure as the driving force for the CH_4–CO_2 replacement had a greater effect on the large $5^{12}6^2$ cages.

Fig. 3.1.7 illustrates the CO_2 compositions in the hydrate phase after the replacement (i.e., the extent of replacement) obtained from the PXRD and GC. The CO_2 compositions calculated from the CO_2 cage occupancies (63.1% at PCO_2 = 2.2 MPa and 68.1 % at PCO_2 = 3.5 MPa) were in good agreement with those measured by GC (64.2% at PCO_2 = 2.2 MPa and 69.1% at PCO_2 = 3.5 MPa). The slightly higher extent of replacement at a higher PCO_2 (3.5 MPa) was mainly caused by the higher CO_2 occupancy in the large $5^{12}6^2$ cages of the sI as revealed by NMR, Raman, and PXRD.

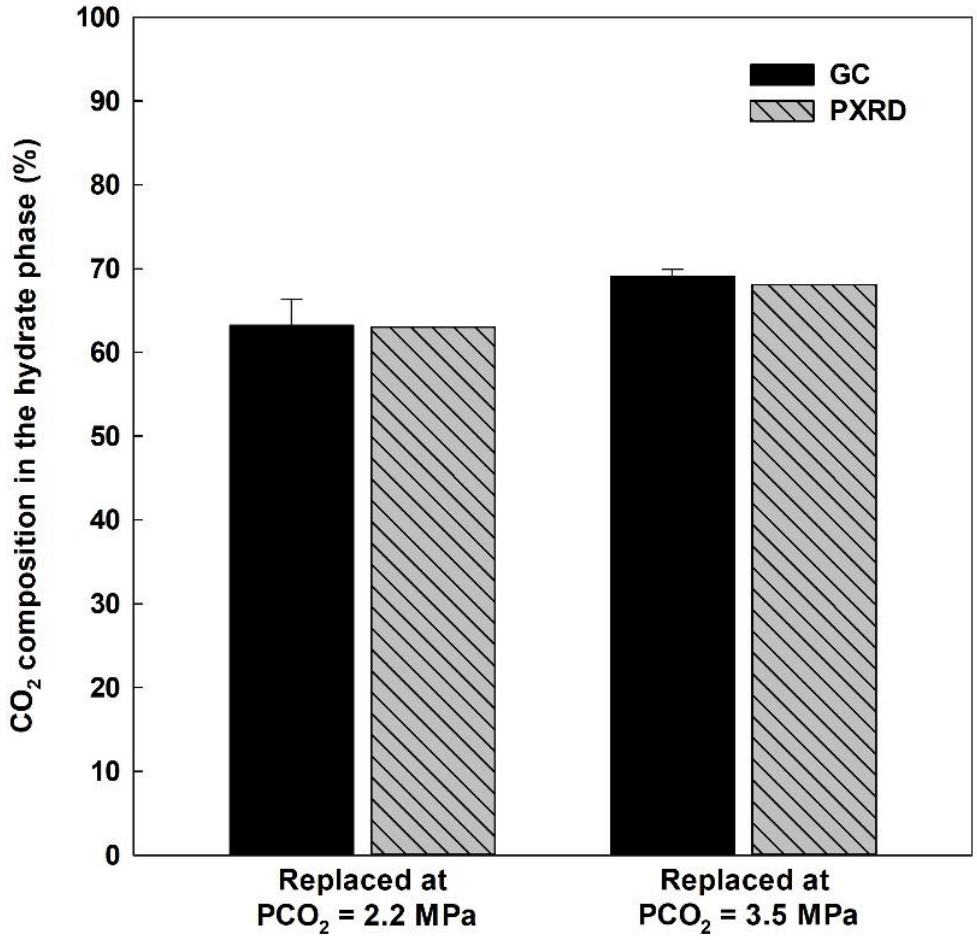


Fig. 3.1.7. Extent of the replacement obtained from GC and PXRD at PCO_2 = 2.2 and 3.5 MPa.

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3.1.4. Multi-methodological analysis of guest exchange behavior

In this study, the time-dependent guest exchange behaviors during the $CH_4 - CO_2$ replacement were observed using the in-situ Raman spectroscopy, and the cage occupancy of CH_4 and CO_2 molecules in the small 5^{12} and large $5^{12}6^2$ cages before and after the replacement were obtained from the Rietveld refinement of PXRD patterns. Furthermore, the accuracy of the cage occupancy of the CH_4 molecules in the sI hydrate was also cross-checked from the ^{13}C NMR spectra because the two resonance peaks from the CH_4 molecules enclathrated in the large $5^{12}6^2$ and small 5^{12} cages of sI hydrate were completely split. As shown in Fig. 3.1.8, the $\theta_L/\theta_{S,\,CH4}$ values were almost identical to one another irrespective of the analytical method, suggesting that a multi-methodological method, which refers to the combination of the in-situ Raman spectroscopy and PXRD analysis methods, could be used to observe the time-dependent exchange behavior in the sI hydrates.

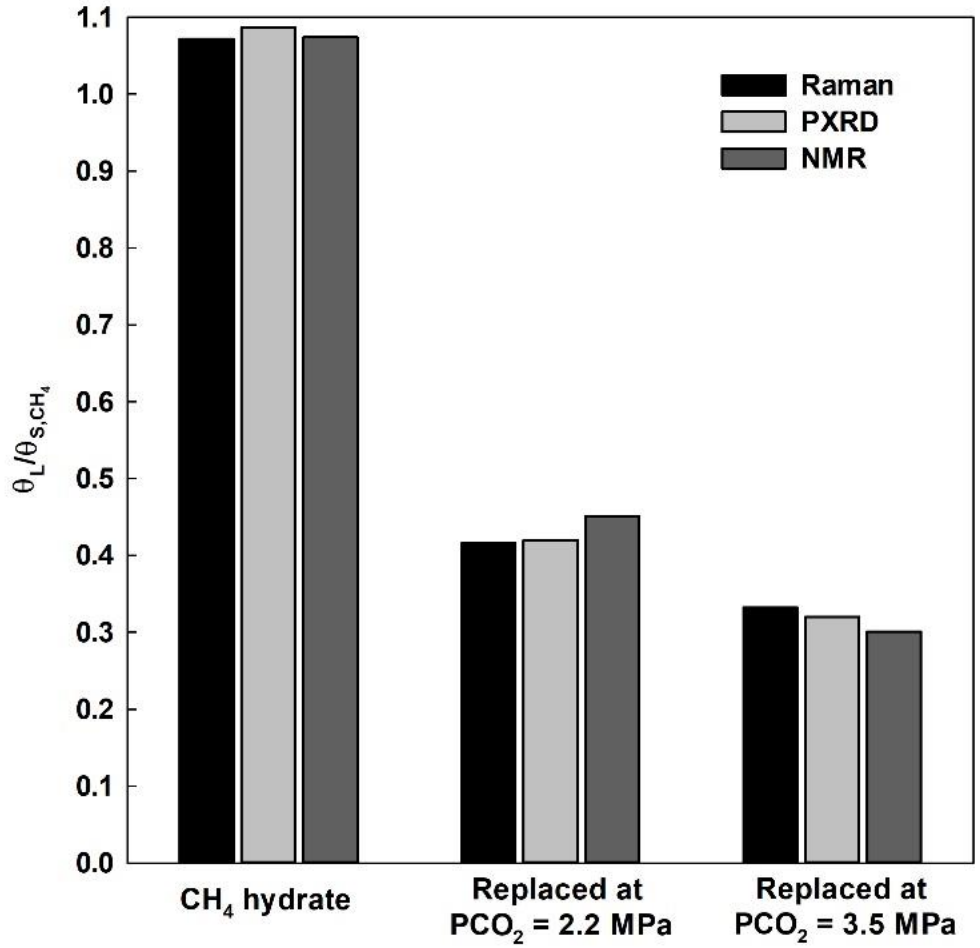


Fig. 3.1.8. Cage occupancy ratio of the enclathrated CH_4 molecules in the large $5^{12}6^2$ cages and small 5^{12} cages of the initial CH_4 hydrate and the replaced hydrates at PCO_2 = 2.2 and 3.5 MPa.

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The changes in the cage occupancy of the CH_4 molecules in the large $5^{12}6^2$ and small 5^{12} cages at PCO_2 = 2.2 and 3.5 MPa are shown in Fig. 3.1.9. The time-dependent cage occupancy of CH_4 molecules in each hydrate cage was calculated by Eq. (1).

$$\theta_t = \frac{A_t - A_f}{A_i - A_f}(\theta_i - \theta_f) + \theta_f \quad (1)$$

where A_i, A_f and A_t are Raman peak area at 0 h, 72 h and t h, respectively, and θ_i, θ_f and θ_t are the cage occupancy at 0 h, 72 h and t h, respectively. The θ_L and θ_S values of the CH_4 molecules before and after the replacement were obtained from the PXRD analysis, and the changes in the θ_L and θ_S values during the replacement were calculated from the time-dependent changes in the Raman peak areas from the enclathrated CH_4 molecules in the large $5^{12}6^2$ cages and the small 5^{12} cages. As shown in Fig. 3.1.9, the CO_2 inclusion and resultant CH_4 production occurred mainly at the early stage of the replacement, and the CH_4 occupancy in both the small 5^{12} and large $5^{12}6^2$ cages were lower at a higher PCO_2. Note that for the large $5^{12}6^2$ cages, a marked difference in the $\theta_{L,CH4}$ values just after the replacement (0.5 h) was observed at two different PCO_2s (θ_L = 0.735 at PCO_2 = 2.2 MPa and 0.634 at PCO_2 = 3.5 MPa), and the difference was almost maintained until the end of the replacement (72 h) (θ_L = 0.264 at PCO_2 = 2.2 MPa and 0.203 at PCO_2 = 3.5 MPa). However, for the small 5^{12} cages, the difference in the $\theta_{S,CH4}$ values was very small throughout the replacement (θ_S = 0.867 at PCO_2 = 2.2 MPa and 0.865 at PCO_2 = 3.5 MPa at 0.5 h, θ_S = 0.635 at PCO_2 = 2.2 MPa, and 0.626 at PCO_2 = 3.5 MPa at 72 h). This distinguishable replacement kinetics depending on the cage type at different PCO_2 indicates that the CO_2 injecting pressure, which is the driving force of the replacement, can affect the cage-dependent guest exchange in the sI hydrates.

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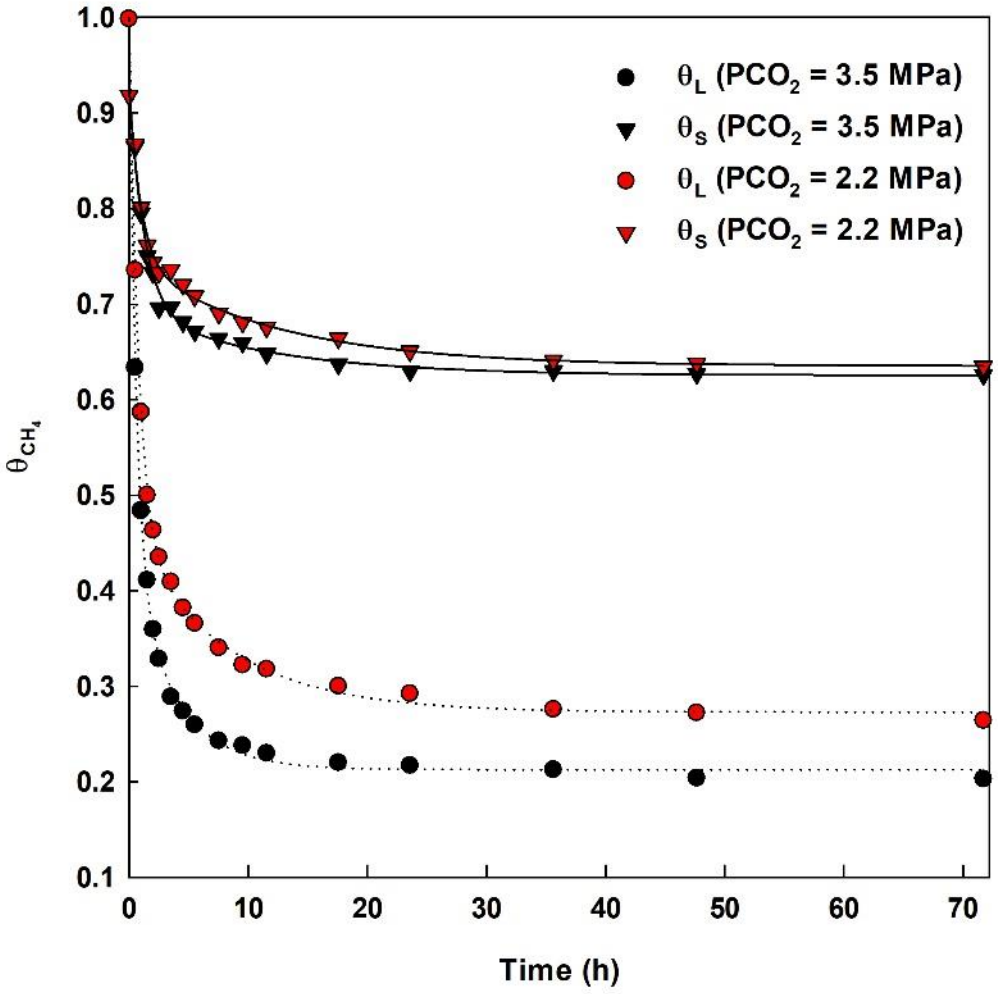


Fig. 3.1.9. Time-dependent cage occupancy of the CH_4 molecules in the large $5^{12}6^2$ cages and small 5^{12} cages at PCO_2 = 2.2 and 3.5 MPa.

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To examine accurately the mechanism of the $CH_4 - CO_2$ replacement and the effect of PCO_2 on the replacement behavior, the cage occupancy of the CO_2 molecules in the small 5^{12} and large $5^{12}6^2$ cages during the replacement should be analyzed at PCO_2 = 2.2 and 3.5 MPa. However, conducting a quantitative analysis of the time-dependent CO_2 enclathration in each cage of the sI hydrates using Raman spectroscopy is difficult because the Raman peaks from the CO_2 molecules entrapped in the small 5^{12} and large $5^{12}6^2$ cages are not cage-specific. Therefore, in this study, the changes in $\theta_{S,CO2}$ and $\theta_{L,CO2}$ were calculated by assuming a one-to-one exchange between the CH_4 and CO_2 molecules in the sI hydrates; an increment in $\theta_{S,CO2}$ and $\theta_{L,CO2}$ is equal to a decrement in $\theta_{S,CH4}$ and $\theta_{L,CH4}$, and the values were re-fitted with the results of PXRD analysis. Fig. 3.1.10 presents the changes in the CO_2 compositions in each hydrate cage along with the time-dependent extent of replacement at PCO_2 = 2.2 and 3.5 MPa. At a higher PCO_2 (3.5 MPa), a more rapid increase in the extent of replacement at the early stage and a slightly higher CO_2 composition throughout the replacement were observed, also indicating that the driving force for the replacement affected not only the extent of the replacement but also the kinetics of the $CH_4 - CO_2$ replacement. Furthermore, it is clear that a guest exchange occurring in the large $5^{12}6^2$ cages had a dominant effect on the replacement efficiency at both PCO_2 conditions.

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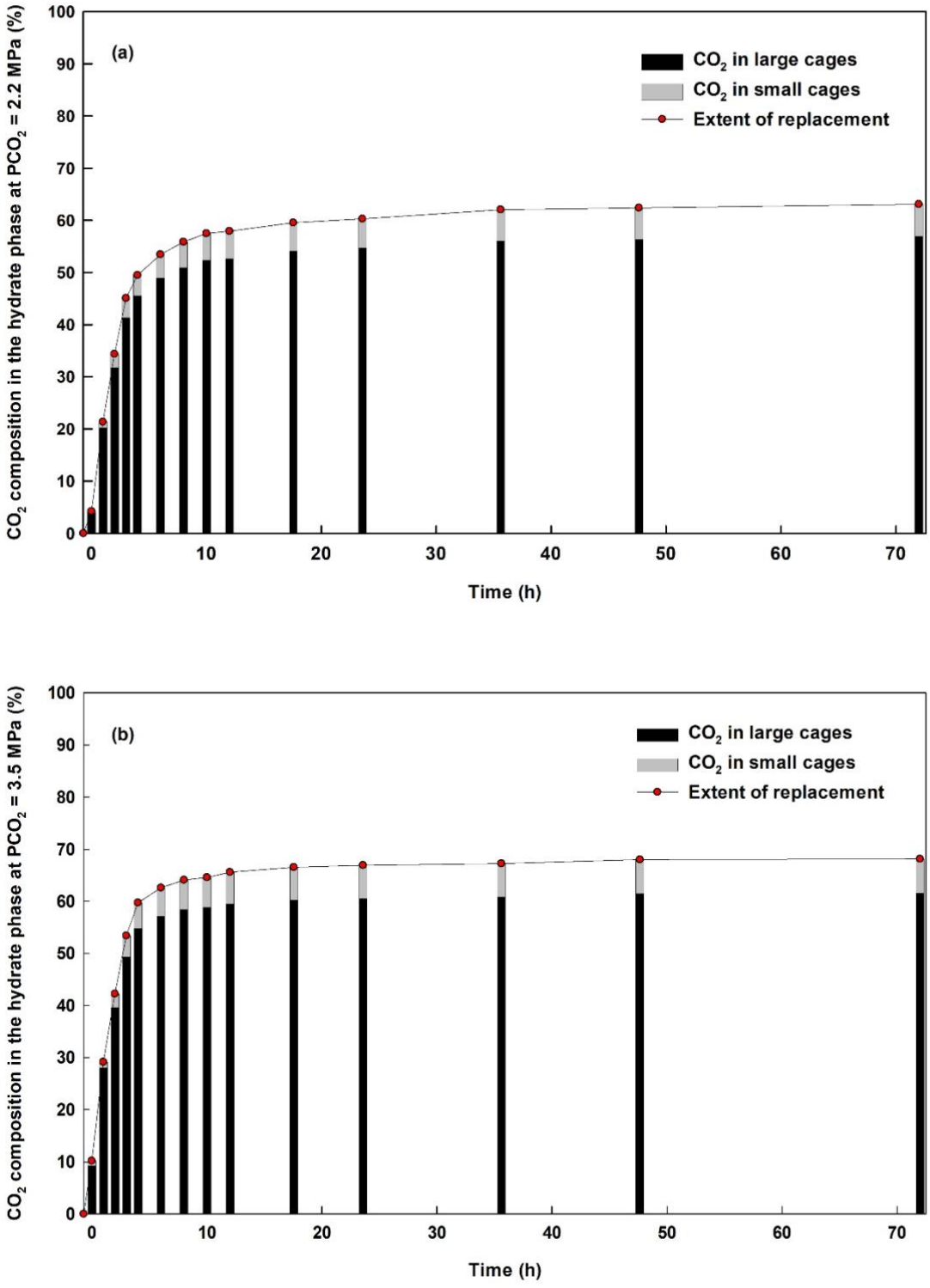


Fig. 3.1.10. Changes in the CO_2 compositions in each hydrate cage along with the time-dependent extent of replacement at PCO_2 = 2.2 and 3.5 MPa.

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To examine the effect of PCO_2 on the cage-dependent CO_2 enclathration, the changes in $\theta_L/\theta_{S,CO2}$ during the replacement are presented in Fig. 3.1.11. The rapid decline and the subsequent gradual decrease in the $\theta_L/\theta_{S,CO2}$ values at both PCO_2 conditions (2.2 and 3.5 MPa) indicated that the CO_2 inclusion in the large $5^{12}6^2$ cages was predominant at the early stage of the replacement.

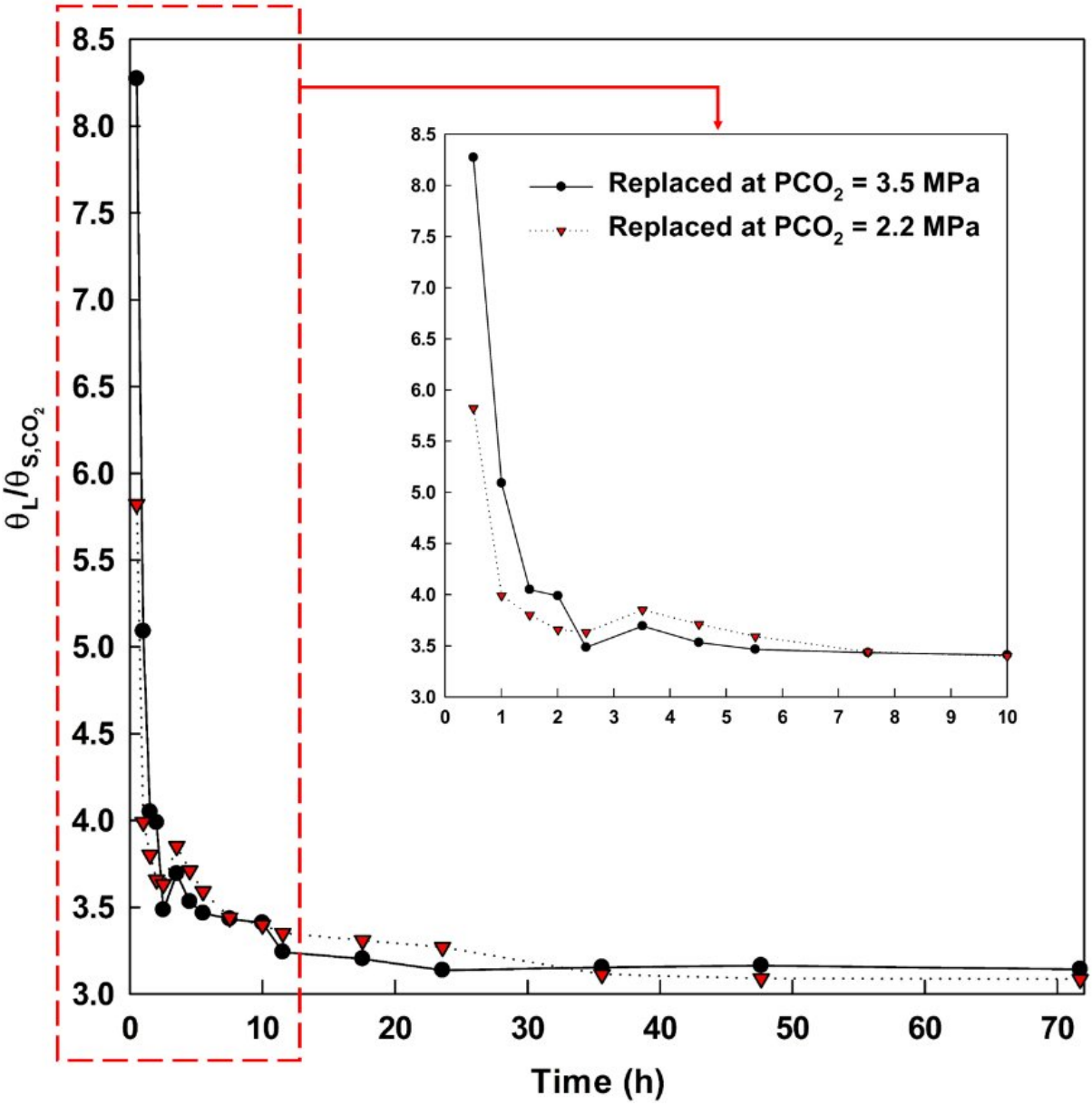


Fig. 3.1.11. Time-dependent cage occupancy ratio of the enclathrated CO_2 molecules in the large $5^{12}6^2$ cages and small 5^{12} cages at PCO_2 = 2.2 and 3.5 MPa. The inset figure shows the changes for the initial 10 h.

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In addition, the inset figure in Fig. 3.1.11, which enlarges the changes in the $\theta_L/\theta_{S,\,CO2}$ values for the initial 10 h, shows a remarkable difference in the declining behavior of $\theta_L/\theta_{S,\,CO2}$ at two PCO_2 conditions. Immediately after the CO_2 injection for the replacement, the $\theta_L/\theta_{S,\,CO2}$ value at PCO_2 = 3.5 MPa (8.27 at 0.5 h) was much higher than that at PCO_2 = 2.2 MPa (5.82 at 0.5 h), whereas the $\theta_L/\theta_{S,\,CO2}$ values at the end of the replacement were similar at both PCO_2 conditions ($\theta_L/\theta_{S,\,CO2}$ = 3.08 at PCO_2 = 2.2 MPa and $\theta_L/\theta_{S,\,CO2}$ = 3.14 at PCO_2 = 3.5 MPa). Figs. 3.1.10 and 3.1.11 clearly demonstrated that the extent of replacement was more influenced by CO_2 replacement in the large $5^{12}6^2$ cages than that in the small 5^{12} cages. Accordingly, the faster guest exchange kinetics and the slightly higher extent of the replacement occurred at a higher PCO_2 because of the higher cage occupancy of CO_2 in the large $5^{12}6^2$ cages.

3.1.5. Conclusions

In this study, the time-dependent guest exchange in the CH_4–CO_2 replacement in a closed system was investigated using GC, PXRD, NMR, and in-situ Raman spectroscopy at PCO_2 = 2.2 and 3.5 MPa. The in-situ Raman spectra demonstrated that the rate of CO_2 inclusion and the resultant CH_4 depletion in the hydrate phase during the replacement was faster at a higher PCO_2, and thus the driving force for the replacement could affect the guest exchange behavior. The ^{13}C NMR spectra confirmed that the $\theta_L/\theta_{S,CH4}$ value after the replacement was significantly smaller than that before the replacement and became much smaller at a higher PCO_2, thus indicating that the CO_2 enclathration during the replacement occurred primarily in the large $5^{12}6^2$ cages. The Rietveld refinement of the PXRD patterns offered a quantitative cage occupancy of the CH_4 and CO_2 molecules before and after the replacement. After the replacement, the $\theta_{L,CO2}$ value was higher at a higher PCO_2 (3.5 MPa), whereas the $\theta_{S,CO2}$ value was not significantly dependent on PCO_2. The time-dependent θ_L and θ_S values of the guest molecules, which were obtained by the combination of the PXRD analysis and time-dependent changes in the Raman peak areas, clearly demonstrated that the predominant CO_2 replacement in the large $5^{12}6^2$ cages had a greater effect on the extent of replacement and that the guest exchange kinetics was faster at a higher PCO_2. The experimental results obtained in this study provide valuable insights into the mechanism of the CH_4–CO_2 replacement and the effect of the CO_2 injecting pressure on the kinetics and equilibrium of the guest exchange in gas hydrates.

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3.2. N_2-assisted CH_4 hydrate dissociation and replacement

This chapter includes the published contents:

Mok, J., Choi, W., & Seo, Y. (2021). The dual-functional roles of N_2 gas for the exploitation of natural gas hydrates: An inhibitor for dissociation and an external guest for replacement. Energy, 232, 121054. Reproduced with permission. Copyright © 2021 Elservier Inc.

3.2.1. Abstract

In this study, the dissociation behavior and the time-dependent guest distributions of methane (CH_4) hydrate after the injection of gaseous N_2 were closely investigated using gas chromatography, *in situ* Raman spectroscopy, and nuclear magnetic resonance (NMR) spectroscopy for two different N_2 gas injecting pressures (4.0 and 8.0 MPa) at three different temperatures (268.8, 274.2, and 278.2 K). The dissociation kinetics of the CH_4 hydrate were accelerated at a higher temperature and a lower N_2 injecting pressure. Time-dependent Raman spectra confirmed that the N_2 molecules began to be captured in the hydrate cages immediately after the N_2 gas injection at 268.8 K, and the extent of N_2 incorporation in the hydrate phase increased at a higher N_2 gas injecting pressure. The ^{13}C NMR spectra revealed that an increase in the N_2 composition in the hydrate phase induced a structural transition of the CH_4 + N_2 hydrates from sI to sII and that N_2 molecules were preferentially captured in the small (5^{12}) cages of the CH_4 + N_2 hydrates. The overall results provide deep insight into the exact role of N_2 molecules in the inhibitor injection method and CH_4 – CO_2 + N_2 replacement methods for the exploitation of natural gas hydrates.

3.2.2. Kinetics of the CH_4 hydrate dissociation induced by gaseous N_2 injection

To examine the effect of the temperature and N_2 injecting pressure (P_{N2}) on the dissociation behavior of the CH_4 hydrate, in situ GC measurements were conducted at two different P_{N2} (4.0 and 8.0 MPa) and three different temperatures (268.8, 274.2, and 278.2 K), as shown in Fig. 3.2.1. As hydrate dissociation accompanies the release of the entrapped guest gas molecules, an increase in the system pressure is a critical indicator in detecting whether the gas hydrates dissociate or not. As shown in Figs. 3.2.1(a) and (b), for P_{N2} = 4.0 MPa and 8.0 MPa at 278.2 K, the pressure of the system gradually increased immediately after the injection of the N_2 gas and finally reached saturation, indicating that the dissociation of the CH_4 hydrate was initiated by the N_2 gas injection. When there was no further increase in the pressure, the vapor phase of the reactor was immediately evacuated, and the reactor was heated up to 293.2 K, but no residual hydrate phase was observed. This shows that the stabilization of the pressure in Figs. 3.2.1(a) and (b) was caused by the full decomposition of the CH_4 hydrate. The CH_4

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composition in the vapor phase also increased over time and reached saturation after the full decomposition of the CH_4 hydrate.

The full decomposition of the CH_4 hydrate was also observed in the case of P_{N2} = 4.0 MPa and 8.0 MPa at 274.2 K, as shown in Figs. 3.2.1 (c) and (d). The pressure and the CH_4 composition in the vapor phase converged into approximately 5.7 MPa and 35% for P_{N2} = 4.0 MPa, respectively, and approximately 9.4 MPa and 21% for P_{N2} = 8.0 MPa, respectively, at both 274.2 K and 278.2 K, indicating that the N_2 gas injection at both 274.2 K and 278.2 K resulted in the full decomposition of the CH_4 hydrate in the closed system. However, it should be noted that at 268.8 K, the pressure and the CH_4 composition in the vapor phase did not reach these values after the N_2 gas injection. As an increase in the system pressure in a closed system is a good indicator of the dissociation of gas hydrates, as mentioned earlier, the conversion to a relatively lower system pressure and CH_4 composition in the vapor phase implied the incomplete dissociation of the CH_4 hydrate and the consequent existence of the residual hydrate phase in the reactor.

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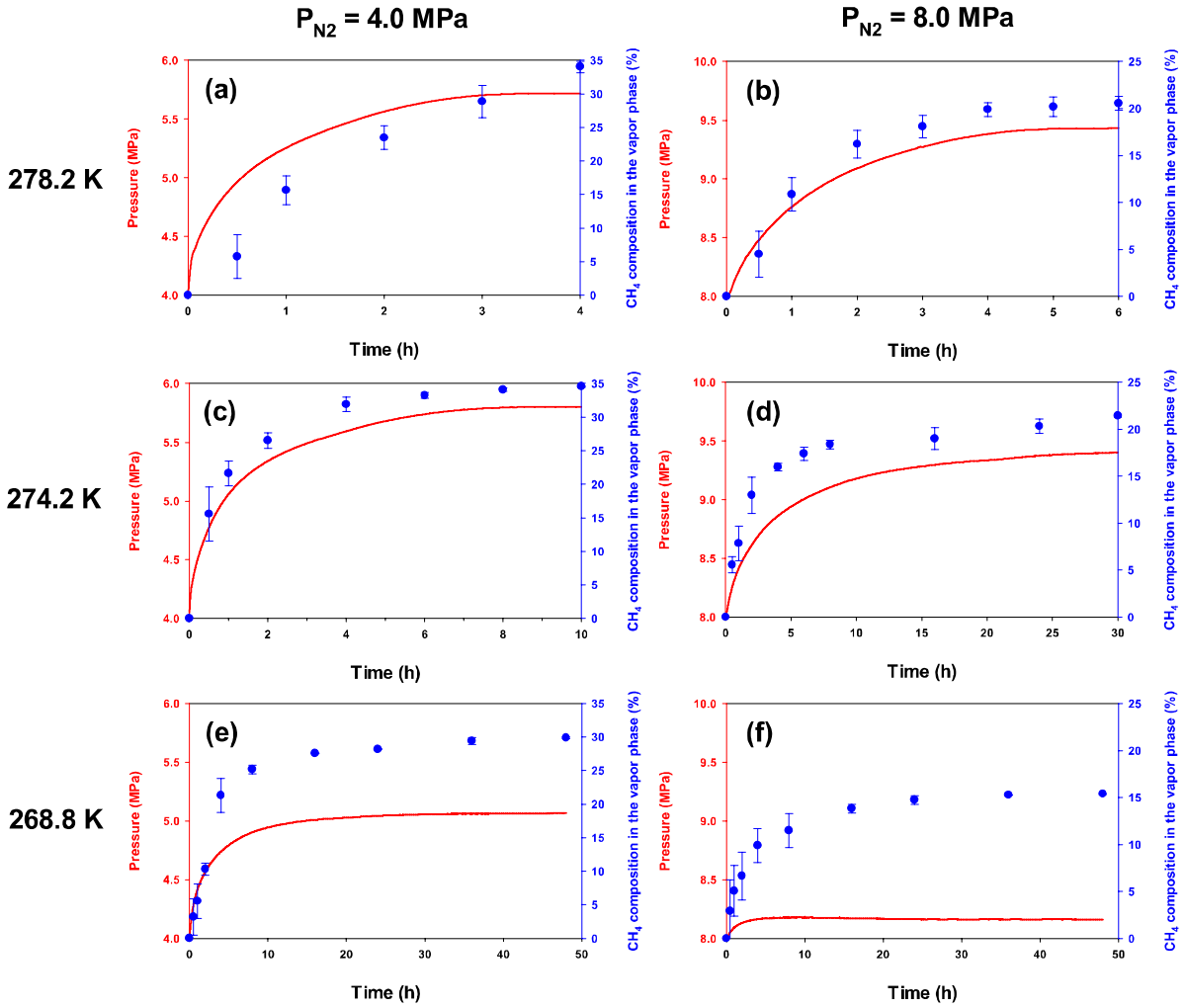


Fig. 3.2.1 Changes in pressure and CH_4 composition in the vapor phase for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K, 274.2 K, and 278.2 K

To examine the quantitative dissociation behavior of the CH_4 hydrate by N_2 gas injection, the normalized residual amounts of CH_4 in the hydrate phase in each case were obtained based on the results from the *in situ* GC measurements (Fig. 3.2.2). The changes in the normalized residual CH_4 in the hydrate phase under each N_2 injecting condition showed that the dissociation behavior of the CH_4 hydrate varied depending on the temperature and the P_{N2} conditions. The dissociation rate of the CH_4 hydrate was accelerated with an increase in temperature, and the termination of the CH_4 hydrate dissociation was faster at P_{N2} = 4.0 MPa than at P_{N2} = 8.0 MPa (4 h [P_{N2} = 4.0 MPa] and 6 h [P_{N2} = 8.0 MPa] at 278.2 K; 10 h [P_{N2} = 4.0 MPa] and 30 h [P_{N2} = 8.0 MPa] at 274.2 K). The faster kinetics of the CH_4 hydrate dissociation induced by the N_2 gas injection were observed at a higher temperature and a

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lower P_{N2}. Conversely, the CH_4 captured in the hydrate phase was not completely released at 268.8 K even after a long time, and 22% and 39% of the initial CH_4 in the hydrate phase was still trapped in the hydrate cages for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, respectively. This indicates that a higher extent of CH_4 production could be obtained at a lower P_{N2} when incomplete hydrate dissociation occurred at the N_2-induced CH_4 production. Kang et al. already reported incomplete hydrate dissociation through air injection in a closed system and demonstrated that the dissociation of the CH_4 hydrate in a high-pressure reactor through the injection of air and air + CO_2 mixtures was terminated when the vapor phase composition reached a certain CH_4 concentration, which is called the critical methane concentration (CMC)[24].

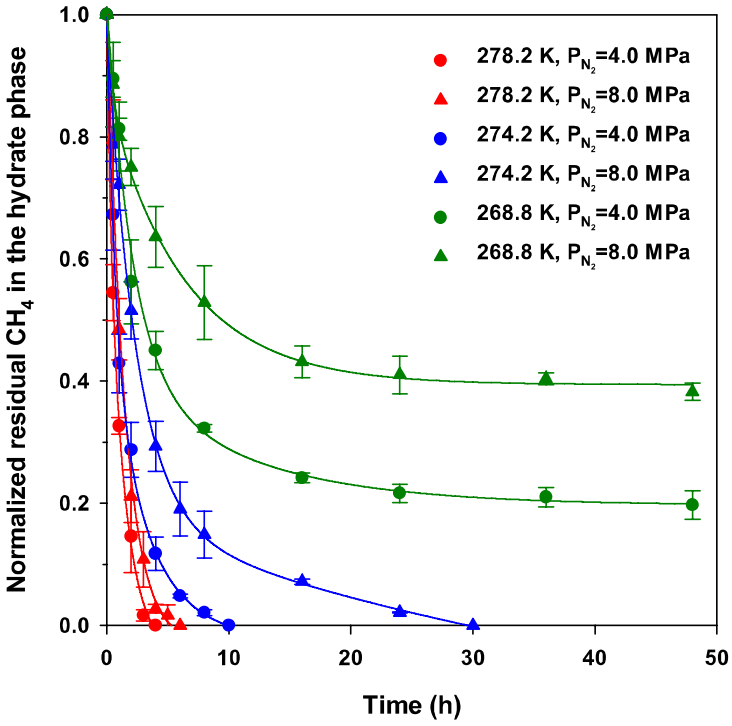


Fig. 3.2.2. Changes in the normalized amount of residual CH_4 in the hydrate phase for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K, 274.2 K, and 278.2 K

The dissociation of gas hydrates is caused by the thermodynamic instability of the system, as shown in the conventional methods for NGH production, such as depressurization and thermal stimulation. As inhibitor injection is also closely related to the thermodynamic instability of the system, the retention of the thermodynamic driving force for hydrate dissociation is the core issue to avoid the decline in the rate and extent of hydrate dissociation. In the case of gaseous N_2 injection, hydrate dissociation is triggered by the disequilibrium state of the vapor and hydrate phases. As shown in Fig. 3.2.1, the

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thermodynamic driving force for hydrate dissociation could not be maintained during hydrate melting in the closed system due to an increase in the pressure and CH_4 composition in the vapor phase.

Fig. 3.2.3 illustrates the changes in the thermodynamic environment and the paths that CH_4 hydrates go through during dissociation at each temperature for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa. As previously mentioned, at both 274.2 K and 278.2 K, the pressure and the CH_4 composition in the vapor phase converged into 5.7 MPa and 35%, and 9.4 MPa and 21% for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, respectively, after the full decomposition of the CH_4 hydrate. As shown in Fig. 3.2.3 (a), in the case of P_{N2} = 4.0 MPa at 274.2 K, the dissociation of the CH_4 hydrate was initiated immediately after N_2 gas injection (red solid circle), and the pressure and the gas composition in the vapor phase at that point were 4.0 MPa and 100% N_2, respectively. The pressure and the CH_4 composition in the vapor phase gradually increased because of hydrate dissociation, and the changes were terminated by the full decomposition of the hydrate phase (red solid diamond: 5.7 MPa and CH_4 (35%) + N_2 (65%)). However, this point was still located below the equilibrium pressure of the CH_4 (35%) + N_2 (65%) hydrate (blue solid diamond) at 274.2 K, indicating that the possibility of full decomposition of the CH_4 hydrate by N_2 gas injection could be predicted from the hydrate phase equilibria. Therefore, at both 274.2 K and 278.2 K, the CH_4 hydrates for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa were expected to be fully decomposed from the H – L_w – V equilibria of the CH_4 + N_2 hydrates. This prediction is consistent with the results presented in Fig. 3.2.1. However, at 268.8 K (P_{N2} = 4.0 MPa), the equilibrium pressure (4.3 MPa, blue empty diamond) of the CH_4 (35%) + N_2 (65%) hydrate was lower than 5.7 MPa, as shown in Fig. 3.2.3 (a). This implies that the CH_4 hydrate at 268.8 K might not be fully dissociated by N_2 gas injection and that a certain fraction of the hydrate phase might eventually remain. As shown in Fig. 3.2.1(e), the terminal pressure and vapor phase composition were 5.1 MPa and CH_4 (30%) + N_2 (70%), respectively, and 30% CH_4 was considered the CMC at 268.8 K for P_{N2} = 4.0 MPa with our experimental setup. Note that in the case of P_{N2} = 8.0 MPa, at 268.8 K, the N_2 gas injecting pressure (8.0 MPa) was initially higher than the equilibrium pressure (6.2 MPa) of the CH_4 (21%) + N_2 (79%) hydrate (Fig. 3.2.3 (b)), and the CH_4 composition in the vapor phase slightly and gradually increased without a significant increase in pressure, as shown in Fig. 3.2.1(f).

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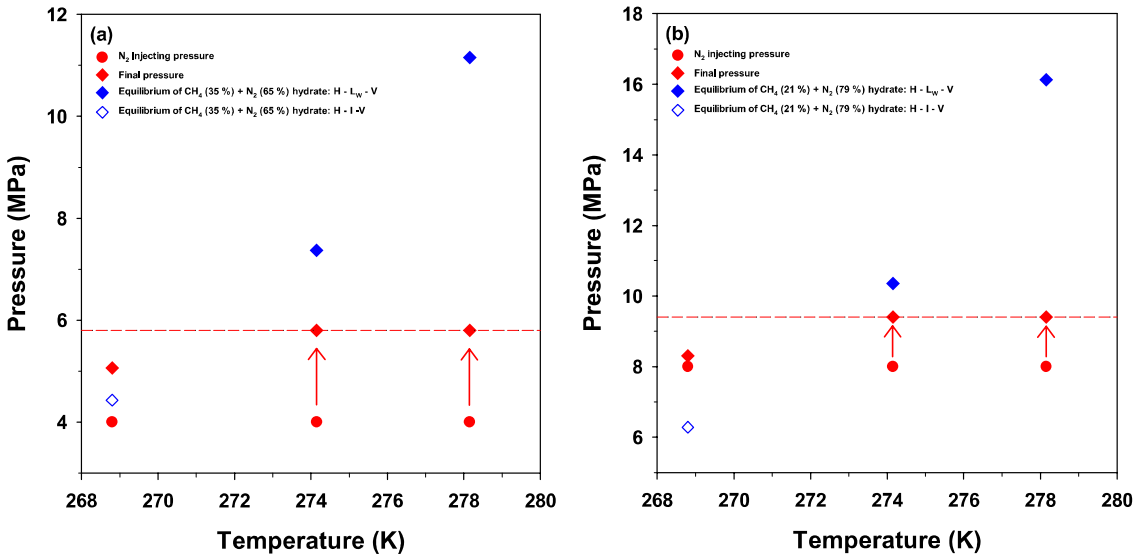


Fig. 3.2.3. The three-phase equilibria of $CH_4 + N_2$ hydrates and traces of pressure changes after N_2 gas injection for (a) P_{N2} = 4.0 MPa and (b) P_{N2} = 8.0 MPa

The guest composition and saturation of the residual hydrate phase 48 h after the N_2 gas injection (P_{N2} = 4.0 MPa and 8.0 MPa) at 268.8 K are shown in Fig. 3.2.4. The saturation of the residual hydrate (S_r) was estimated using the following equation:

$$S_r = \frac{N_r}{N_T C} \cdot 100$$

where N_r is the number of moles of the residual CH_4 in the final hydrate phase, N_T is the number of moles of the total CH_4 in the initial hydrate phase, and C is the mole fraction of CH_4 in the hydrate phase obtained using GC after dissociating the residual hydrate phase. For simplicity, the total cage occupancy of the CH_4 hydrate was assumed to be equal to that of the $CH_4 + N_2$ hydrate. As shown in Fig. 3.2.4, a remarkable fraction of N_2 in the residual hydrate was observed for both P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, indicating that N_2 participated in the hydrate lattices as a hydrate guest during the hydrate dissociation process. This is not surprising because, with the residual hydrate phase, the compositions of the vapor and hydrate phases would be rearranged to achieve a new three-phase equilibrium of the $CH_4 + N_2$ hydrate at a given thermodynamic circumstance, and thus, N_2 should be included in the hydrate phase. The saturations of the residual hydrates were 41% and 77%, and the N_2 compositions in the residual hydrate phase were 47% and 49% for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, respectively. This indicates that as P_{N2} increased, hydrate dissociation declined sharply, whereas the extent of N_2 storage in the hydrate lattices increased.

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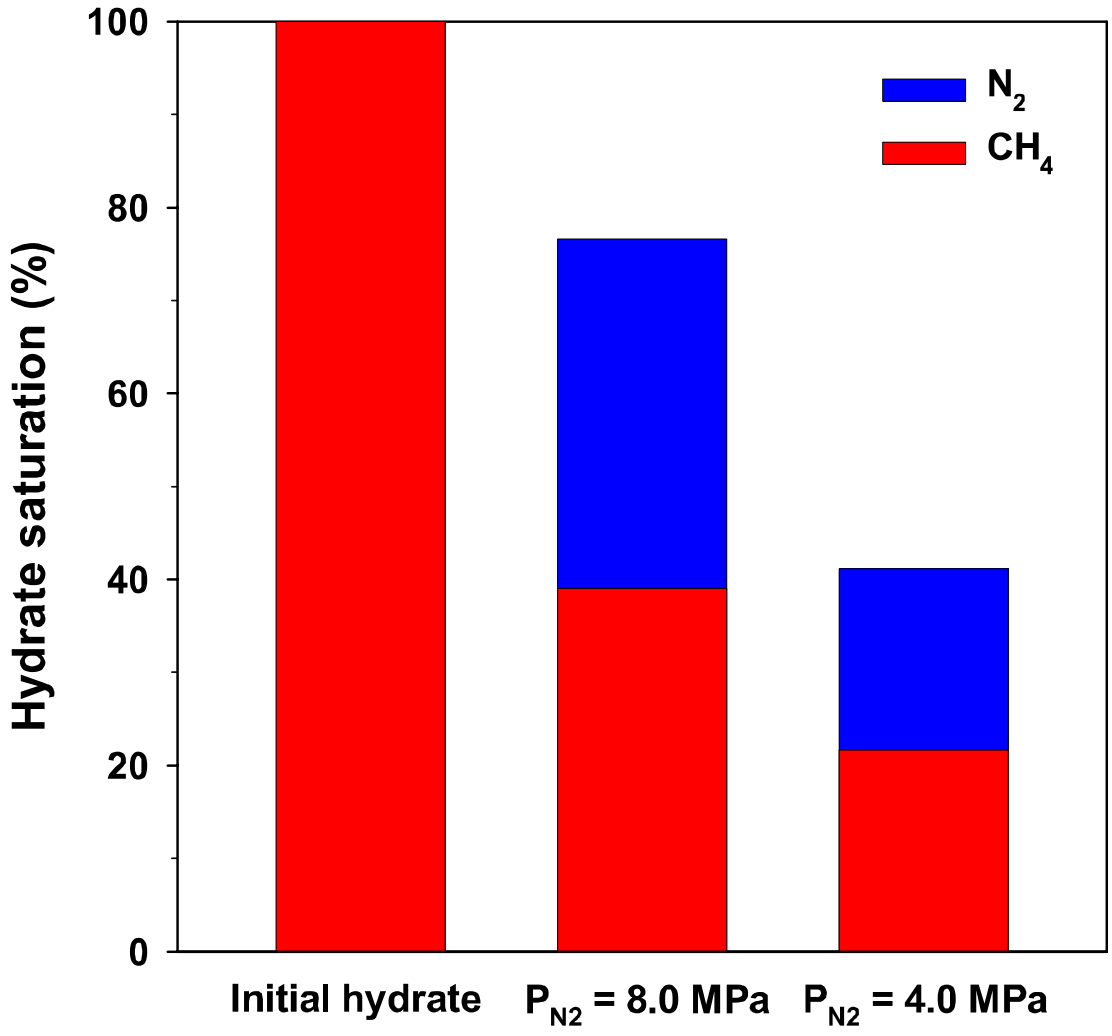


Fig. 3.2.4. Guest composition and saturation of the residual hydrate phase for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K after 48 h

3.2.3. Guest distributions observed by Raman spectroscopy and NMR

To observe the CH_4 release and the N_2 occupation in the hydrate phase after the N_2 gas injection, the time-dependent Raman spectra of the enclathrated CH_4 and N_2 molecules were measured at 268.8 K for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, as shown in Fig. 3.2.5. In situ Raman spectroscopy is a nondestructive analytical tool that can explore the guest molecules in the hydrate phase[79]. Due to the difference in the chemical environment between the free gas molecules and the enclathrated guest molecules, the Raman

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signals of the guest molecules entrapped in the hydrate cages can offer information about the hydrate structure and cage types. The Raman signals for the symmetric C–H stretching mode of the enclathrated CH_4 molecules in the large ($5^{12}6^2$) cages (at 2905 cm^{-1}) and small (5^{12}) cages (at 2915 cm^{-1}) of the sI hydrate were distinguishable from that of the free CH_4 gas molecules (2917 cm^{-1})[73]. The CH_4 hydrate is known as sI, and the N_2 hydrate is known as sII. In addition, the structural transition of the CH_4 + N_2 hydrates from sI to sII was reported to occur when the CH_4 composition in the hydrate phase was lower than 28%[80]. As shown in Fig. 3.2.5, the Raman signal positions of the CH_4 molecules entrapped in the large ($5^{12}6^2$) and small (5^{12}) cages of the sI hydrate did not shift during hydrate dissociation and N_2 enclathration, indicating that the dissociation of the CH_4 hydrate by the N_2 gas injection was an iso-structural process at 268.8 K for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa. For the N_2 molecules, the Raman signal at 2,323 cm^{-1} was assigned to the N_2 molecules in the hydrate phase, and it was distinguishable from the signal at 2,329 cm^{-1} that was assigned to the free N_2 gas molecules. However, as each Raman signal for the N_2 molecules entrapped in the large ($5^{12}6^2$) and small (5^{12}) cages of the sI hydrate was not distinctly separable in this study, the Raman band for the enclathrated N_2 molecules was considered a single signal.

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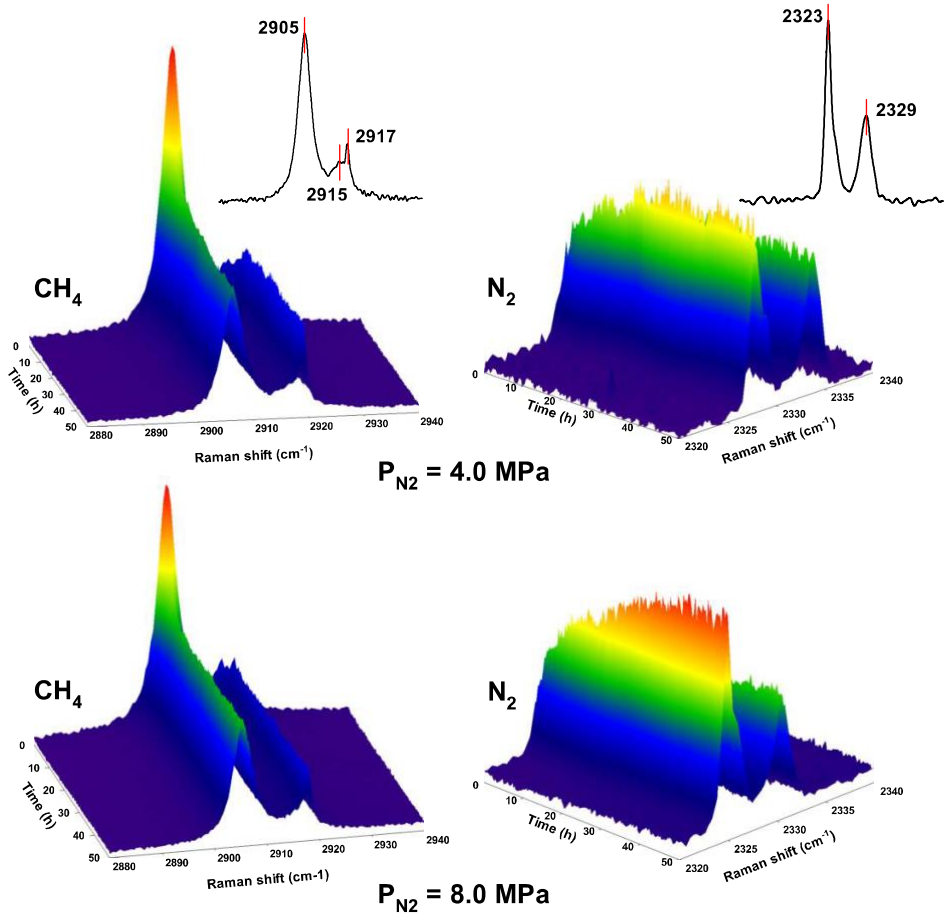


Fig. 3.2.5 *In-situ* Raman spectra of the CH_4 and N_2 molecules in the hydrate and vapor phases for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K

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The normalized intensities of the Raman signal for the enclathrated N_2 molecules during the N_2 gas injection for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K are depicted in Fig. 3.2.6. The normalization was based on the largest Raman signal observed at 2,323 cm^{-1} for P_{N2} = 8.0 MPa, and the quantitative composition ratio of N_2 molecules in the hydrate phase shown in Fig. 3.2.4 was used for the normalization of P_{N2} = 4.0 MPa. Fig. 3.2.6 shows that the N_2 molecules were continuously captured in the hydrate phase and that the N_2 occupation in the hydrate phase predominated at P_{N2} = 8.0 MPa versus at P_{N2} = 4.0 MPa. Although the continuous and gradual increase in the pressure was not observed after an abrupt pressure increase at the initial stage by the N_2 gas injection for both P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K, as shown in Figs. 3.2.1 (e) and (f), the incorporation of N_2 molecules into the hydrate phase was observed. Thus, it is reasonable to assume that both hydrate dissociation and hydrate replacement after the N_2 gas injection occurred for both P_{N2} = 4.0 MPa and 8.0 MPa at 268.8 K. The lesser pressure change after the N_2 gas injection for P_{N2} = 8.0 MPa indicates that hydrate replacement was more dominant than hydrate dissociation at a higher $P_{N2.}$

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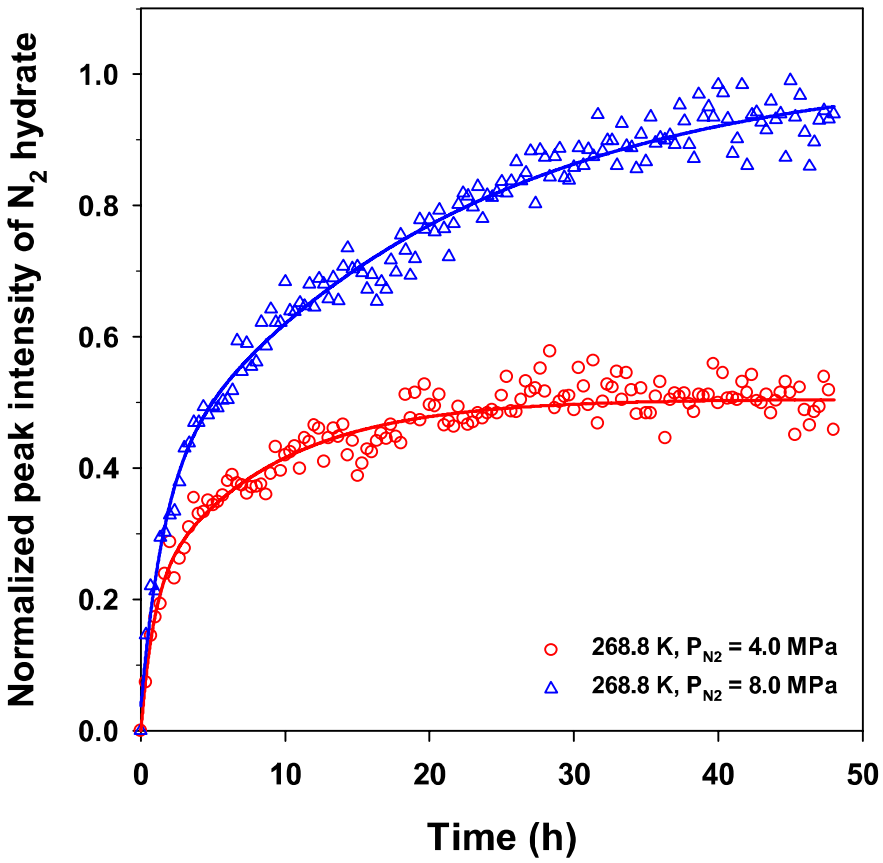


Fig. 3.2.6 Changes in the normalized signal intensity of the N_2 hydrate for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K

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A higher P_{N2} is not favored for hydrate dissociation because it not only decelerates hydrate dissociation but also interrupts the collapse of the crystal frameworks by increasing the participation of N_2 molecules as guest molecules in the hydrate lattices. Conversely, in terms of CH_4 production from NGHs by hydrate replacement, at a higher P_{N2}, N_2 molecules can act as a more favorable guest to push out CH_4 molecules from the hydrate cages. It is generally accepted that the guest-swapping process in gas hydrates occurs only when external guest gases are injected into the gas hydrates in their hydrate-stable regions (thermodynamically stable pressure and temperature conditions of gas hydrates formed from external guest gases). However, the initial injecting pressures of gaseous N_2 (4.0 and 8.0 MPa) were lower than the corresponding equilibrium pressure of the pure N_2 hydrate at 268.8 K (10.8 MPa)[81]. Nevertheless, as shown in Figs. 3.2.5 and 3.2.6, the N_2 molecules enclathrated in the sI hydrate were clearly observed from the early stage of the experiment. This meta-stability during the CH_4-N_2 exchange in the gas hydrates was also examined computationally by Matsui et al.[82]. They revealed that the injected N_2 molecules mainly acted as hydrate inhibitors to dissociate the CH_4 hydrate, and at the same time, the N_2 molecules also took part in the guest exchange, thus forming the $CH_4 + N_2$ hydrate. Therefore, the experimental results in this study suggest that the replacement-like behavior observed by the N_2 gas injection at 268.8 K was mainly caused by the reconstruction of the $CH_4 + N_2$ hydrate accompanied by the rapid collapse and subsequent reformation of the hydrate structures rather than the structure-maintained replacement. After the $CH_4 + N_2$ hydrate was thermodynamically stabilized by the increased pressure and CH_4 composition in the vapor phase due to rapid CH_4 production (after 5 h for P_{N2} = 8.0 MPa at 268.8 K, as shown in Fig. 3.2.1(f)), guest swapping between CH_4 and N_2 occurred to meet the thermodynamic equilibrium compositions of the hydrate and vapor phases at a given pressure and temperature without further hydrate dissociation (no further increase in the system pressure).

The cage-dependent guest distributions in the hydrate phase during guest exchange are important for understanding the hydrate replacement mechanism, as each guest has a different preference for hydrate cages. To observe cage-dependent replacement behavior, the changes in the intensity ratio of the Raman signals for the CH_4 molecules in the large ($5^{12}6^2$) and small (5^{12}) cages (I_L/I_S) for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K are illustrated in Fig. 3.2.7. For both P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa, the I_L/I_S ratios increased gradually by approximately 0.2 over the first 12 h, without a significant difference between the two P_{N2} conditions. As both hydrate dissociation and hydrate replacement occurred simultaneously at an early stage after the N_2 gas injection at 268.8 K, it is reasonable to say that the changes in the I_L/I_S values were induced by the participation of the N_2 molecules in the hydrate cages and their slight preference for small (5^{12}) cages of the sI hydrate during the CH_4–N_2 swapping.

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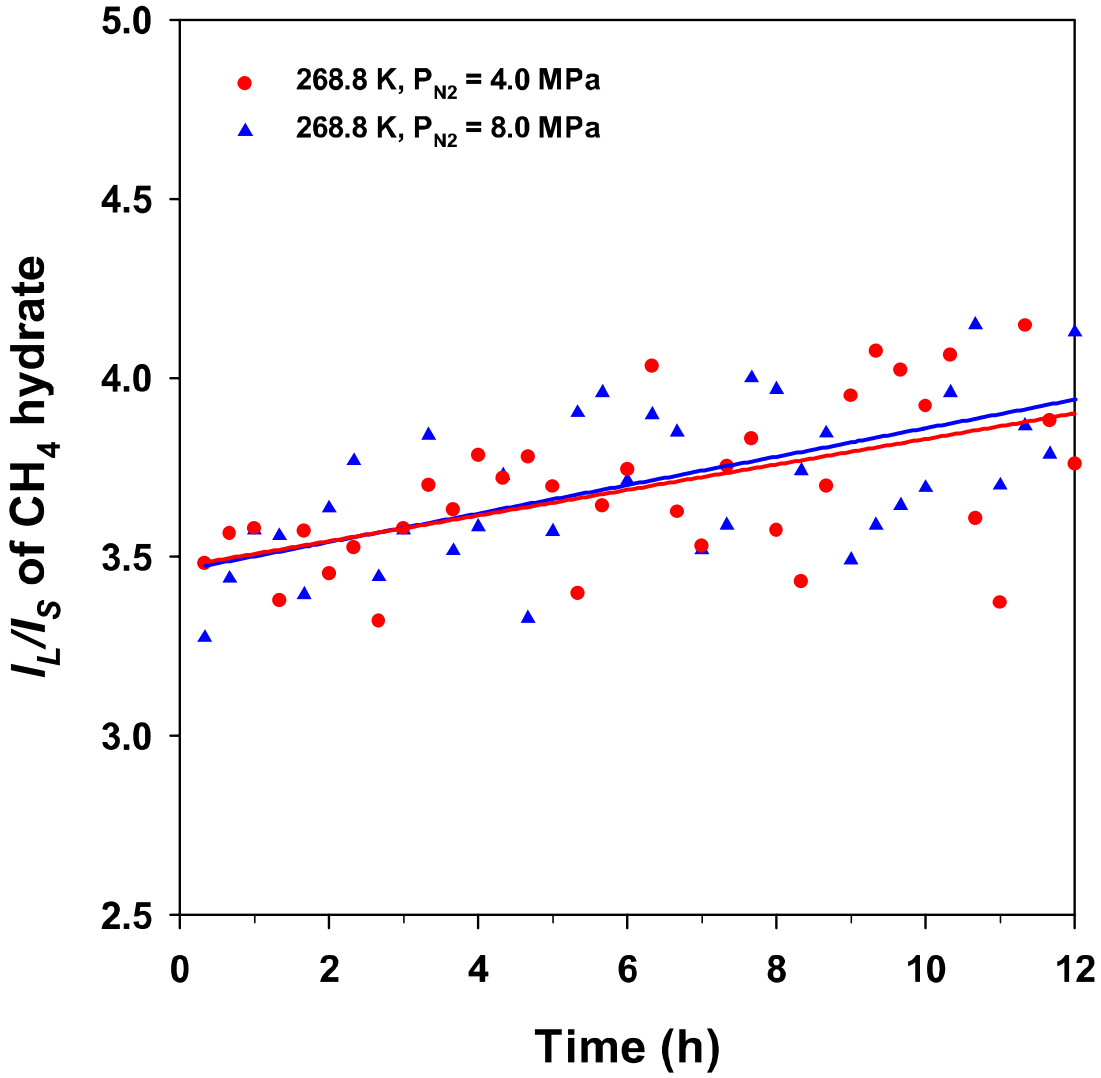


Fig. 3.2.7. Changes in the intensity ratio of the Raman signals for CH_4 molecules in the large ($5^{12}6^2$) and small (5^{12}) cages of the sI hydrate for P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K

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Unfortunately, it was difficult to separate the Raman signal of the CH_4 molecules enclathrated in the small (5^{12}) cages of the sI hydrate (at 2,915 cm^{-1}) from that of free gaseous CH_4 (at 2,917 cm^{-1}) after 12 h because of the increased amount of gaseous CH_4 released from the hydrate phase. To examine the structural transformation and the cage preference of the guest molecules in the $CH_4 + N_2$ hydrates, the ^{13}C NMR spectra of the $CH_4 + N_2$ hydrates with various guest compositions were measured, and the cage occupancy ratios of the CH_4 molecules in the large and small cages ($\theta_L/\theta_{S,\,CH4}$) were obtained, as shown in Fig. 3.2.8. The cage-dependent ^{13}C NMR chemical shifts of the enclathrated CH_4 molecules were effective in determining the structural types of gas hydrates[83]. As shown in Fig. 3.2.8 (a), the ^{13}C NMR spectra of the $CH_4 + N_2$ hydrates with CH_4 compositions over 30% in the hydrate phase exhibited only two signals at – 6.6 ppm and – 4.3 ppm, which corresponded to the CH_4 molecules in the large ($5^{12}6^2$) and small (5^{12}) cages of the sI hydrate, respectively[84]. However, aside from very weak signals at – 6.6 ppm and – 4.3 ppm, the CH_4 (25%) + N_2 (75%) hydrate had two additional signals at – 8.2 ppm and – 4.5 ppm, which were assigned to the CH_4 molecules in the large ($5^{12}6^4$) and small (5^{12}) cages of the sII hydrate, respectively, indicating the coexistence of the sI (minor) and sII (dominant) hydrates[85]. In the case of the CH_4 (17%) + N_2 (83%) hydrate, only two signals at – 8.2 ppm and – 4.5 ppm from the sII hydrate were detected, which confirmed that an increase in the N_2 composition (i.e., a decrease in the CH_4 compositions) in the $CH_4 + N_2$ hydrates induced the structural transformation of sI to sII[86]. The composition of CH_4 (25%) + N_2 (75%) in the hydrate phase was found to be the transitional one for the structural change in the $CH_4 + N_2$ hydrates. The ^{13}C NMR results were consistent with the Raman spectroscopic results in that the iso-structural $CH_4 – N_2$ replacement was observed using in situ Raman spectroscopy for the $CH_4 + N_2$ hydrates with CH_4 compositions over 50% for both P_{N2} = 4.0 MPa and 8.0 MPa at 268.8 K.

The $\theta_L/\theta_{S,CH4}$ values were calculated by dividing the area ratio of the ^{13}C NMR signals for the CH_4 molecules in the large and small cages by the stoichiometric ratio of the large and small cages in the unit cell of each hydrate structure (Fig. 3.2.8 (b)). All the $\theta_L/\theta_{S,CH4}$ values of the $CH_4 + N_2$ hydrates were larger than those of the pure CH_4 hydrate (1.07), indicating that the N_2 molecules have a preference for the small (5^{12}) cages for both the sI and sII hydrates because of the smaller molecular size of N_2. In addition, the $\theta_L/\theta_{S,CH4}$ values increased as the N_2 composition in the $CH_4 + N_2$ hydrates increased for both sI and sII hydrates. This implies that the preferable occupation of N_2 molecules in the small (5^{12}) cages was enhanced with an increase in the N_2 composition in the hydrate phase.

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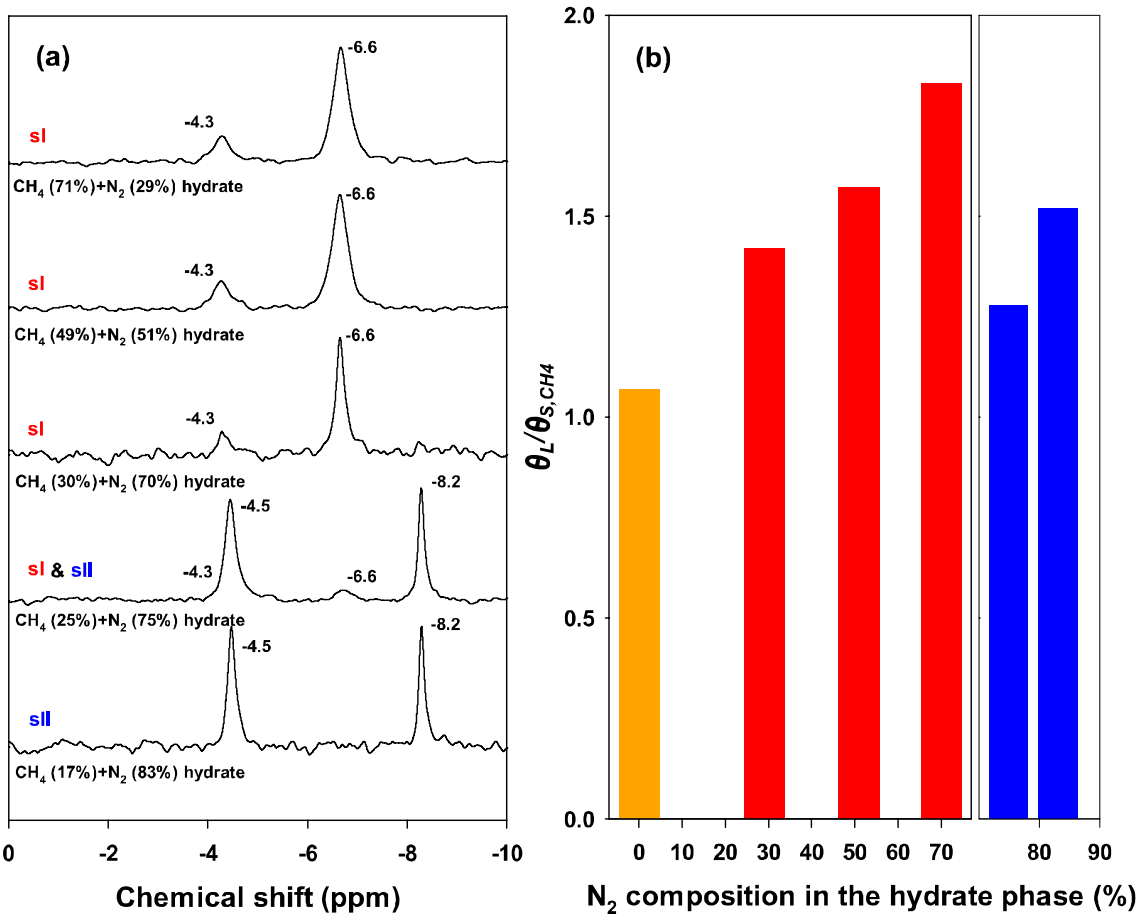


Fig. 3.2.8. (a) The ^{13}C NMR spectra of the enclathrated CH_4 molecules and (b) cage occupancy ratios of the CH_4 molecules in the large ($5^{12}6^2$) and small (5^{12}) cages in the $CH_4 + N_2$ hydrates

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The cage-dependent replacement mechanism has been well known for the CH_4–CO_2 replacement in the sI hydrate[87,88]. The guest exchange between external CO_2 and enclathrated CH_4 occurred primarily in the large ($5^{12}6^2$) cages of the sI hydrate because of the larger molecular diameter of CO_2 (5.12 Å) than that of CH_4 (4.36 Å)[89]. The predominant enclathration of the CO_2 molecules in the large ($5^{12}6^2$) cages was more significant at a higher CO_2 composition in the hydrate phase. Furthermore, the concept of the $CH_4 - CO_2 + N_2$ replacement has also been proposed to increase the replacement efficiency and economic feasibility of hydrate replacement technology. Additional guest exchange between the N_2 and CH_4 molecules in the small (5^{12}) cages was achieved because of the relatively smaller molecular diameter of N_2 (4.10 Å) than that of CO_2[90]. Recently, beyond the conventional concept of simply injecting flue gas with a fixed composition (e.g., CO_2 (10%) + N_2 (90%)), several attempts have been made to optimize the CO_2/N_2 composition of the injection gas for replacement and thus to enhance the replacement efficiency under a given thermodynamic condition of the target NGH layer[91].

In this study, the composition-dependent cage preference of the CH_4 and N_2 molecules in the hydrate phase was experimentally investigated, and it could be a key factor for the development and establishment of the $CH_4 - CO_2 + N_2$ replacement technology. In real NGH sediment, not only guest exchange but also various other phenomena, such as hydrate formation, hydrate dissociation, hydrate re-arrangement, and wellbore blockages, may occur simultaneously when injecting the $CO_2 + N_2$ gas mixture into the NGH layer because diverse components, including free gas, free water, gas hydrates, various chemicals, and particles of different sizes, can form physically and chemically complex systems[92]. In fact, according to the reported results of the Iġnik Sikumi gas hydrate exchange field trial in 2012 through the injection of the $CO_2 + N_2$ gas mixture, the reaction was primarily dominated by CH_4 and N_2 in the deeper reservoir away from the injection well because CO_2 was almost depleted in the region nearer the well[93].

In such an isolated system of the CH_4 hydrate and N_2 gas, the dissociation and subsequent reconstruction of the gas hydrates as well as the guest exchange may occur simultaneously depending on the given thermodynamic conditions, as shown in this study[94]. Therefore, the dissociation kinetics of the CH_4 hydrate by the injection of N_2 and the $CH_4 - N_2$ replacement behavior under various thermodynamic conditions can broaden our understanding of the guest-releasing behavior for the inhibitor injection and the guest exchange mechanism for hydrate replacement to exploit natural gas from the NGH reservoirs in complex environments.

3.2.4. Conclusions

The dissociation kinetics and guest exchange behavior of the CH_4 hydrate through N_2 gas injection were investigated at various temperatures (268.8, 274.2, and 278.2 K) and pressure (P_{N2} = 4.0 MPa and P_{N2}

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= 8.0 MPa) conditions. The experimental results for the CH_4 production through N_2 injection demonstrate that the dissociation kinetics of the CH_4 hydrate were faster at a higher temperature and a lower P_{N2}. For both P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K, the earlier termination of the CH_4 release and the higher final pressure than the equilibrium pressure of the CH_4 + N_2 hydrates after N_2 injection indicated incomplete dissociation of the initial CH_4 hydrate. The guest composition analysis using GC also supported the existence of residual hydrate and the occupation of N_2 molecules in the hydrate phase for both P_{N2} = 4.0 MPa and P_{N2} = 8.0 MPa at 268.8 K. In situ Raman spectroscopy was used to examine the time-dependent guest distributions in the hydrate phase and the enclathration of N_2 molecules after N_2 gas injection at 268.8 K. The more significant occupation of the N_2 molecules and the longer duration of the $CH_4 - N_2$ replacement were observed at a higher P_{N2}. The experimental results clearly indicated the dual functional roles of N_2 as an inhibitor to dissociate the CH_4 hydrate and as an external guest to replace the CH_4 molecules in the hydrate cages. The ^{13}C NMR spectra showed that the structural transition of the CH_4 + N_2 hydrates could occur from sI to sII with an increase in the N_2 composition in the hydrate phase and that the N_2 molecules were preferentially enclathrated in the small (5^{12}) cages of the CH_4 + N_2 hydrates. The overall results covering the multifunctional roles of the injected N_2 gas molecules and the different cage preferences of the CH_4 and N_2 molecules can provide deep insights into the guest exchange, guest release, gas storage, and stability of cage structures associated with $CH_4 - CO_2 + N_2$ replacement as well as the inhibitor injection for the exploitation of natural gas from the NGH reservoirs with complex systems.

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3.3. Condition-dependent CH_4 - flue gas replacement behavior

This chapter includes the published contents:

Mok, J., Choi, W., Lee, J., & Seo, Y. (2022). Effects of pressure and temperature conditions on thermodynamic and kinetic guest exchange behaviors of CH_4− CO_2+ N_2 replacement for energy recovery and greenhouse gas storage. Energy, 239, 122153.
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3.3.1. Abstract

Both natural gas production and CO_2 sequestration can be simultaneously achieved in natural gas hydrates (NGHs) by using a guest swapping technique. In this study, firstly, the effects of replacement pressure (10.0 – 18.5 MPa) and temperature (274.2 – 277.2 K) conditions on the guest exchange behaviors of CH_4 − CO_2 + N_2 replacement were investigated, focusing on extent of replacement and replacement kinetics. At 274.2 K, the extent of replacement increased with injection pressure of CO_2 (20%) + N_2 (80%) gas, which is mainly attributed to a larger N_2 inclusion at a higher pressure. At a higher temperature, the extent of replacement did not change, but CO_2/N_2 ratio in the replaced hydrates decreased slightly. An increase in the pressure led to an accelerated CO_2 inclusion rate in the large ($5^{12}6^2$) cages at the initial stage and an enhanced N_2 inclusion in the small (5^{12}) cages at the final stage. The enhanced replacement kinetics at a higher temperature is attributable to the increased inclusion rates of both CO_2 and N_2 at the initial stage of replacement. Secondly, the effect of CO_2/N_2 ratio of injecting gas on CH_4 − CO_2 + N_2 replacement behavior were investigated, focusing on the extent of replacement and guest behavior in the hydrate phase after the replacement. The extent of replacement was increased as N2 composition in the injecting gas increased at same partial pressure of injecting CO_2, but the CO_2 composition in the hydrate phase after the replacement was decreased. The cage occupancy of each guest molecules obtained by PXRD analysis revealed that additional participation of N2 was occurred in both large and small cages, which reduce the storage amount of CO2. The results provide valuable insights into the guest swapping mechanism of CH_4 − CO_2 + N_2 replacement occurring in NGH reservoirs with various locations and environments.

3.3.2. Effects of pressure and temperature on the extent of CH_4 − CO_2 + N_2 replacement

To confirm the effect of replacement pressure on the extent of CH_4 production and CO_2 sequestration, the guest compositions in the hydrate phase after the replacement were investigated in the pressure range of 10.0 – 18.5 MPa using a CO_2 (20%) + N_2 (80%) gas mixture at 274.2 K. It was assumed that a one-to-one guest exchange occurs in CH_4 − CO_2 + N_2 replacement, and accordingly, the extent of

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replacement (i.e., CH_4 recovery) was considered as the sum of CO_2 and N_2 compositions in the replaced hydrates. As shown in Fig. 3.3.1, the extent of replacement gradually increased with the injection pressure of the CO_2 (20%) + N_2 (80%) gas mixture at 274.2 K. In our previous study, which had an identical experimental setup to this study, at 274.2 K, 64% and 69% of initial CH_4 were observed to be replaced by CO_2 under the injection pressures of 2.2 and 3.5 MPa (pure CO_2), respectively, in CH_4 − CO_2 replacement[74]. In CH_4 − CO_2 + N_2 replacement, the extents of replacement were 76% and 81% at injection pressures of 10.0 and 18.5 MPa (CO_2 (20%) + N_2 (80%)), respectively, which had similar CO_2 partial pressures to those in CH_4 − CO_2 replacement. This indicates that the extent of replacement was improved by approximately 12% with the addition of N_2 to the injection gas having a similar CO_2 partial pressure.

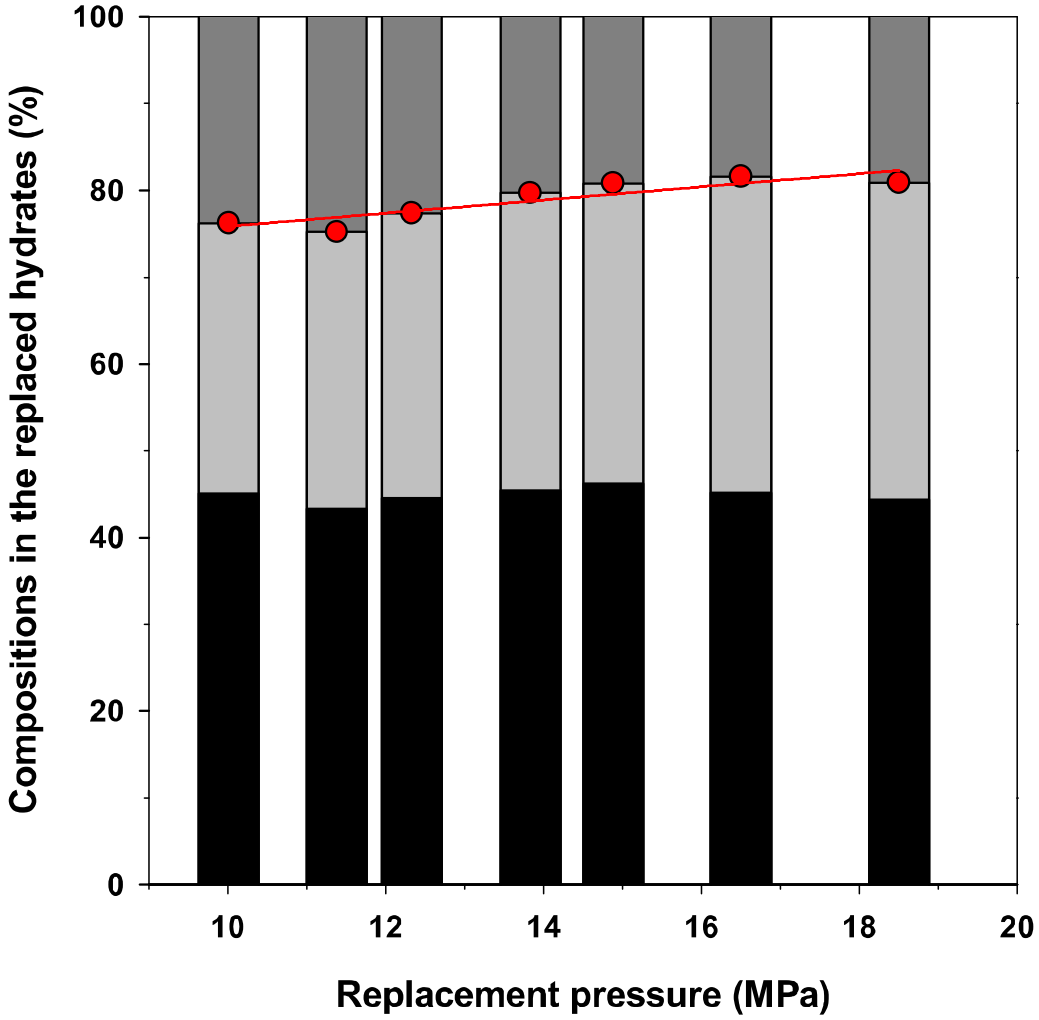


Fig. 3.3.1. Guest compositions in the replaced hydrates at different replacement pressures (at 274.2 K) (Dark gray: CH_4, light gray: N_2, black: CO_2, red dot: extent of replacement).

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The compositions of CO_2 and N_2 molecules in the replaced hydrates at different replacement pressures are depicted in Fig. 3.3.2(a). As the injection pressure of CO_2 (20%) + N_2 (80%) was increased, the CO_2 composition increased very slightly, whereas the increase in N_2 composition was more pronounced, indicating that an increased N_2 inclusion in the hydrate phase mainly increased the extent of replacement at the increased replacement pressure. Both $CH_4 - CO_2$ and $CH_4 - CO_2 + N_2$ replacements are known to occur iso-structurally, meaning that the initial sI hydrate lattices are not broken or dissociated significantly during the replacement. In $CH_4 - CO_2$ replacement, due to the large molecular size of CO_2 (5.12 Å), CO_2 molecules preferentially replaced CH_4 molecules in the large ($5^{12}6^2$) cages of sI. Consequently, approximately 30% of CH_4 molecules in the small (5^{12}) cages and 75% of CH_4 molecules in the large ($5^{12}6^2$) cages were replaced by CO_2 molecules in $CH_4 - CO_2$ replacement. In contrast, because N_2 has a small molecular diameter (4.10 Å), a preferential occupation of N_2 molecules in the small (5^{12}) cages of $CH_4 + N_2$ hydrates was observed, and this tendency was enhanced as the N_2 composition in the hydrate phase was increased[95]. As shown in Fig. 3(a), in $CH_4 - CO_2 + N_2$ replacement, an increased N_2 composition in the replaced hydrates with an increase in the replacement pressure was also attributable to the increased N_2 inclusion in the small (5^{12}) cages, which were relatively less attacked by CO_2 molecules at the increased replacement pressure.

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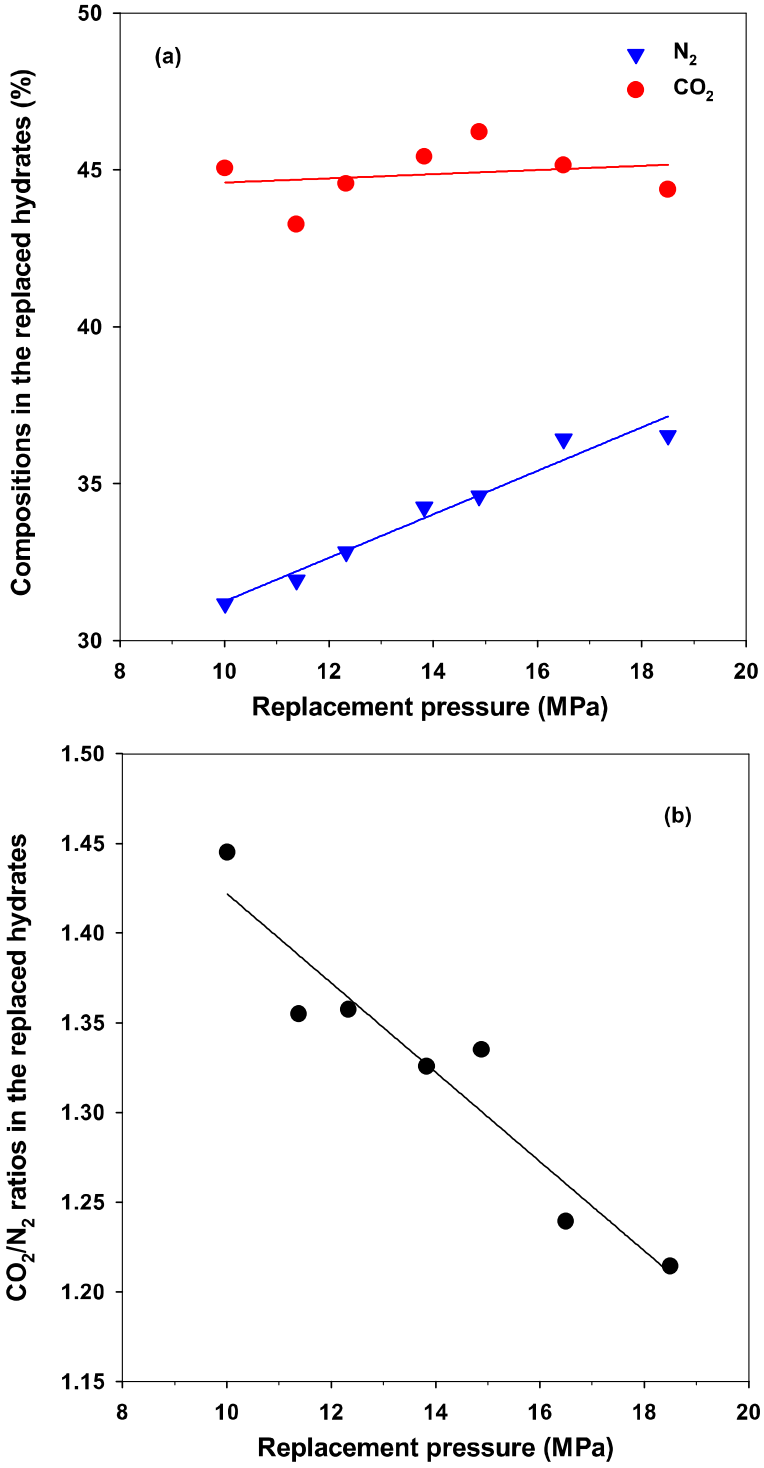


Fig. 3.3.2. (a) CO_2 and N_2 compositions and (b) CO_2/N_2 ratios in the replaced hydrates at different replacement pressures (at 274.2 K).

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Fig. 3.3.2(b) presents the CO_2/N_2 ratios in the replaced hydrates at different replacement pressures. The CO_2/N_2 ratios within all pressure ranges exceeded 1.0, which indicated that, despite the lower CO_2 concentration in the injection gas (CO_2/N_2 ratio in the injecting gas: 0.25), CH_4 molecules in the hydrate cages were replaced more by CO_2 molecules than by N_2 molecules. This can be explained by the fact that CO_2 exhibits a lower hydrate equilibrium pressure than N_2 at any given temperature and that the sI hydrate has three times as many large ($5^{12}6^2$) cages as small (5^{12}) cages in a unit cell: $6(5^{12}6^2)\cdot 2(5^{12})\cdot 46(H_2O)$. The CO_2/N_2 ratios in the replaced hydrates decreased with an increase in the injection pressure of CO_2 (20%) + N_2 (80%), which was caused by the increased N_2 occupation in the hydrate phase at a higher injection pressure. Fig. 3.3.2(a) and (b) clearly demonstrates that the enhanced extent of replacement at higher injection pressures of CO_2 (20%) + N_2 (80%) was mainly induced by increased guest exchange between CH_4 and N_2 without lowering the CO_2 storage in the hydrate phase.

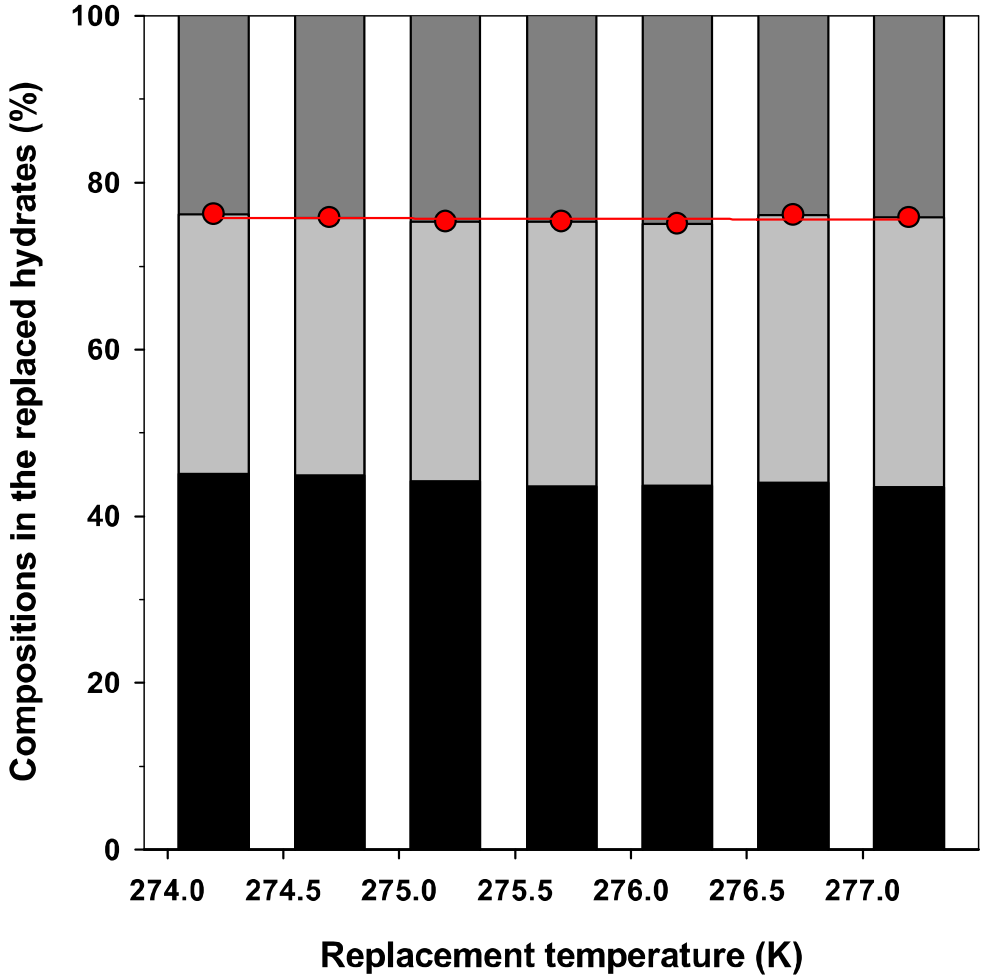


Fig. 3.3.3. Guest compositions in the replaced hydrates at different replacement temperatures (at 10.0 MPa) (Dark gray: CH_4, light gray: N_2, black: CO_2, red dot: extent of replacement).

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To observe the effect of replacement temperature on CH_4 recovery, the extent of replacement was examined in the temperature range of 274.2 – 277.2 K at the injection gas pressure of 10.0 MPa. Fig. 3.3.3 shows that the extent of replacement was almost constant in the experimental temperature ranges, which is clearly distinct from the result of the pressure change experiment shown in Fig. 3.3.1. The CO_2 and N_2 compositions in the replaced hydrates are depicted in Fig. 3.3.4(a). With an increase in the replacement temperature, the N_2 composition increased slightly, whereas the CO_2 composition decreased slightly. Such guest behavior was consistent with the thermodynamic calculation results obtained for ternary $CH_4 + CO_2 + N_2$ hydrates. The thermodynamic calculations demonstrated that the N_2/CO_2 ratio in the hydrate phase was higher at a higher temperature and that an increase in this ratio was more pronounced in the small (5^{12}) cages with an increase in N_2 concentration[96]. Therefore, the guest distributions in the replaced hydrates were rearranged in accordance with the adjacent pressure and temperature conditions to reach a corresponding equilibrium state of ternary $CH_4 + CO_2 + N_2$ hydrates through guest exchange in $CH_4 - CO_2 + N_2$ replacement. As shown in Fig. 3.3.4(b), the CO_2/N_2 ratios in the replaced hydrates exceeded 1.0 in all replacement temperature ranges and decreased with an increase in the replacement temperature, which is similar to the trend observed in the pressure change experiment, as shown in Fig. 3.3.2(b). However, unlike the pressure change experiment, with an increase in the replacement temperature, the amount of CO_2 storage decreased slightly, whereas that of CH_4 production remained constant. Therefore, in $CH_4 - CO_2 + N_2$ replacement, a lower replacement temperature is more favorable for a larger CO_2 storage.

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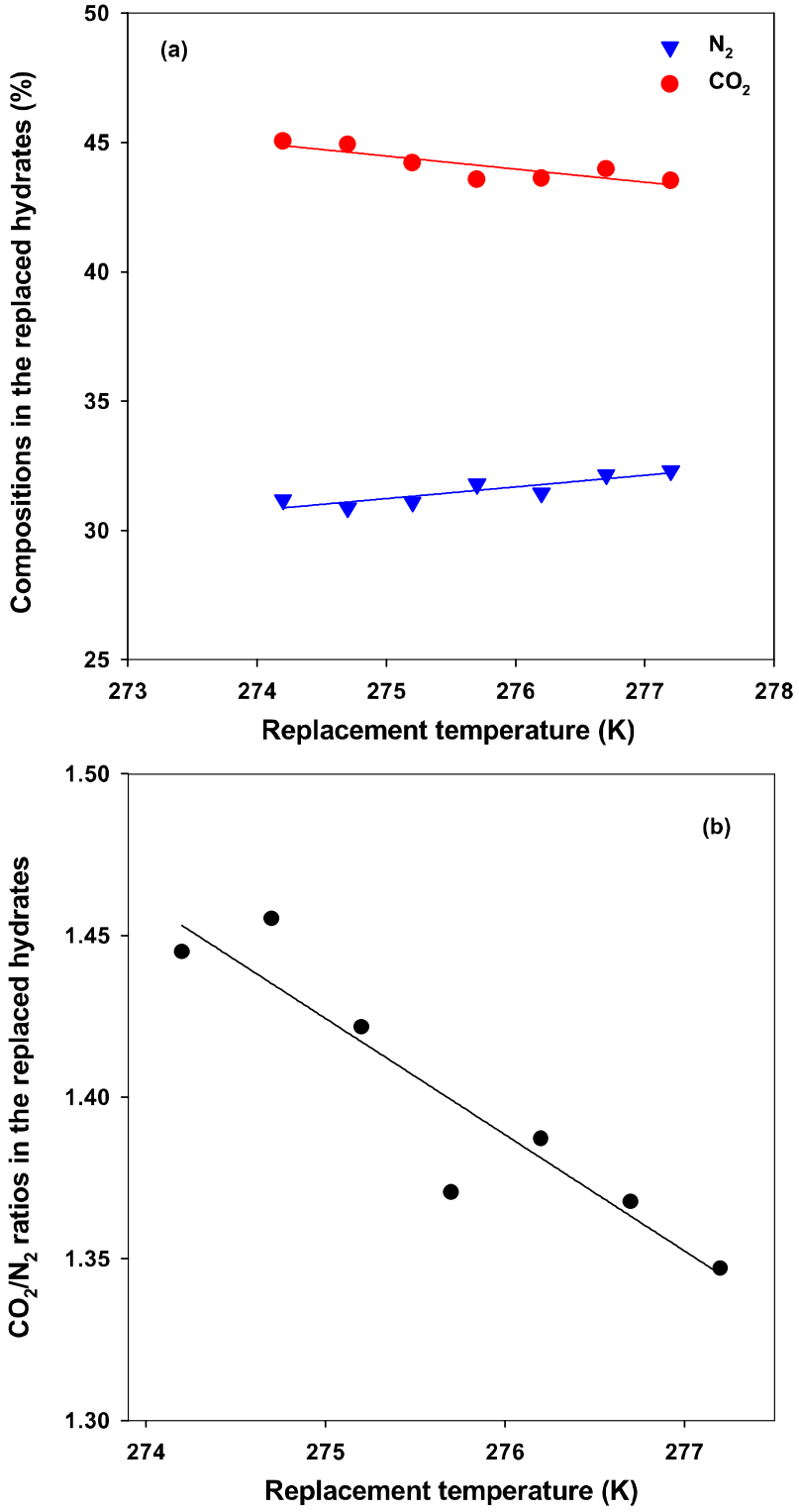


Fig. 3.3.4. (a) CO_2 and N_2 compositions and (b) CO_2/N_2 ratios in the replaced hydrates at different replacement temperatures (at 10.0 MPa).

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3.3.3. Effects of pressure and temperature on the kinetics of $CH_4 - CO_2 + N_2$ replacement

Besides the extent of replacement, the replacement kinetics is also an important factor for estimating the replacement efficiency and observing the guest exchange behavior of $CH_4 - CO_2 + N_2$ replacement. To examine the effect of replacement pressure on the kinetics of $CH_4 - CO_2 + N_2$ replacement, the time-dependent extent of the replacement was observed at two gas injection pressures of 10.0 and 18.5 MPa (at 274.2 K). Due to the large surface area of the hydrate-saturated porous silica gels, the replacement reaction was almost completed within 30 h; the changes in the extent of replacement and in guest compositions for the initial 30 h are depicted in Fig. 3.3.5. At a higher replacement pressure, faster replacement kinetics are observed in Fig. 3.3.5(a). To achieve 50% of the final extent of replacement, the replacement at 10.0 MPa took 1.9 h, whereas that at 18.5 MPa took 1.4 h. This indicates that both the replacement kinetics and the extent of $CH_4 - CO_2 + N_2$ replacement were enhanced with an increase in the replacement pressure. To examine the guest exchange kinetics, the time-dependent compositions of CO_2 and N_2 in the hydrate phase during the replacement were analyzed (Fig. 3.3.5(b)). At the early stage of the replacement, the increasing rate of N_2 composition was almost identical at the two pressures of 10.0 and 18.5 MPa, whereas the change in CO_2 composition was more significant at 18.5 MPa than at 10.0 MPa. However, the N_2 composition at 18.5 MPa was much higher than that at 10.0 MPa after 10 h, whereas no significant difference in CO_2 composition was observed at the two pressures after 10 h. Therefore, at a higher pressure, an increased CH_4 recovery rate at the initial stage is mainly attributed to the increased CO_2 inclusion rate, whereas an enhanced CH_4 recovery at the final stage is attributed to the increased N_2 inclusion. In $CH_4 - CO_2$ replacement, an accelerated CO_2 inclusion was also reported at a higher CO_2 injecting pressure[84]. Therefore, a faster CO_2 depletion at the early stage of CH_4 production should be considered for a real-time application of the $CH_4 - CO_2 + N_2$ replacement technology at a higher pressure. Fig. 3.3.5(c) shows the changes in the CO_2/N_2 ratio in the hydrate phase during the replacement. At 10.0 MPa, the CO_2/N_2 ratio was initially high, then decreased slowly, and later remained almost constant. The ratio exceeded the value of 1.0 during the replacement, which indicates the preferential inclusion of CO_2 over N_2 in the hydrate phase. Due to the accelerated CO_2 inclusion for the initial 10 h at a higher pressure, the CO_2/N_2 ratio at the early stage of replacement was much higher at 18.5 MPa than at 10.0 MPa. However, as observed in Figs. 3.3.2(a) and 3.3.5(b), the higher final extent of replacement at a higher pressure was mainly attributed to the larger N_2 inclusion in the hydrate phase, and thus, the final CO_2/N_2 ratio at 18.5 MPa was lower than that at 10.0 MPa.

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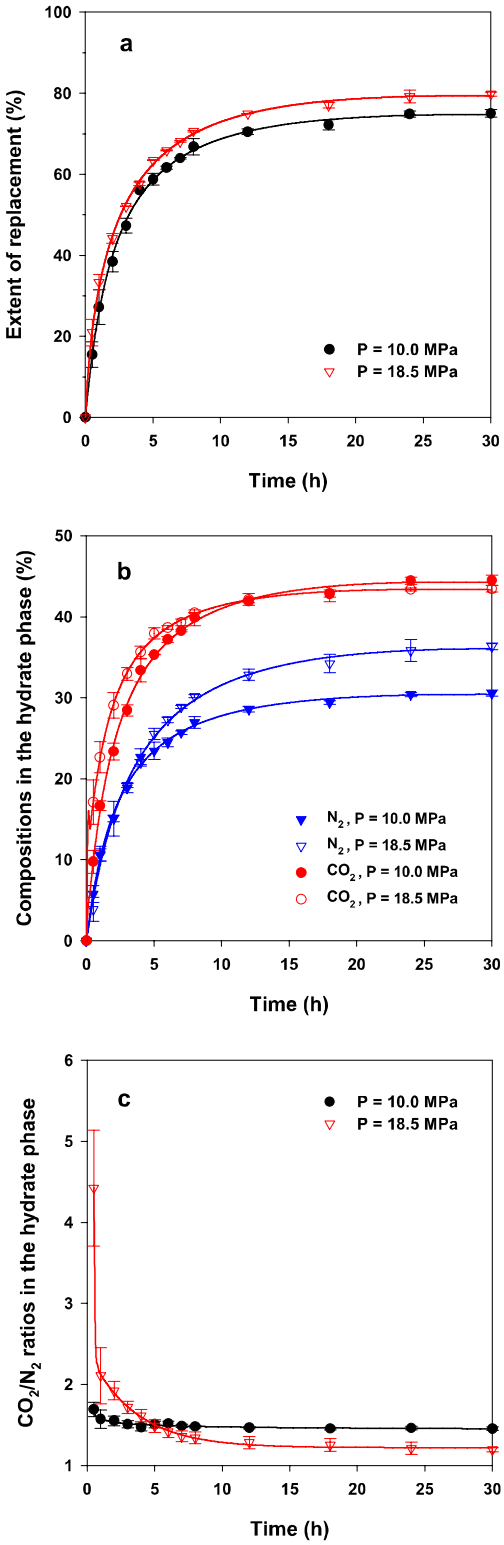


Fig. 3.3.5. Changes in (a) extent of replacement, (b) CO_2 and N_2 compositions, and (c) CO_2/N_2 ratios in the replaced hydrates during the replacement at pressures of 10.0 and 18.6 MPa (at 274.2 K).

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To confirm the effect of the replacement temperature on the kinetics of $CH_4 - CO_2 + N_2$ replacement, the time-dependent extent of replacement was examined at two replacement temperatures of 274.2 and 277.2 K (at the injection gas pressure of 10.0 MP). As shown in Fig. 3.3.6(a), the extent of replacement at the end of the experiment was similar for both cases, but faster replacement kinetics were observed at the higher replacement temperature (277.2 K). To achieve 50% of the final extent of replacement, the replacement at 277.2 K took 1.0 h, whereas that at 274.2 K took 1.9 h. The changes in CO_2 and N_2 compositions in the hydrate phase (Fig. 3.3.6(b)) demonstrated that the inclusion rates of both CO_2 and N_2 were higher at the higher replacement temperature, indicating that the enhanced CH_4 production kinetics at the early stage at a higher temperature is attributable to the accelerated inclusion rates of both CO_2 and N_2. Fig. 3.3.6(c) presents the time-dependent CO_2/N_2 ratio in the hydrate phase at two different temperatures (274.2 and 277.2 K). At 277.2 K, the CO_2/N_2 ratio was initially low, then increased, and later became almost constant, which was markedly different from the change in the CO_2/N_2 ratio at 274.2 K. Due to a slight increase in N_2 storage and a slight decrease in CO_2 storage at higher replacement temperatures, as shown in Fig. 3.3.4, the final CO_2/N_2 ratio at 277.2 K was slightly lower than that at 274.2 K. Note that despite the enhanced N_2 inclusion at the higher temperature, the CO_2/N_2 ratio exceeded 1.0 throughout the replacement process at the two temperatures, which demonstrates that CO_2 was thermodynamically more selective than N_2 in the hydrate phase during the replacement.

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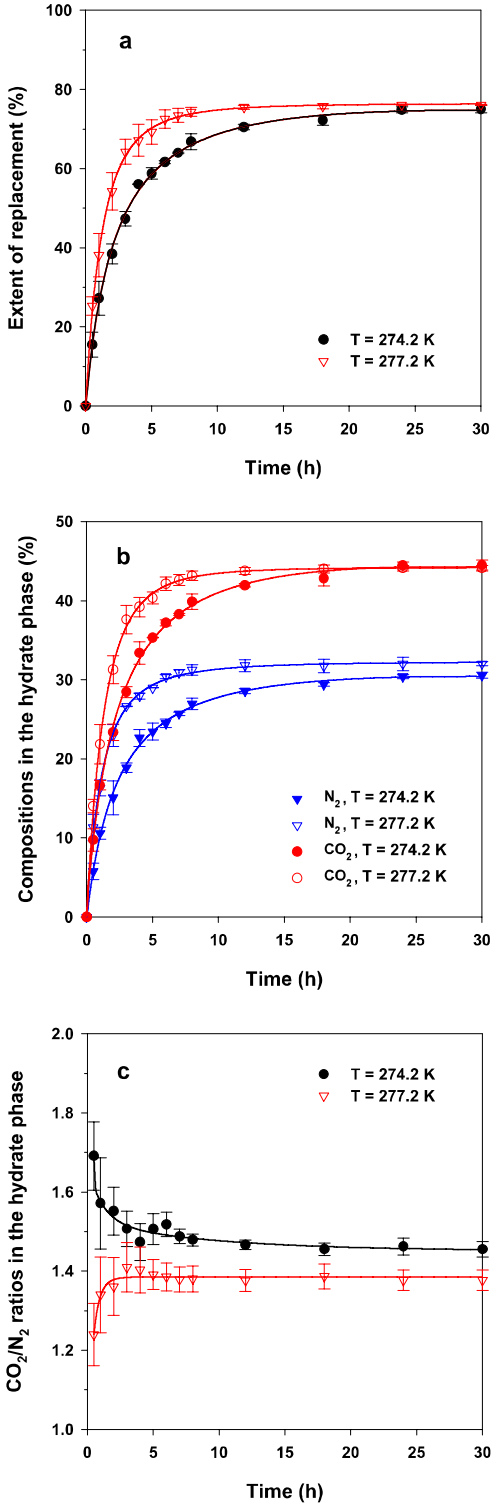


Fig. 3.3.6. Changes in (a) the extent of replacement, (b) CO_2 and N_2 compositions, and (c) CO_2/N_2 ratios in the replaced hydrates during the replacement at temperatures of 274.2 and 277.2 K (at 10.0 MPa).

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The guest distributions in the hydrate phase during the replacement can provide a deeper insight into cage-specific replacement behavior. The cage-dependent ^{13}C NMR resonance peaks of enclathrated CH_4 molecules can be effectively used to distinguish the types of hydrate structures and cages and to examine the guest distributions in the gas hydrates[97]. To observe the effects of replacement pressure and temperature on the time-dependent CH_4 production behavior, the ^{13}C NMR spectra of the replaced hydrates using CO_2 (20%) + N_2 (80%) were measured under the following three replacement conditions: (1) 10.0 MPa and 274.2 K, (2) 18.6 MPa and 274.2 K, and (3) 10.0 MPa and 277.2 K. As shown in Figs. 6 and 7, because almost half of the replacement was achieved after the initial 2 h, the hydrate samples in each experimental set were extracted at 2 h and 72 h after the injection of the replacing gas for the NMR measurement. As shown in Fig. 3.3.7(a) – (c), all replaced hydrate samples exhibited two resonance signals at −6.6 and −4.3 ppm, which correspond to the CH_4 molecules captured in the large ($5^{12}6^2$) and small (5^{12}) cages of the sI hydrate, respectively. This clearly indicates that the CH_4 − CO_2 + N_2 replacement under the given conditions was an iso-structural process.

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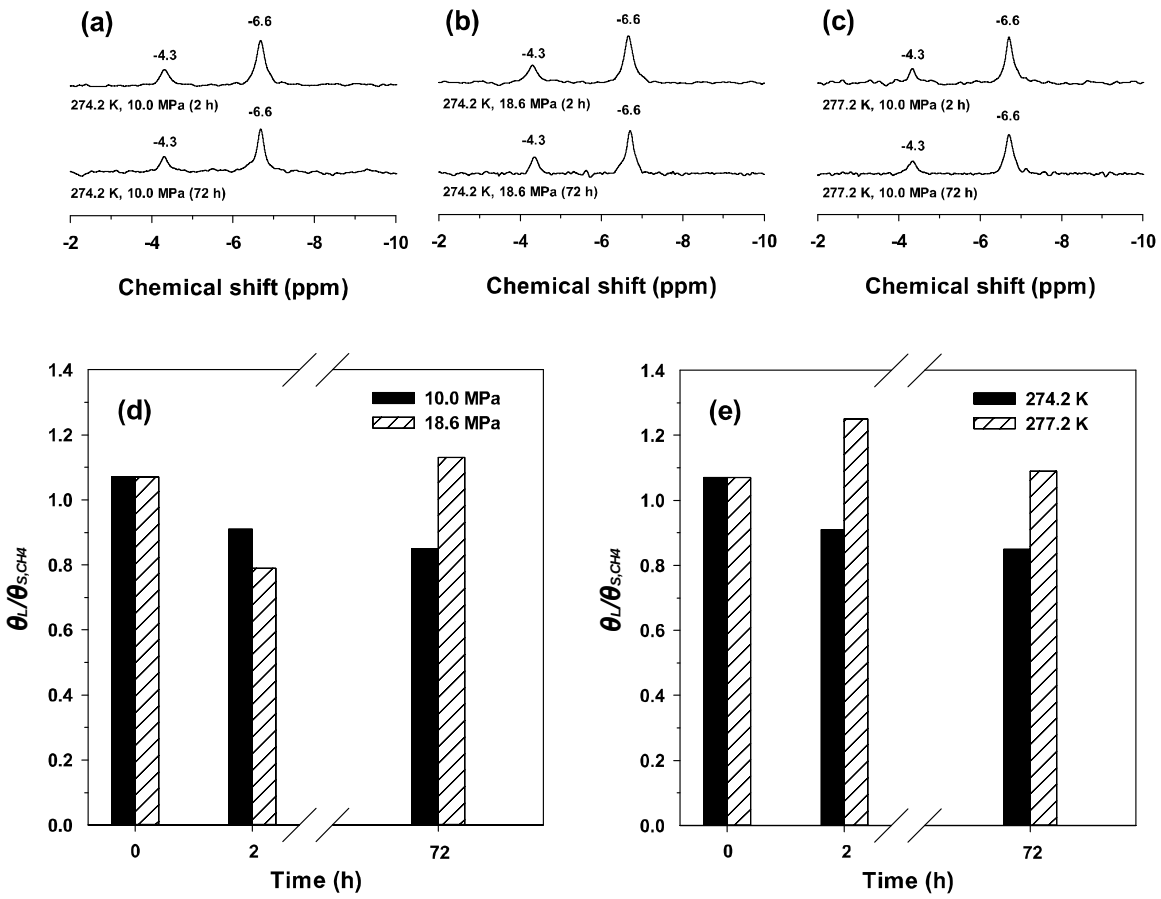


Fig. 3.3.7. ^{13}C NMR spectra of the replaced hydrates sampled at 2 and 72 h under (a) 274.2 K and 10.0 MPa, (b) 274.2 K and 18.6 MPa, and (c) 277.2 K and 10.0 MPa, as well as changes in $\theta_L/\theta_{S,CH4}$ values of the replaced hydrates sampled at 0 h, 2 h, and 72 h under (d) 10.0 and 18.6 MPa pressures (at 274.2 K) and (e) 274.2 K and 277.2 K temperatures (at 10.0 MPa).

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The cage occupancy ratio of the enclathrated CH_4 molecules ($\theta_L/\theta_{S,CH4}$) can be calculated by dividing the area ratio of the resonance signals for the CH_4 molecules captured in the large ($5^{12}6^2$) and small (5^{12}) cages by the stoichiometric ratio of these cages in a unit cell of the sI hydrate. Fig. 3.3.7(d) depicts the $\theta_L/\theta_{S,CH4}$ values of the replaced hydrates sampled at 0, 2, and 72 h at the two pressures of 10.0 and 18.6 MPa (at 274.2 K). The $\theta_L/\theta_{S,CH4}$ value of the initial CH_4 hydrate was 1.07. For the replaced hydrate at 10.0 MPa, the $\theta_L/\theta_{S,CH4}$ value decreased gradually as the replacement progressed (0.91 at 2 h and 0.85 at 72 h). However, for the replaced hydrate at 18.6 MPa, the $\theta_L/\theta_{S,CH4}$ value decreased once at 2 h (0.79) and then increased at 72 h (1.13). A decrease in the $\theta_L/\theta_{S,CH4}$ value means that the CH_4 molecules in the large ($5^{12}6^2$) cages were relatively more replaced by the external guest molecules than those in the small (5^{12}) cages. The temporary decrease in the $\theta_L/\theta_{S,CH4}$ value at 2 h was caused by the enhanced CO_2 inclusion in the large ($5^{12}6^2$) cages at a higher pressure, as shown in Fig. 3.3.5(b). In addition, a lower $\theta_L/\theta_{S,CH4}$ value at the early stage of the replacement due to the higher CO_2 occupancy in the large ($5^{12}6^2$) cages was also observed at higher CO_2 injection pressure in $CH_4 - CO_2$ replacement. As shown in Fig. 3.3.5, because an increase in the final extent of replacement was primarily induced by an increased N_2 inclusion in the small (5^{12}) cages at the higher pressure, the $\theta_L/\theta_{S,CH4}$ value at 72 h was higher at 18.6 MPa than that at 10.0 MPa.

In addition, to examine the effect of replacement temperature on the cage-specific guest exchange behavior of $CH_4 - CO_2 + N_2$ replacement, the $\theta_L/\theta_{S,CH4}$ values of the replaced hydrates sampled at 0, 2, and 72 h at the two temperatures of 274.2 and 277.2 K (at 10.0 MPa) were observed (Fig. 3.3.7(e)). For the replaced hydrate at 277.2 K, the $\theta_L/\theta_{S,CH4}$ value increased temporarily at 2 h (1.25) and then decreased at 72 h (1.09), whereas for the replaced hydrate at 274.2 K, the $\theta_L/\theta_{S,CH4}$ value decreased gradually. As shown Fig. 3.3.6(b) and (c), at the higher replacement temperature (274.2 K), the N_2 inclusion in the hydrate phase at the initial stage was more accelerated than the CO_2 inclusion. Therefore, the accelerated replacement by N_2 molecules at the initial stage was thought to occur preferentially in the small (5^{12}) cages due to the smaller molecular size of N_2. Because of the slightly lower CO_2/N_2 ratio in the replaced hydrate at 277.2 K than that at 274.2 K, as shown in Fig. 3.3.4(b), the final $\theta_L/\theta_{S,CH4}$ value at 277.2 K was higher than that at 274.2 K.

The experimental results clearly demonstrated that the pressure and temperature changes could affect the extent of replacement and the replacement kinetics through cage-specific guest exchange in the $CH_4 - CO_2 + N_2$ replacement. This indicated that the locations and environments of NGH reservoirs should be carefully considered to understand the replacement mechanism and to estimate the productivity of natural gas when the $CH_4 - CO_2 + N_2$ replacement method is actually applied for production. In particular, when $CO_2 + N_2$ gas is injected into the NGH reservoirs located in the high temperature region, dissociation of initial hydrates and subsequent formation of new mixed hydrates can inevitably occur.

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Therefore, further research into optimization of CO_2/N_2 ratio in the injecting gas should be conducted to increase the replacement efficiency under various pressure and temperature conditions.

3.3.4. Effects of injecting gas composition on the extent of $CH_4 - CO_2 + N_2$ replacement

To confirm the effect of CO_2/N_2 ratio of injecting gas on the extent of CH_4 production and CO_2 sequestration, the guest compositions in the hydrate phase after the replacement were investigated with various at 274.2 K (Fig. 3.3.8). To observe the effect of the addition of N_2 to the injecting gas, the partial pressure of CO_2 in the injecting gas was unified to 3.5 MPa.

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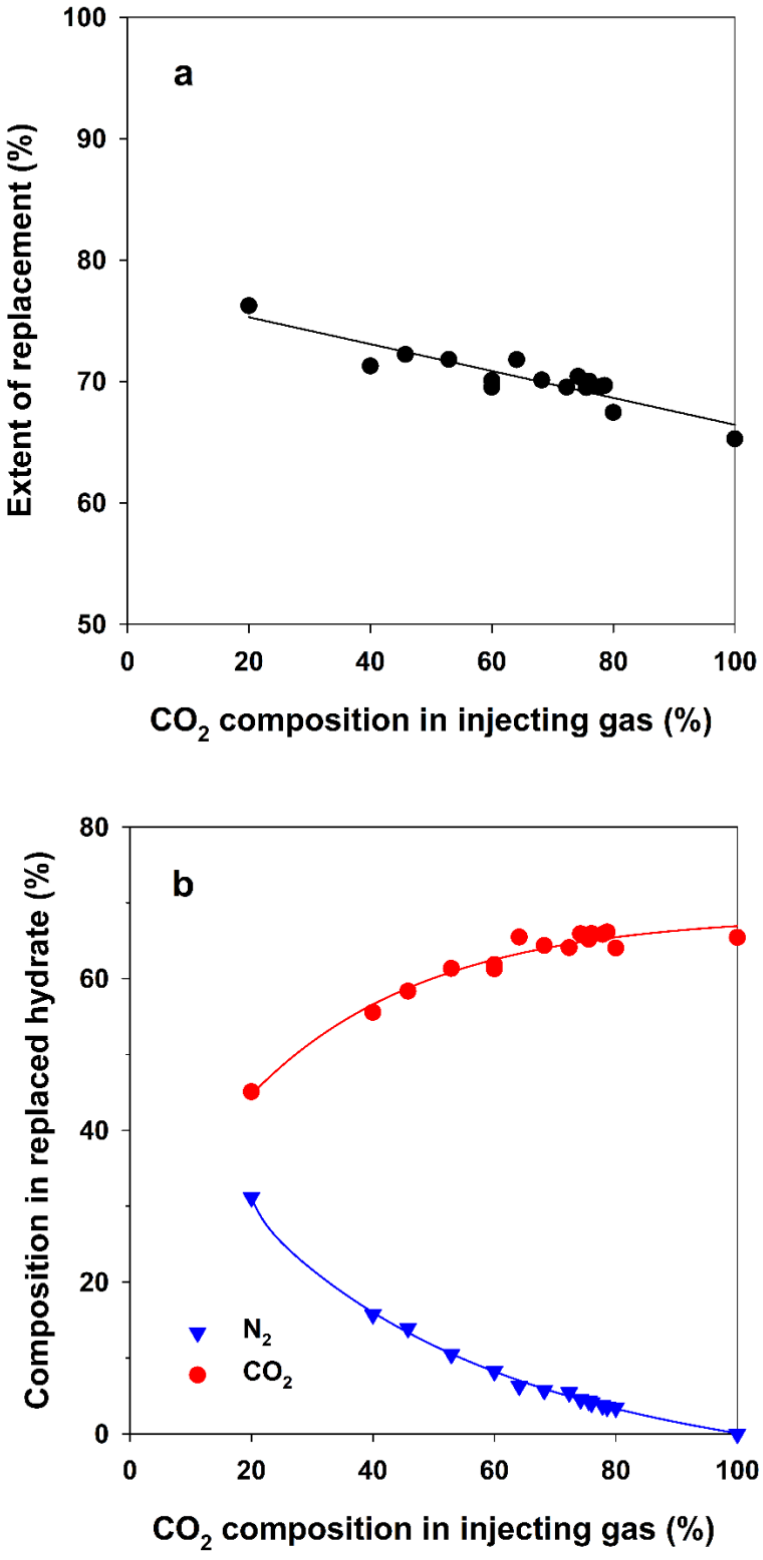


Fig. 3.3.8. (a) Extent of replacement and (b) CO_2 and N_2 compositions in the replaced hydrates by injecting CO_2 (20~100%) + N_2 balanced mixed gas at 274.2 K.

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The extent of replacement waw increased as N_2 composition in the injecting gas increased (Fig. 3.3.8a). As shown in Fig. 3.3.8b, this is because the participation of N_2 into the hydrate phase increased as N_2 composition in the injecting gas increased. However, as the N_2 concentration in hydrate phase increased, that of CO_2 decreased, which implies that the added N_2 additionally participates in the production of CH_4 whereas some N_2 were enclathrated instead of CO_2. One interesting point here is that the CO_2/N_2 ratio in the injecting gas shows a linear correlation with that in the replaced hydrate (Fig. 3.3.9). In this study, the replacement reaction rate was increased using porous silica which can maximize the reaction rate by increasing surface are of the hydrates, and the reaction was terminated after the equilibrium state was reached. Therefore, this result suggests that the CO_2/N_2 ratio in the replaced hydrate, which is ideally replaced to equilibrium state, can be inferred as the CO_2/N_2 ratio in the injecting gas. The reported results in which have similar experimental conditions (the replacement reaction was carried out to equilibrium state using porous silica gels) indicated also similar tendency. Therefore, it is considered necessary to further verify the trend of this experimental result and whether the fitted linear equation is universal.

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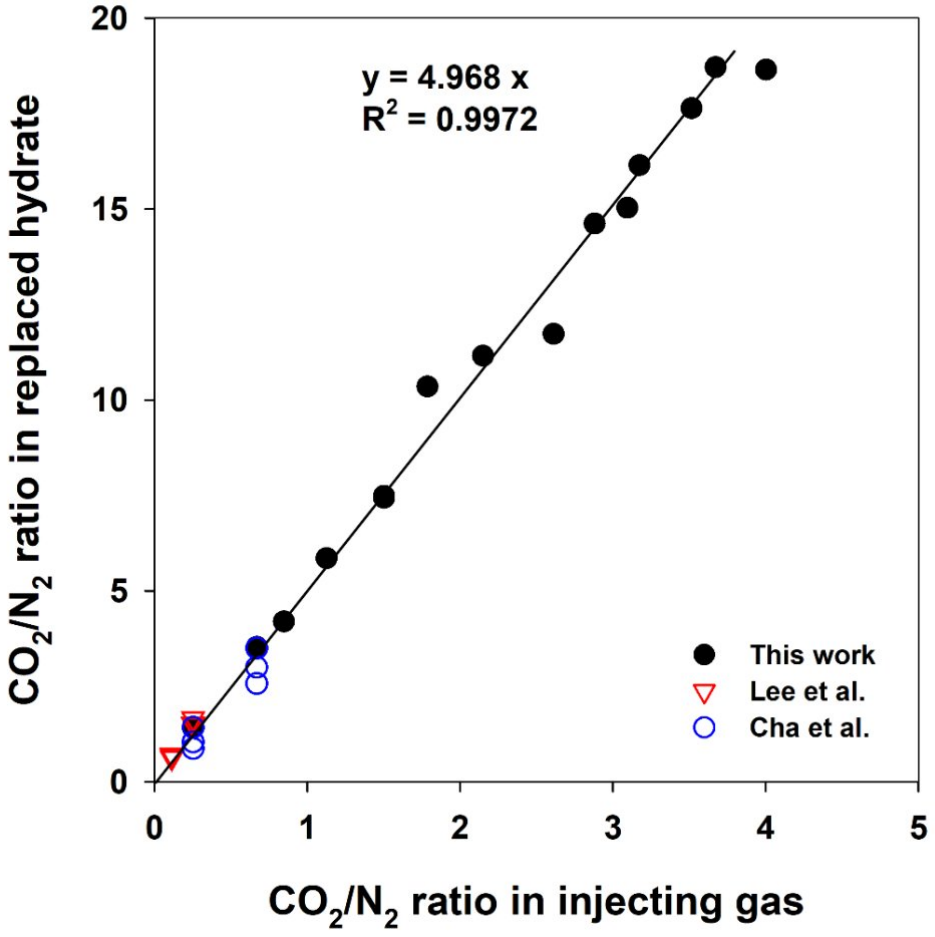


Fig. 3.3.9. CO_2/N_2 ratio in the replaced hydrates by injecting CO_2 (20~100%) + N_2 balanced mixed gas at 274.2 K.

To examine the guest distribution in the replaced hydrate phase, the quantitative cage occupancy of each guest molecules after the $CH_4 - CO_2$ (20%) + N_2 (80%) replacement were obtained by Rietveld analysis of PXRD results (Fig. 3.3.10 and Table. 3.3.1). Compared to $CH_4 - CO_2$ replacement, additional CH_4 production due to the new participation of N_2 occurred actively in the small cages (46% increase), while it was minimal in the large cages (5% increase), which is well consistent with the hypothesis in the literatures that N_2 participates in the small cages during N_2 injection and the extent of replacement increases. However in terms of CO_2 storage, the perspective became different: N_2 takes up not only CH_4 but also places where CO_2 can be stored, reducing the extent of CO_2 storage in both large and small cages compared to $CH_4 - CO_2$ replacement. At the $CH_4 - CO_2$ (20%) + N_2 (80%) replacement, the share of CO_2 decreased 18% and 90% in the large and small cages, respectively. In terms of entire

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guest distribution, the extent of CO_2 storage decreased by 25%, and 65% of this decrease was due to a decrease in CO_2 storage in the large cages. Therefore, when selecting the CO_2/N_2 ratio of the injecting gas, the trade-off between an increase in CH_4 production and a decrease in CO_2 storage by the additional participation of N_2 should be considered for the $CH_4 - CO_2 + N_2$ replacement. In addition, because the composition of the injecting gas will change due to the replacement reaction and additional hydrate formation/dissociation during the process of flue gas injection into the NGH layer, modeling and experimental design in consideration of changes in the $CH_4 - CO_2 + N_2$ replacement behavior according to changes in composition of the injecting gas should be established in large-scale replacement conditions.

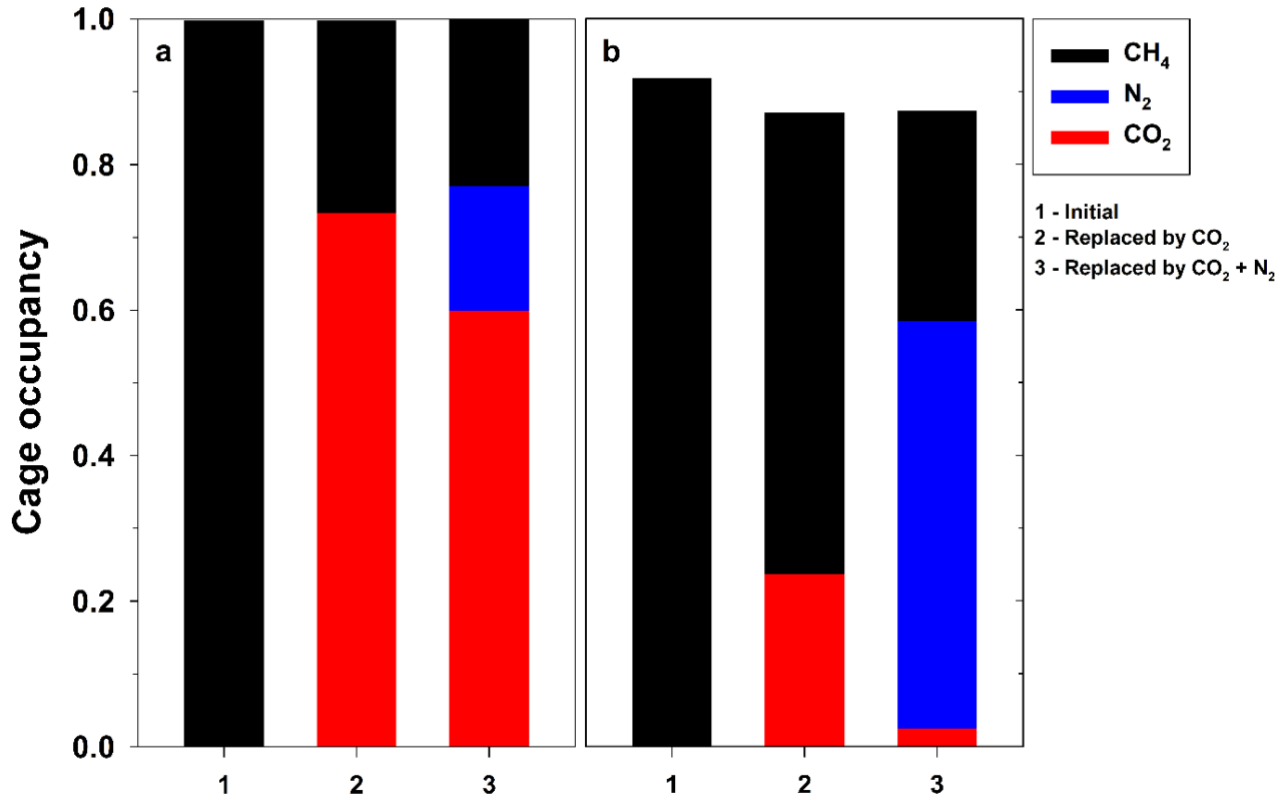


Fig. 3.3.10. Cage occupancy of CH_4, CO_2, and N_2 molecules in the initial CH_4 hydrate and the replaced hydrates (a) in large cages and (b) small cages with pure CO2 gas and $CO_2 + N_2$ mixed gas (PCO_2 = 3.5 MPa).

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Table 3.3.1. Cage occupancy of CH_4, CO_2, and N_2 molecules in the initial CH_4 hydrate and the replaced hydrates with pure CO2 gas and $CO_2 + N_2$ mixed gas (PCO_2 = 3.5 MPa) with R_{wp} value of the Rietveld analysis

	θ_L in sI hydrate			θ_S in sI hydrate			Rwp (%)
	CO_2	CH_4	N_2	CO_2	CH_4	N_2	
Initial	-	0.998(4)	-	-	0.918(2)	-	7.17
Replaced by CO_2	0.733(9)	0.264(3)	-	0.237(7)	0.634(8)	-	9.41
Replaced by $CO_2 + N_2$	0.599(6)	0.228(5)	0.171(1)	0.024(5)	0.289(1)	0.560(4)	10.2

3.3.5. Conclusions

In this study, the guest exchange behavior of $CH_4 - CO_2$ (20%) + N_2 (80%) replacement was investigated, with a primary focus on the extent of replacement and the replacement kinetics under different temperature and pressure conditions. At 274.2 K, the extent of replacement was increased from 76% at 10.0 MPa to 81% at 18.5 MPa, which was mainly attributed to the increase in N_2 storage at a higher pressure. In contrast, the extent of replacement did not change with an increase in temperature from 274.2 K to 277.2 K at 10.0 MPa, but the CO_2/N_2 ratio in the replaced hydrates decreased slightly as the replacement temperature increased. The kinetics of $CH_4 - CO_2 + N_2$ replacement were also affected by the replacement pressure and temperature conditions. An increase in the pressure caused an accelerated CO_2 inclusion in the large ($5^{12}6^2$) cages at the initial stage and an increased N_2 inclusion in the small (5^{12}) cages at the final stage, which resulted in the higher final extent of replacement at higher pressure. At the higher temperature, faster replacement kinetics were induced by the increased inclusion rates of both CO_2 and N_2 at the initial stage of replacement. As N_2 concentration in the injecting gas increased, the CH_4 recovery was increased whereas extent of CO_2 storage in the hydrate phase was decreased due to the competitive inclusion of CO_2 and N_2 in both large and small cages. The experimental results obtained for the different replacement behaviors of the guest molecules according to the temperature, pressure, and composition of injecting gas changes can provide meaningful insights into the replacement mechanism and estimation of the productivity of natural gas from NGH reservoirs in various locations and environments when the $CH_4 - CO_2 + N_2$ replacement technology is applied for production.

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3.4. CO_2 enclathration behavior during mixed hydrate formation for CO_2 sequestration

This chapter includes the published contents:

Mok, J., Choi, W., Kim, S., Lee, J., & Seo, Y. (2022). NaCl-induced enhancement of thermodynamic and kinetic CO_2 selectivity in CO_2 + N_2 hydrate formation and its significance for CO_2 sequestration. Chemical Engineering Journal, 451, 138633.
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3.4.1. Abstract

The importance of carbon capture and storage (CCS) has recently been emphasized owing to the ever-increasing global warming. Direct CO_2 sequestration in marine sediments is an attractive option for CO_2 storage, and some amount of injected CO_2 can be stored in the form of solid gas hydrates. In this study, the CO2 enclathration behavior in various mixed gas conditions are observed for CO2 sequestration. Firstly, the effects of NaCl on the hydrate phase equilibria, thermodynamic and kinetic CO_2 selectivity, and time-dependent growth behaviors of CO_2 + N_2 hydrates were experimentally investigated to elucidate their implications for hydrate-based CO_2 sequestration. The presence of NaCl shifted the equilibrium conditions of CO_2 + N_2 hydrates to the higher-pressure or lower-temperature region, whereas it increased thermodynamic CO_2 selectivity at a specified temperature and pressure, which was also confirmed by the pressure-composition diagram. As revealed by powder X-ray diffraction analysis, the conversion into CO_2 + N_2 hydrate was lower in the saline water system because of the lower initial driving force and gradual salt enrichment in the residual solution during hydrate growth. *In situ* Raman spectroscopic measurements demonstrated that CO_2 was kinetically selective at the early stage of CO_2 + N_2 hydrate formation and that kinetic CO_2 selectivity was more noticeable in the saline water system. Secondly, the guest enclathration behavior was experimentally observed in different composition of CH_4 + CO_2 + N_2 mixed hydrate formation. In same CO_2/CH_4 ratio of injecting gas, the time-dependent CO_2/CH_4 ratio in the hydrate phase during the hydrate formation was almost similar regardless of additional N_2. The overall results provide an in-depth understanding of the role of salts and N_2 in CO_2-mixed hydrate formation and thus offer valuable insights into hydrate-based CO_2 storage and geological CO_2 sequestration.

3.4.2. Thermodynamic CO_2 selectivity in pure and saline water systems

Considering that hydrate formation behavior is strongly affected by the thermodynamic driving force,

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an accurate understanding and information about the thermodynamic phase equilibria of the target gas hydrate systems are essential. The three-phase (H–L_W–V) equilibria of the CO_2 (20%) + N_2 (80%) + NaCl (3.5 wt%) + water mixture were measured and are depicted in Fig. 3.4.1 and Table 3.4.1. Compared with the hydrate phase equilibrium curve of the pure water system, that of the saline water system shifted to the thermodynamically harsher region, that is, a lower temperature at any given pressure or a higher pressure at any given temperature, owing to the thermodynamic inhibition effect of salts[6,98-100]. This indicates that the thermodynamic driving force for hydrate formation in marine sediments is smaller than that in pure water; thus, the growth behavior of the CO_2 + N_2 hydrates in saline water might differ from that in pure water.

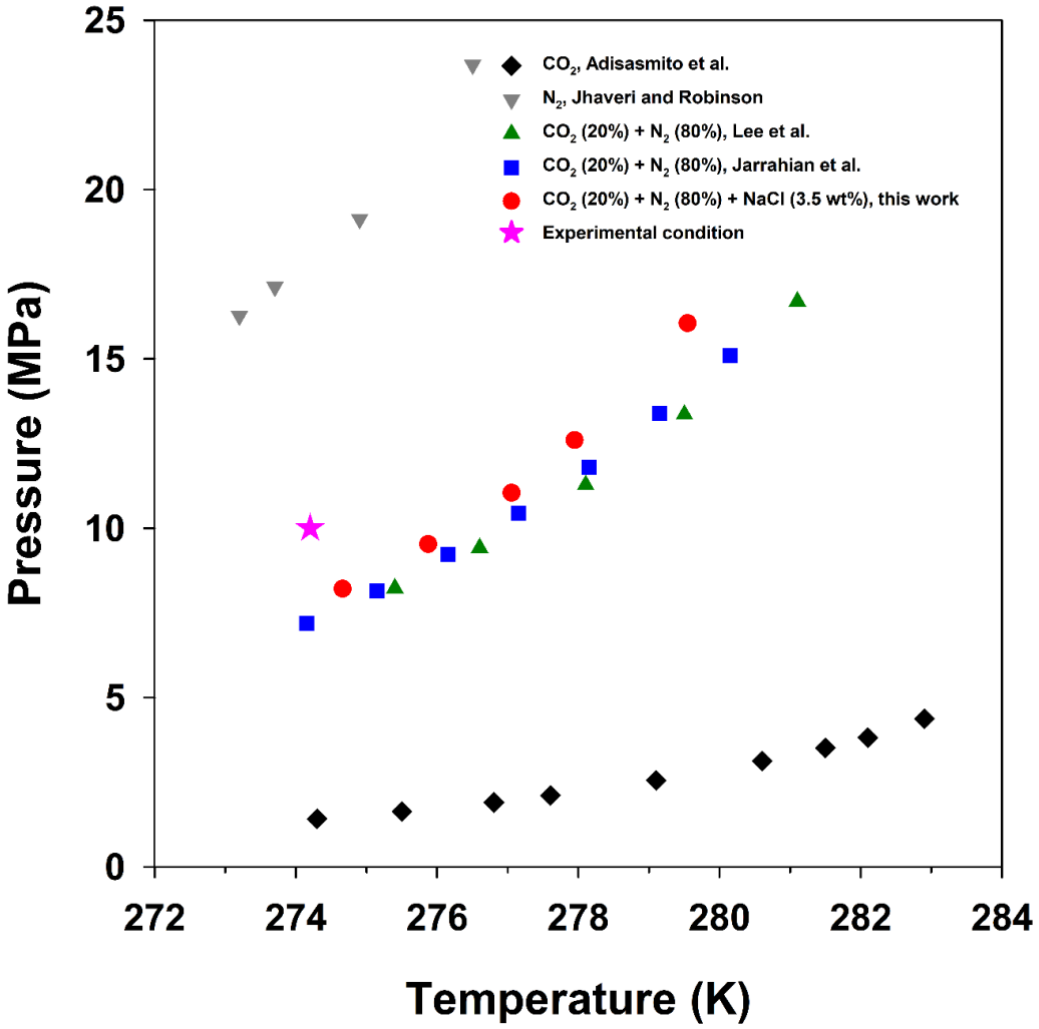


Figure 3.4.1. Three-phase equilibria of CO_2 (20%) + N_2 (80%) + NaCl (0 and 3.5 wt%) + water mixtures

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Table 3.4.1. Three-phase (H–Lw–V) equilibrium data of the CO_2 (20%) + N_2 (80%) + NaCl (3.5 wt%) + water system

CO_2 (20%) + N_2 (80%) + NaCl (3.5 wt%) + water system	
T (K)	**P (MPa)**
274.7	8.2
275.9	9.5
277.0	11.0
277.9	12.6
279.5	16.1

To observe the effect of salt (NaCl) on hydrate growth, the experiments for CO_2 + N_2 hydrate formation were conducted at a specified temperature and pressure condition (274.2 K and 10.0 MPa) for the pure water and NaCl (3.5 wt%) solution systems. Fig. 3.4.2 shows the changes in the vapor phase composition and accumulated gas consumption during the CO_2 + N_2 hydrate growth in pure water and NaCl (3.5 wt%) solution at 274.2 K and 10.0 MPa. During hydrate growth, the CO_2 concentration in the vapor phase decreased gradually in both cases, indicating that CO_2 was preferentially incorporated in the hydrate phase. However, at the end of the experiment, the CO_2 concentration in the vapor phase for the pure water system (14%) was lower than that for the saline water system (17%), and the total gas consumption for the pure water system was 2.4 times larger than that for the saline water system. This difference was attributable to the smaller driving force and increased inhibition effect for the saline water system. As both experiments were conducted at 274.2 K and 10.0 MPa, a smaller initial driving force for hydrate formation was given for the saline water system due to the equilibrium curve shift to the thermodynamically harsher region in the presence of NaCl, as shown in Fig. 3.4.1. In addition, the enrichment of both NaCl and N_2 during hydrate formation exerted a double-inhibition effect on the saline water system. In both the pure water and saline water systems, the N_2 concentration in the vapor phase during hydrate growth was increased gradually by the preferential occupation of CO_2 molecules in the hydrate phase, which resulted in a gradual increase in the equilibrium pressure of the CO_2 + N_2 hydrate at a specified temperature and a consequent gradual decrease in the driving force. Furthermore, for the saline water system, the NaCl concentration in the residual solution and N_2 concentration in the vapor phase increased during hydrate formation, which led to an additionally increased thermodynamic inhibition effect on the hydrate phase equilibria. Although the N_2 enrichment in the vapor phase was less significant in the saline water system, the double-thermodynamic inhibition effect accelerated the depletion of the driving force during hydrate growth and finally caused the smaller hydrate conversion for the saline water system at 274.2 K and 10.0 MPa.

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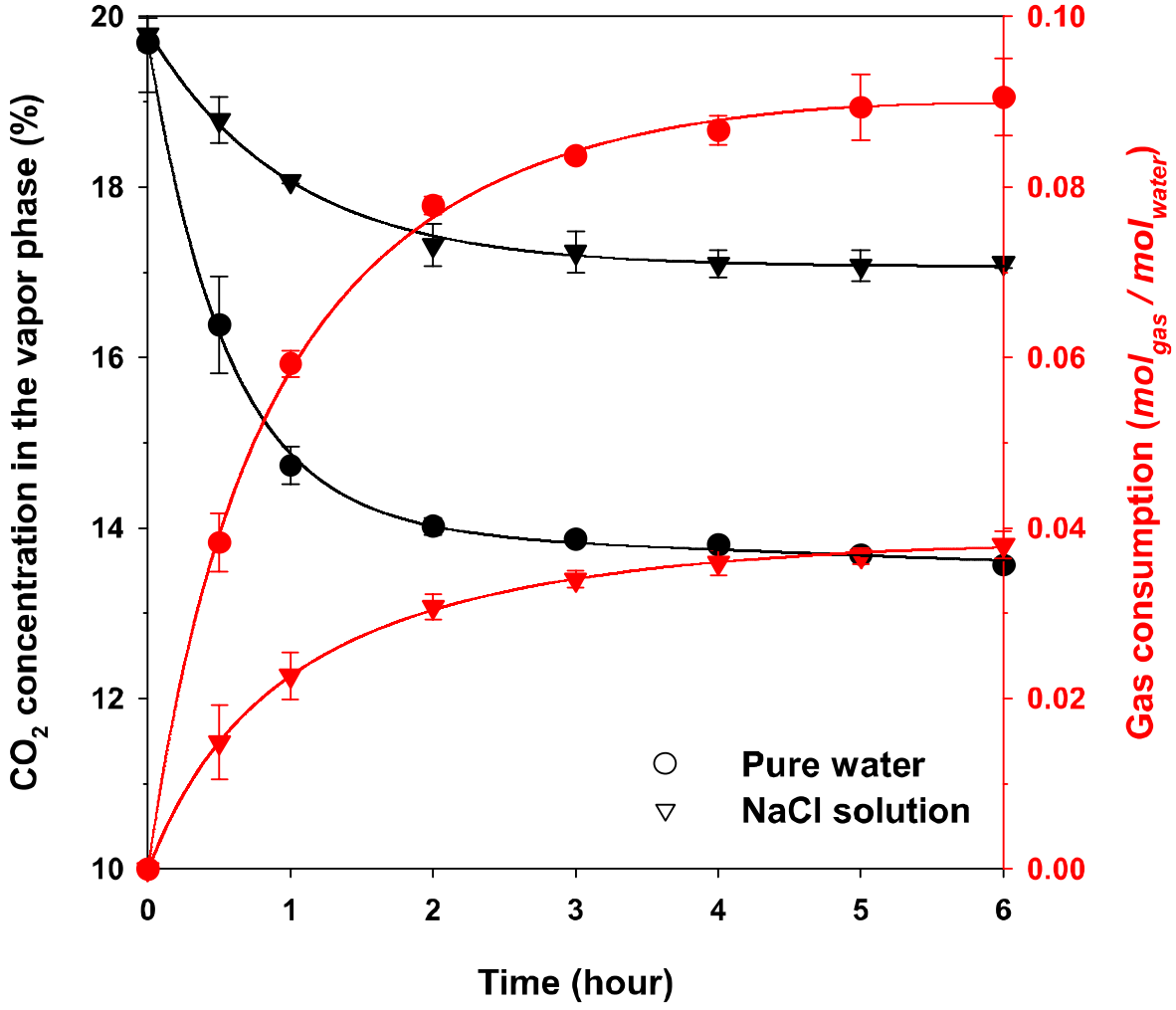


Figure 3.4.2. Changes in CO_2 concentration in the vapor phase and accumulated gas consumption during CO_2 (20%) + N_2 (80%) mixed hydrate formation in pure water and NaCl (3.5 wt%) solution at P = 10.0 MPa and T = 274.2 K

The PXRD measurement and subsequent Rietveld refinement of the PXRD patterns were performed to observe the crystal structure and quantify the hydrate conversion after hydrate formation (Fig. 3.4.3). CO_2 + N_2 gas mixtures are known to form sI hydrates except in the very low CO_2 concentration range (<1%), even though the crystal structure of pure CO_2 hydrate is sI and the pure N_2 hydrate is sII[101]. Fig. 3.4.3 clearly shows that the crystal structure of both CO_2 (20%) + N_2 (80%) hydrates formed with pure water and a NaCl (3.5 wt%) solution was sI with a space group of cubic $Pm\bar{3}n$ (lattice parameter of

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11.8873(7) Å for pure water and 11.8893(4) Å for the NaCl solution, which means that the presence of salt did not affect the hydrate structure. NaCl dihydrate ($NaCl \cdot 2H_2O$) was additionally observed in the $CO_2 + N_2$ + NaCl hydrate and was thought to be formed from the residual NaCl solution in the cooling process using liquid nitrogen for the PXRD sampling[102]. The Rietveld refinement of the PXRD patterns showed that the hydrate conversion in the saline water system was much lower than that in the pure water system (Table 3.4.2), consistent with the results of the gas consumption experiment.

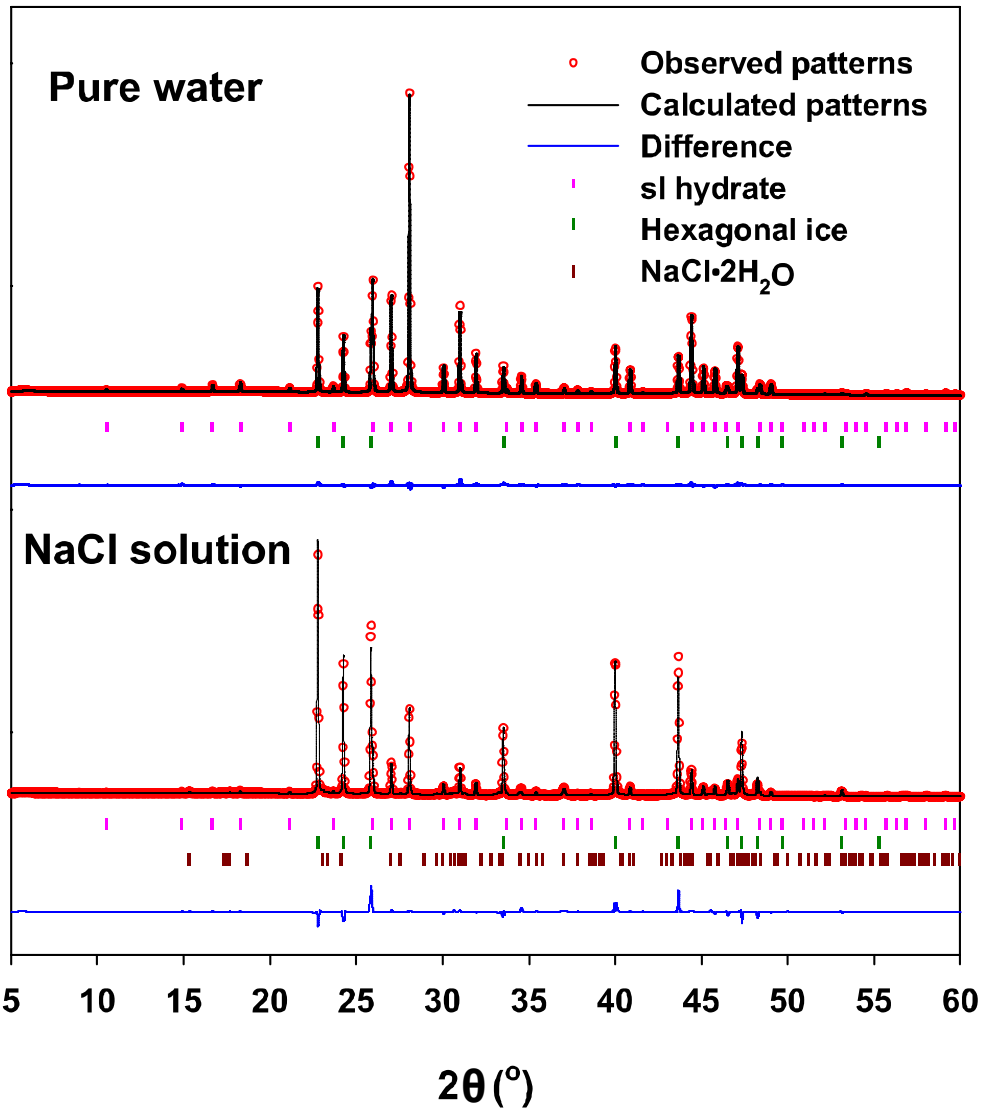


Figure 3.4.3. PXRD patterns of the CO_2 (20%) + N_2 (80%) mixed hydrates formed with pure water and NaCl (3.5 wt%) solution at P = 10.0 MPa and T = 274.2 K (baseline subtracted)

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Table 3.4.2. Hydrate structure and phase fraction of CO_2 (20%) + N_2 (80%) hydrates formed with pure water (R_{wp} = 6.02%) and 3.5-wt% NaCl solution (R_{wp} = 14.7%)

	Hydrate structure	Phase fraction (wt%)		
		Hydrate	Hexagonal ice	sodium chloride dihydrate
CO_2 + N_2 hydrate (pure water)	sI	79.1 (0.2)	20.9 (0.1)	-
CO_2 + N_2 hydrate (NaCl 3.5 wt% solution)	sI	26.5 (0.7)	68.1 (1.5)	5.4 (0.5)

To compare the CO_2 selectivity between the pure water and saline water systems, the guest compositions in the vapor and hydrate phases were measured and are shown in Fig. 3.4.4. In both systems, the final CO_2 composition in the hydrate phase was higher than the CO_2 composition in feed gas, demonstrating that CO_2 molecules were thermodynamically selective in the hydrate phase. The final CO_2 composition in the hydrate phase for the saline water system (62%) was higher than that for the pure water system (51%). Fig. 3.4.4 clearly indicates that at a specified temperature and pressure condition (274.2 K and 10.0 MPa), CO_2 molecules were more preferentially captured in the hydrate phase for the saline water system.

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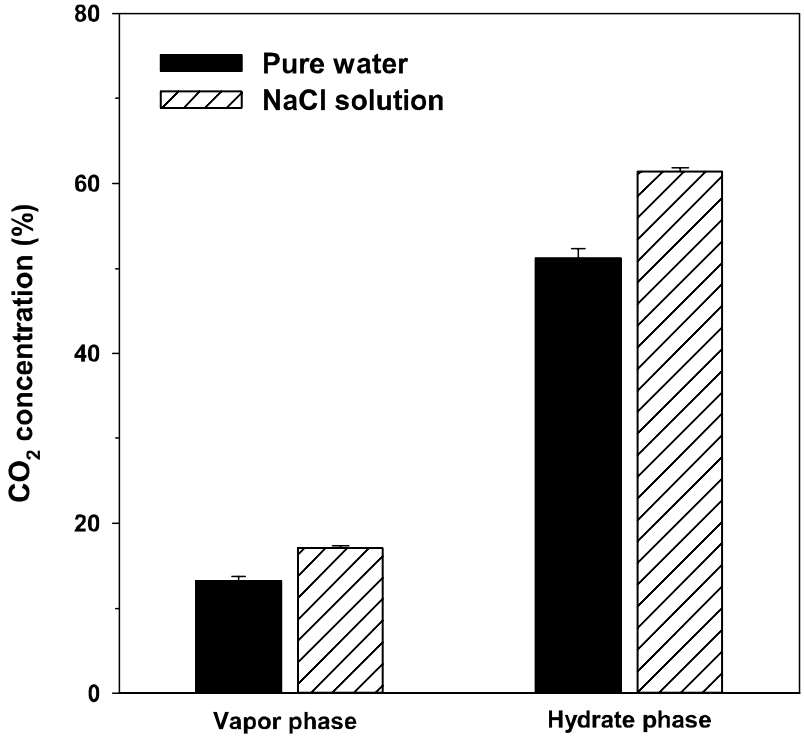


Figure 3.4.4 CO_2 concentration in the vapor and hydrate phases after CO_2 (20%) + N_2 (80%) hydrate formation in pure water and NaCl (3.5 wt%) solution at P = 10.0 MPa and T = 274.2 K

To examine the effect of CO_2 selectivity in the hydrate phase of both the pure water and saline water systems on the CO_2 sequestration efficiency, the guest compositions obtained from the GC measurement are illustrated in Fig. 3.4.5 along with the quantitative hydrate conversion obtained from the PXRD measurement. Fig. 3.4.5 demonstrates that although the CO_2 selectivity in the hydrate phase was higher in the saline water system, the amount of stored CO_2 in the hydrate phase might be smaller in the saline water system because of the lower hydrate conversion.

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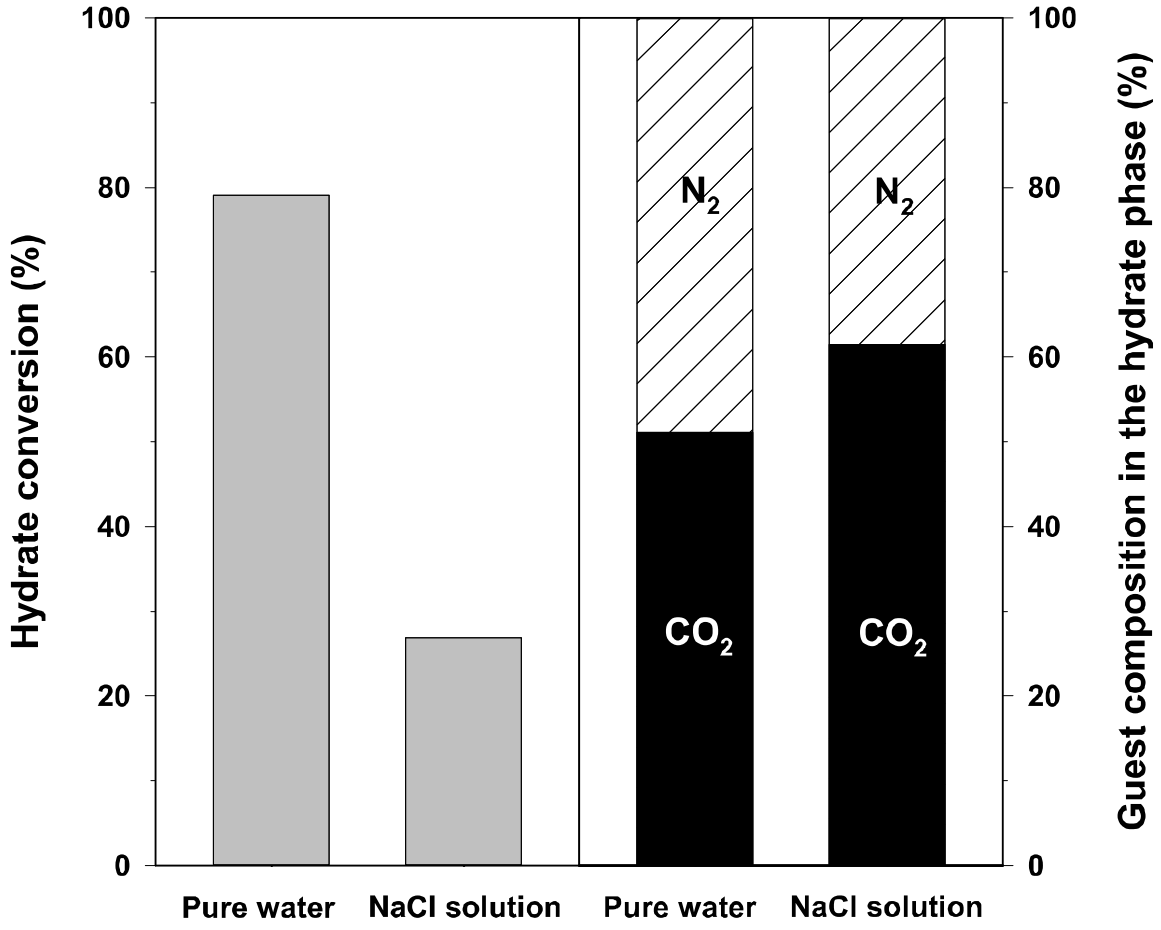


Figure 3.4.5. Quantitative hydrate conversion and guest compositions in the hydrate phase after CO_2 (20%) + N_2 (80%) hydrate formation in pure water and NaCl (3.5 wt%) solution at P = 10.0 MPa and T = 274.2 K

To uncover the reason for the higher CO_2 selectivity in the saline water system, the CO_2 compositions in the hydrate and vapor phases at the three-phase equilibrium state of the CO_2 + N_2 + water and CO_2 + N_2 + NaCl + water systems were determined at 274.2 K using the CSMGem program . The isothermal pressure (P)–composition (x) diagram of the CO_2 + N_2 and CO_2 + N_2 + NaCl hydrates is shown in Fig. 3.4.6. The P-x diagram was drawn within a CO_2 composition range higher than 1%, in which the sI hydrate formation was assured[101]. In Fig. 3.4.6, both the vapor phase lines (solid lines on the left side) and the hydrate phase lines (dotted lines on the right side) moved to the higher-pressure region as the NaCl concentration increased at any given CO_2 composition, consistent with the fact that the addition

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of NaCl to the system increased the three-phase equilibrium pressure, as shown in Fig. 3.4.1. At the vapor phase composition of CO_2 (20%) + N_2 (80%), the corresponding CO_2 compositions in the hydrate phase at the equilibrium state were 69% (at 5.7 MPa), 67% (at 6.9 MPa), and 66% (7.5 MPa) for pure water, the NaCl (3.5 wt%) solution, and the NaCl (4.7 wt%) solution, respectively. This indicates that at a specified temperature and vapor phase composition, the thermodynamic CO_2 selectivity in the hydrate phase decreased slightly as salinity increased.

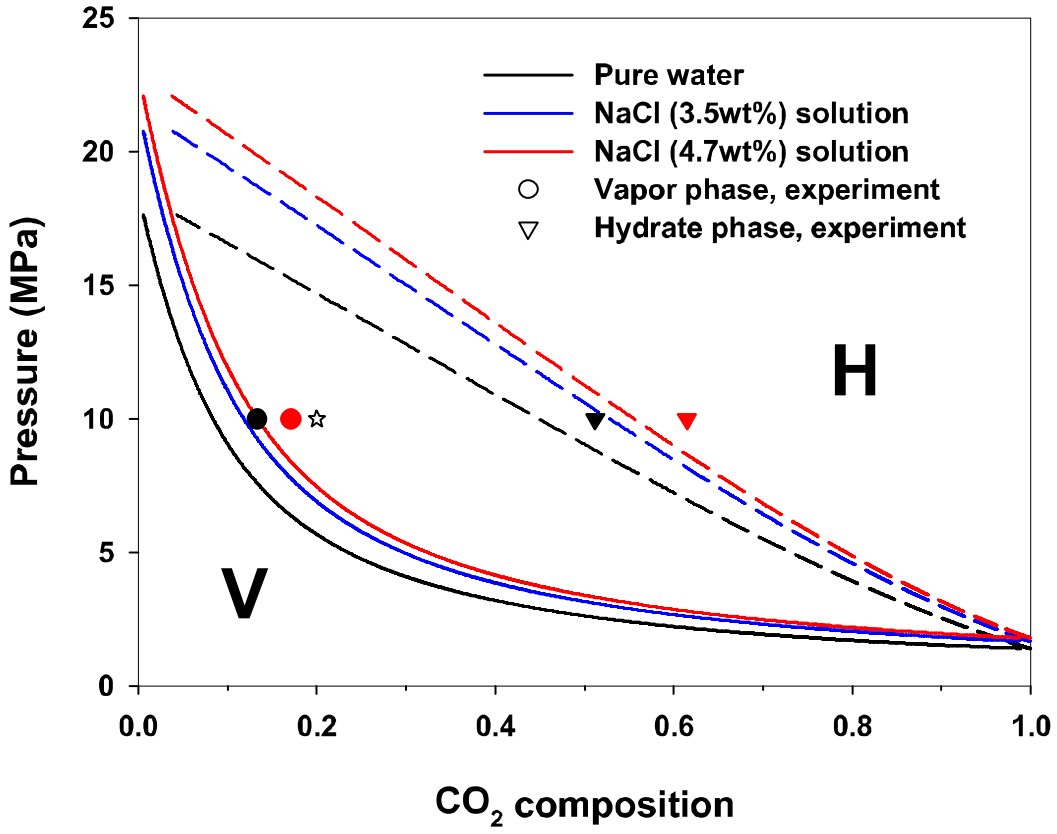


Figure 3.4.6. Pressure-composition diagram of CO_2 + N_2 + NaCl (0, 3.5, and 4.7 wt%) + water mixtures at 274.2 K (solid and dashed lines: calculated using CSMGem, star symbol: initial condition)

However, the temperature and pressure of marine sediments are decided geologically, so the P-x diagram should be analyzed in terms of isobaric conditions, considering the actual CO_2 sequestration into marine sediments. From the hydrate conversion shown in Table 3.4.2 and the mass balance in the saline water system after hydrate formation, the NaCl concentration in the residual solution was

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estimated to be 4.7 wt%. Therefore, the P-x diagram of the $CO_2 + N_2$ + water and $CO_2 + N_2$ + NaCl (4.7 wt%) + water systems at 274.2 K is depicted in Fig. 3.4.6, along with the experimental results. At an isobaric condition, both the hydrate and vapor phase curves for the NaCl (4.7 wt%) solution shifted to the higher CO_2 composition region. The experimentally measured values at 10.0 MPa were in agreement with the predicted vapor and hydrate phase curves in the P-x diagram. It is clear from the P-x diagram in Fig. 3.4.6 that under isobaric conditions, the CO_2 composition in the hydrate phase for the saline water system was much higher than that for the pure water system.

3.4.3. Kinetic CO_2 selectivity in pure and saline water systems

The kinetic CO_2 selectivity in the hydrate phase during the formation of $CO_2 + N_2$ hydrate can provide an understanding of the time-dependent guest-enclathrating behavior in hydrate-based CO_2 sequestration. To observe time-dependent CO_2 selectivity in the hydrate phase, *in situ* Raman spectroscopic measurements were performed during the formation of the $CO_2 + N_2$ hydrate. The Raman peak positions of the incorporated guest molecules (CO_2 and N_2) in the hydrate phase were different from those in the gas phase because of the alteration of the chemical environment of the molecules. In Fig. 3.4.7 (a), the Raman bands of the CO_2 molecules captured in the sI hydrate cages appeared at 1,276 and 1,380 cm^{-1}, which were distinct from the major Raman bands of gaseous CO_2 (at 1,285 and 1,388 cm^{-1})[73]. The Raman peak at 2,323 cm^{-1} was assigned to the N_2 molecules in the hydrate phase, which was distinguishable from that of the free N_2 gas molecules at 2,329 cm^{-1} [103]. The residual air in the Raman probe was also detected at 2,329 cm^{-1}.

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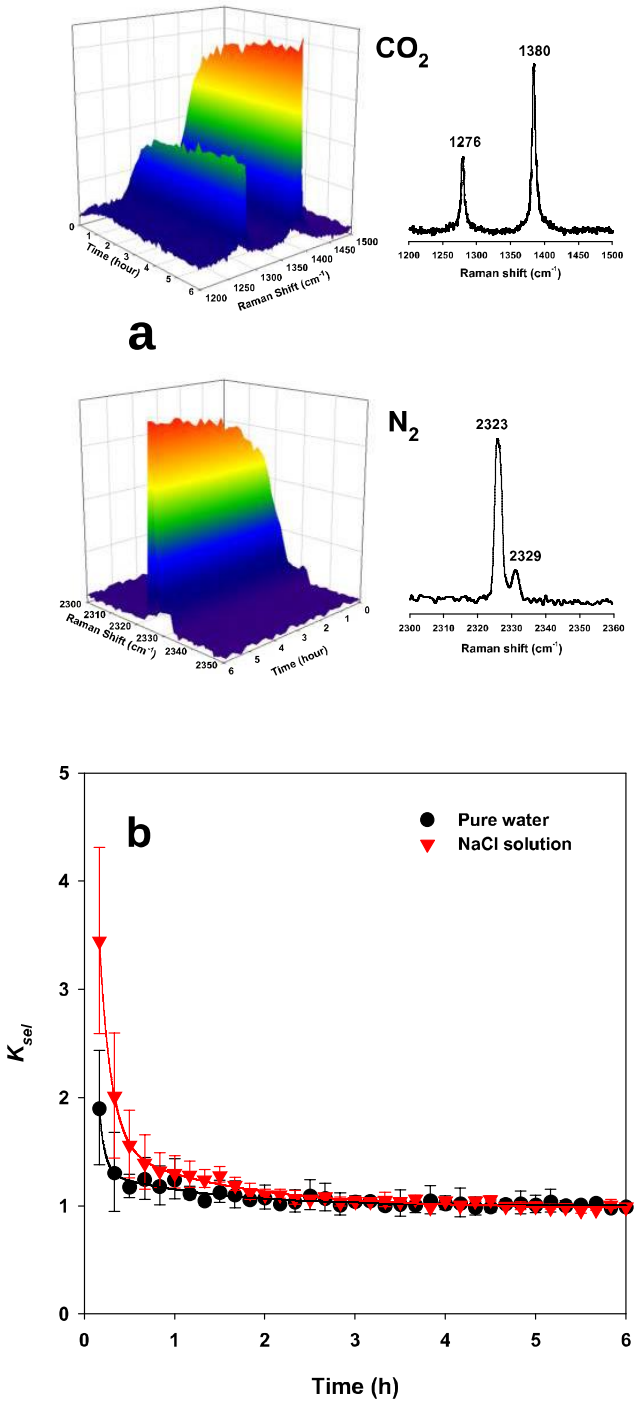


Figure 3.4.7. (a) *In situ* Raman spectra of the CO_2 and N_2 molecules captured in the hydrate phase. (b) Time-dependent K_{sel} values during CO_2 (20%) + N_2 (80%) hydrate formation in pure water and NaCl (3.5 wt%) solution at P = 10.0 MPa and T = 274.2 K

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As shown in Fig. 3.4.7 (a), both the CO_2 and N_2 molecules were clearly captured in the hydrate phase when hydrate formation began. During hydrate growth, the Raman peaks of the CO_2 and N_2 molecules in the hydrate phase did not shift, which indicates that the formation of the sI $CO_2 + N_2$ hydrate occurred iso-structurally. However, it was difficult to observe the composition ratio of the guest molecules in the hydrate phase or to obtain the quantitative cage occupancies of each guest molecule through Raman spectroscopic measurements. Several attempts have been made to observe the composition ratio of enclathrated guest molecules during hydrate formation using Raman spectroscopy. Sum et al. used the calibration curve, which relates the ratio of the Raman peak area for guest molecules to the mole fraction ratio in the vapor phase, to estimate the guest compositions in the hydrate phase on the assumption that the Raman scattering cross sections of the guest molecules in the vapor phase were the same as those in the hydrate phase[73]. Qin and Kuhs suggested F-factors of enclathrated guest molecules, which were obtained using a combination of Raman spectroscopy and PXRD[79,104]. In this study, to compare the enclathrating kinetics of CO_2 and N_2 molecules, the ratio of the normalized Raman peak area for each guest molecule (K_{sel}) at time t was investigated using the following equation[105]:

$$K_{\mathrm{sel}} = \frac{\frac{A_t}{A_{\max}}(CO_2)}{\frac{A_t}{A_{\max}}(N_2)},$$

where A_t and $A_{\max}$ represent the integrated Raman peak area for each guest molecule at time t and at the end of the experiment, respectively. The time-dependent K_{sel} values in pure water and the NaCl (3.5 wt%) solution are depicted in Fig. 3.4.7 (b). In the pure water system, the K_{sel} value was initially higher than 1 but gradually decreased and finally converged to 1, which means that CO_2 molecules were kinetically selective at the early stage of $CO_2 + N_2$ hydrate formation. This was attributable to the fact that the unit cell of sI hydrates has a larger number of large ($5^{12}6^2$) cages, in which CO_2 can be readily captured, than small (5^{12}) cages [unit cell of sI hydrate: 2 (5^{12}) 6 ($5^{12}6^2$) 46 (H_2O)][104]. The CO_2 selectivity at the early stage of hydrate growth was more pronounced in the NaCl solution (Fig. 3.4.7 (b)), which indicates that the presence of the salt increased both the thermodynamic and kinetic CO_2 selectivity during the $CO_2 + N_2$ hydrate formation. These results have important practical implications for hydrate-based CO_2 sequestration. Although the hydrate conversion can be lowered by a decrease in the thermodynamic driving force in the NaCl solution, both thermodynamic and kinetic CO_2 selectivity can be enhanced in the saline water system, which can critically affect the CO_2 sequestration efficiency. In this study, we found that the guest-enclathrating behavior of $CO_2 + N_2$ hydrate in the saline water system was thermodynamically and kinetically different from that in the pure water system. For a real application and deeper understanding of hydrate-based CO_2 storage in marine sediments and NGH-bearing sediments, further studies on hydrate formation behavior using various gas mixtures in a saline

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water-saturated sedimentary environment should be investigated as a next step[106].

3.4.4. Effect of additional N_2 on CO_2 selectivity for $CH_4 + CO_2 + N_2$ hydrate formation

The thermodynamic stability of the gas hydrates are varies with gas composition, and thus the understanding for the hydrate phase equilibrium is important for hydrate formation. To observe the effect of N_2 composition on the $CH_4 + CO_2 + N_2$ mixed hydrates, the composition ratio between CH_4 and CO_2 was maintained for 1:1 and the N_2 was added. The three-phase ($H–L_W–V$) equilibria of the $CH_4 + CO_2 + N_2$ + water mixture in different N_2 compositions were measured and are depicted in Fig. 3.4.8. Due to the harsher equilibrium condition of pure N_2 hydrate, compared with the hydrate phase equilibrium curve of the CH_4 (50%) + CO_2 (50%) system, that of the N_2 added systems shifted to the thermodynamically harsher region, that is, a lower temperature at any given pressure or a higher pressure at any given temperature, and the extent was greater at higher N_2 composition owing to the thermodynamic inhibition effect of N_2.

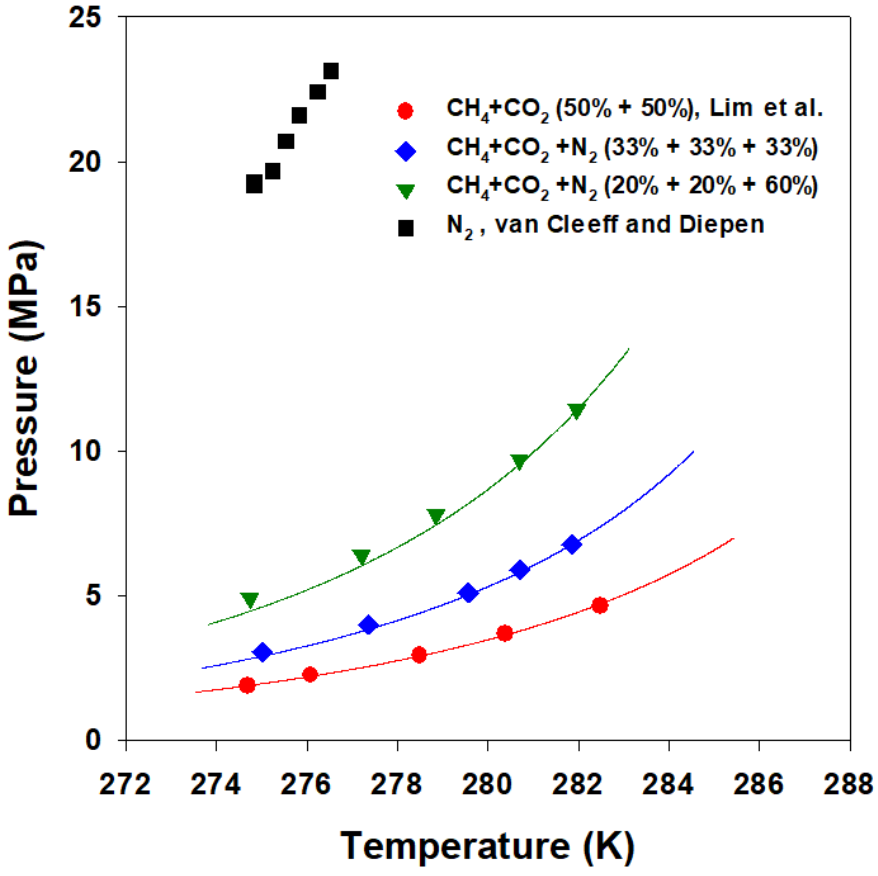


Figure 3.4.8. Three-phase equilibria of $CH_4 + CO_2 + N_2$ + water mixtures in different compositions

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To observe the effect of N_2 composition on the $CH_4 + CO_2 + N_2$ mixed hydrate formation behavior, the gas consumption experiment was conducted on three different gas compositions, (1) CH_4 (50%) + CO_2 (50%), (2) CH_4 (33%) + CO_2 (33%) + N_2 (33%), and (3) CH_4 (20%) + CO_2 (20%) + N_2 (60%) at 278.2 K and 3.9 MPa, 6.0 MPa, and 10.2 MPa, respectively, which is identical partial pressures of $CH_4 + CO_2$ gas contents. As shown in Fig. 3.4.9, the extent of gas consumption is decreased as N_2 composition increased in the initial gas content. This is because during the hydrate formation, the CO_2 composition in the vapor phase is decreased in all cases, which indicates that CO_2 is preferentially occupied in the hydrate phase due to the thermodynamic preference. One interesting thing is that during the mixed hydrate formation, the CO_2/CH_4 ratio in the vapor phase showed similar trends in all three cases regardless of the initial N_2 concentration (Fig. 3.4.10).

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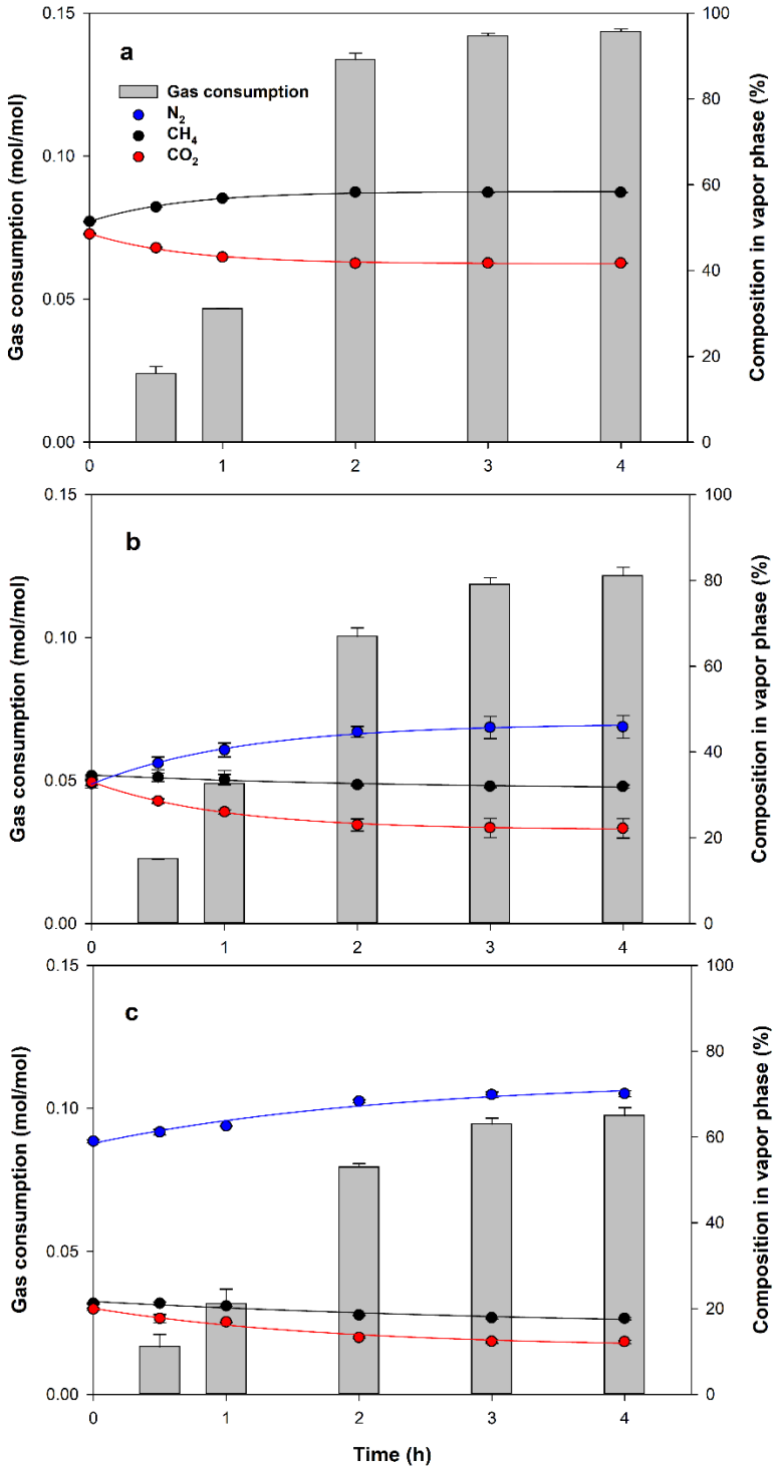


Figure 3.4.9. Composition changes of CH_4, CO_2, and N_2 in the vapor phase and accumulated gas consumption during (a) CH_4 (50%) + CO_2 (50%), (b) CH_4 (33%) + CO_2 (33%) + N_2 (33%), and (c) CH_4 (20%) + CO_2 (20%) + N_2 (60%) mixed hydrate formation at T = 278.2 K

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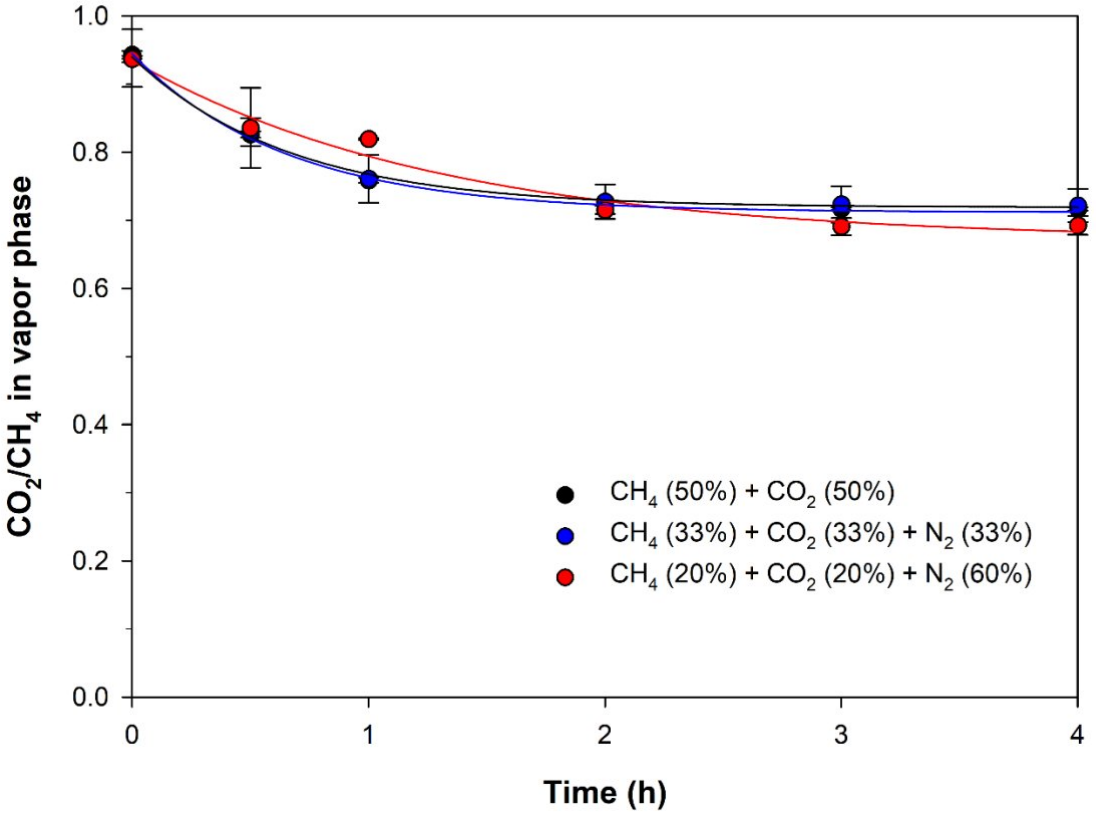


Figure 3.4.10. CO_2/CH_4 ratio in the vapor phase during mixed hydrate formation at $T = 278.2$ K

To verify the reason, the time-dependent guest composition in the hydrate phase during the mixed hydrate formation was measured in each cases. As shown in Fig. 3.4.11, regardless of the N_2 composition in the vapor phase, the time-dependent CO_2/CH_4 ratio in the hydrate phase during the hydrate formation was almost similar in each cases, which indicates that the additional N_2 less affect to the time-dependent enclathration behavior of CO_2 and CH_4, but just reduce the extent of CO_2 and CH_4 storage because of additional participation of N_2 in the hydrate phase and reducing extent of hydrate conversion due to the inhibition effect.

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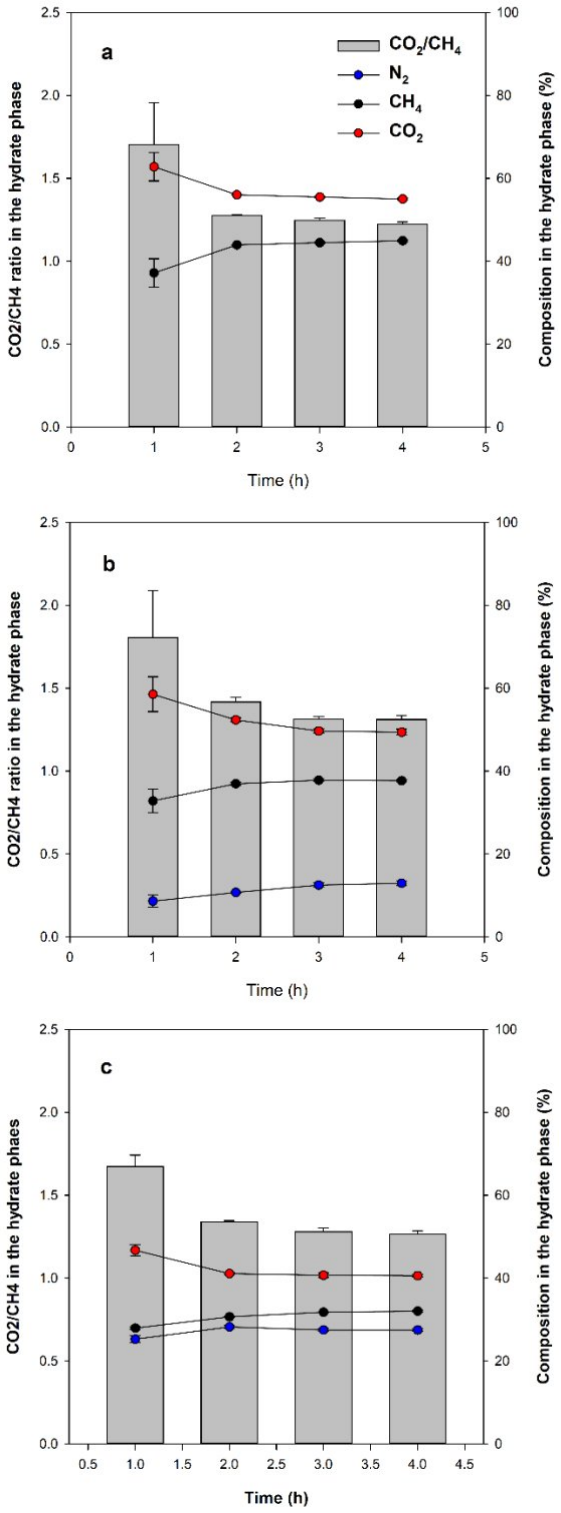


Figure 3.4.11. Composition changes of CH_4, CO_2, and N_2 and CO2/CH4 ratio in the hydrate phase during (a) CH_4 (50%) + CO_2 (50%), (b) CH_4 (33%) + CO_2 (33%) + N_2 (33%), and (c) CH_4 (20%) + CO_2 (20%) + N_2 (60%) mixed hydrate formation at T = 278.2 K

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3.4.5. Conclusions

In this study, the formation behaviors and CO_2 selectivity of CO_2 + N_2 hydrates in both the pure water and saline water systems were investigated in the thermodynamic and kinetic aspects. The three-phase equilibrium curve of the CO_2 (20%) + N_2 (80%) + NaCl (3.5 wt%) + water mixture shifted to the harsher region (higher pressure or lower temperature region) because of the thermodynamic inhibition effect of the salt. The saline water system showed a lower hydrate conversion due to the salt enrichment in the residual solution during the hydrate growth and the consequent gradual decrease in thermodynamic driving force due to the shift of the hydrate equilibrium conditions. The crystal structure and quantitative hydrate conversion of the CO_2 + N_2 hydrates were examined using PXRD analysis. In spite of the lower hydrate conversion, the CO_2 concentration in the hydrate phase was higher in the saline water system. At an isobaric condition, the higher thermodynamic CO_2 selectivity in the saline water system was also confirmed by the P-x diagram of the CO_2 + N_2 + NaCl + water mixtures. The kinetic CO_2 selectivity during the CO_2 + N_2 hydrate formation was investigated by collecting time-dependent Raman spectra. CO_2 was kinetically selective at the early stage of the CO_2 + N_2 hydrate formation, and the kinetic CO_2 selectivity was more pronounced in the saline water system. In case of CH_4 + CO_2 + N_2 mixed hydrate formation, the extent of hydrate formation decreased due to the thermodynamic inhibition effect of additional N_2, and the CO_2 concentration in the hydrate phase also decreased due to the additional participation of N_2. Separately, however, the time-dependent CO_2/CH_4 ratio collected in the hydrate phase during the hydrate growth was similar regardless of the additional N_2 concentration in the vapor phase. The experimental results obtained in this study deepen the understanding of the accurate role of salts in the CO_2 + N_2 hydrate formation and N_2 for CH_4 + CO_2 + N_2 mixed hydrate formation, which provide important insights into hydrate-based CO_2 storage and geological CO_2 sequestration.

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Chapter 4. Efficiency evaluation of guest-dependent hydrate-based desalination

4.1. Thermodynamic properties of gas hydrates in saline water

This chapter includes the published contents:

Mok, J., Choi, W., & Seo, Y. (2021). Evaluation of kinetic salt-enrichment behavior and separation performance of HFC-152a hydrate-based desalination using an experimental measurement and a thermodynamic correlation. Water Research, 193, 116882.
Mok, J., Choi, W., & Seo, Y. (2022). Theoretically achievable efficiency of hydrate-based desalination and its significance for evaluating kinetic desalination performance of gaseous hydrate formers. Desalination, 524, 115487.
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4.1.1. Abstract

Hydrate-based desalination (HBD), a type of freezing-based desalination, can concentrate salts of saline water and produce fresh water via hydrate crystal formation. In this study, the thermodynamic stability, crystallographic information of R152a hydrate were investigated to estimate. The phase equilibria of R152a hydrate in the presence of NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) were measured. The Hu-Lee-Sum correlation was employed to predict the equilibrium shift and hydrate depression temperature The hydration number of the R152a hydrate (sI) was found to be 7.74 through the Rietveld refinement of the powder X-ray diffraction patterns, and it was also used to determine the dissociation enthalpy of the R152a hydrate. In addition, the thermodynamic and kinetic features of propane, R134a, R22, and R152a hydrates in the presence of NaCl were examined to determine the energy-efficient gaseous hydrate former. Structure I hydrate formers (R22 and R152) gave lower hydrate depression temperatures than structure II hydrate formers (propane and R134a) at a given salinity. The overall results indicate that R152a is, potentially, a superior candidate for HBD. The novel approach examined in this study will be useful for assessing the desalination efficiency of the HBD process.

4.1.2. Thermodynamic properties of R152a hydrate

The accurate thermodynamic properties of gas hydrates are important prerequisites for the application of gas hydrates to various industrial processes because they can be fundamental criterion for

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determining the optimal operating conditions of the processes. To evaluate the thermodynamic stability of R152a hydrate in the saline system, the three-phase equilibria of the R152a + NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) + water mixtures were measured using an isochoric method and the results were shown in Fig. 4.1.1 and Table. 4.1.1. The three-phase equilibrium data of pure R152a hydrate were in good agreement with those from the literature[107]. For saline systems, the equilibrium curves of R152a hydrate were shifted to harsher regions (represented by lower temperature and higher pressure) by the inhibition effect of salt ions. As seen in Fig. 4.1.1, the extent of the equilibrium curve shift increased with an increase in the NaCl concentration, indicating that an additional and larger subcooling for gas hydrate formation is required for saline water with a higher NaCl concentration.

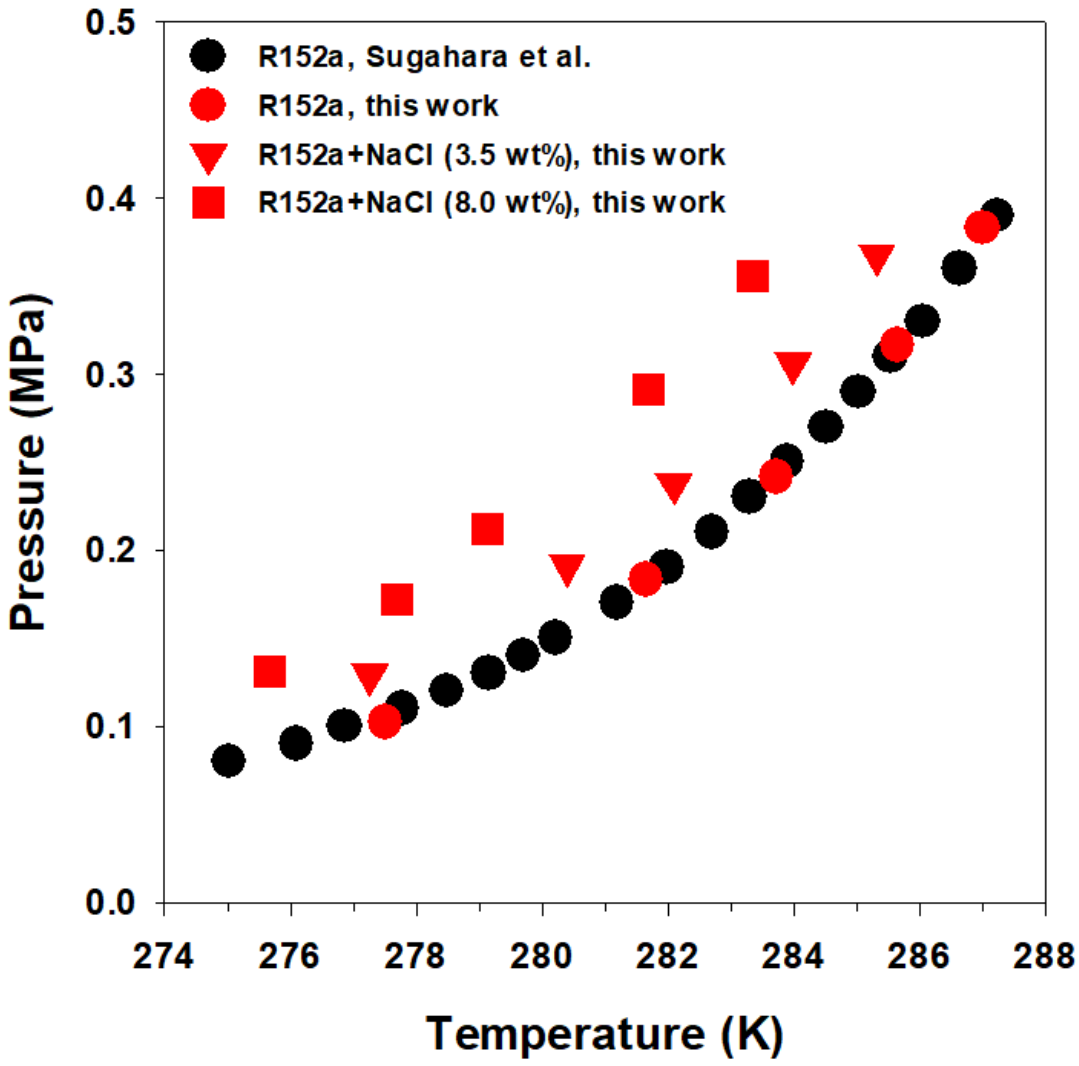


Fig. 4.1.1. Three-phase equilibria of R152a + NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) + water mixtures.

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Table. 4.1.1. Three-phase equilibria of R152a + NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) + water mixtures

Pure water		**NaCl (3.5 wt%) solution**		**NaCl (8.0 wt%) solution**	
T **(K)**	***P*** **(MPa)**	***T*** **(K)**	***P*** **(MPa)**	***T*** **(K)**	***P*** **(MPa)**
277.5	0.10	277.3	0.13	275.7	0.13
281.7	0.18	280.4	0.19	277.7	0.17
283.7	0.24	282.1	0.24	279.1	0.21
285.7	0.32	284.0	0.31	281.7	0.29
287.0	0.38	285.3	0.37	283.4	0.36

The crystallographic information about gas hydrates enabled us to estimate the amount of gas that could be captured in gas hydrates because the cage-to-water ratio in a unit cell of gas hydrates is an intrinsic characteristic of each hydrate structure. To examine the crystalline structure of R152a hydrate, PXRD was employed and the measured diffraction patterns of pure R152a hydrate were provided, as shown in Fig. 4.1.2. There has been a controversy over the crystalline structure of R152a hydrate (sI or sII)[108]. The PXRD patterns in Fig. 4.1.2 clearly demonstrate that the R152a hydrate is sI with a space group of cubic $Pm\bar{3}n$ and that it has a lattice parameter of 11.9888(5) Å (R_{wp} = 10.7% and χ^2 = 8.08). The sI hydrate consists of 6 large ($5^{12}6^2$) cages and 2 small (5^{12}) cages with 46 water molecules, and its ideal unit cell formula is $6(5^{12}6^2)\cdot2(5^{12})\cdot46(H_2O)$. Due to the molecular size of R152a, the Rietveld refinement of the PXRD patterns showed that R152a molecules were captured only in the large ($5^{12}6^2$) cages of sI and the cage occupancy of R152a molecules in the large ($5^{12}6^2$) cages was 0.99, indicating that the hydration number, *n*, which refers to the ratio of the number of host water molecules to the number of entrapped guest molecules in a unit cell, is 7.74 ($R152a\cdot7.74H_2O$). As noted earlier, ions and salts are not included in hydrate crystals, so it is reasonable to say that the crystallographic properties of R152a hydrate are not affected by the surrounding saline environment.

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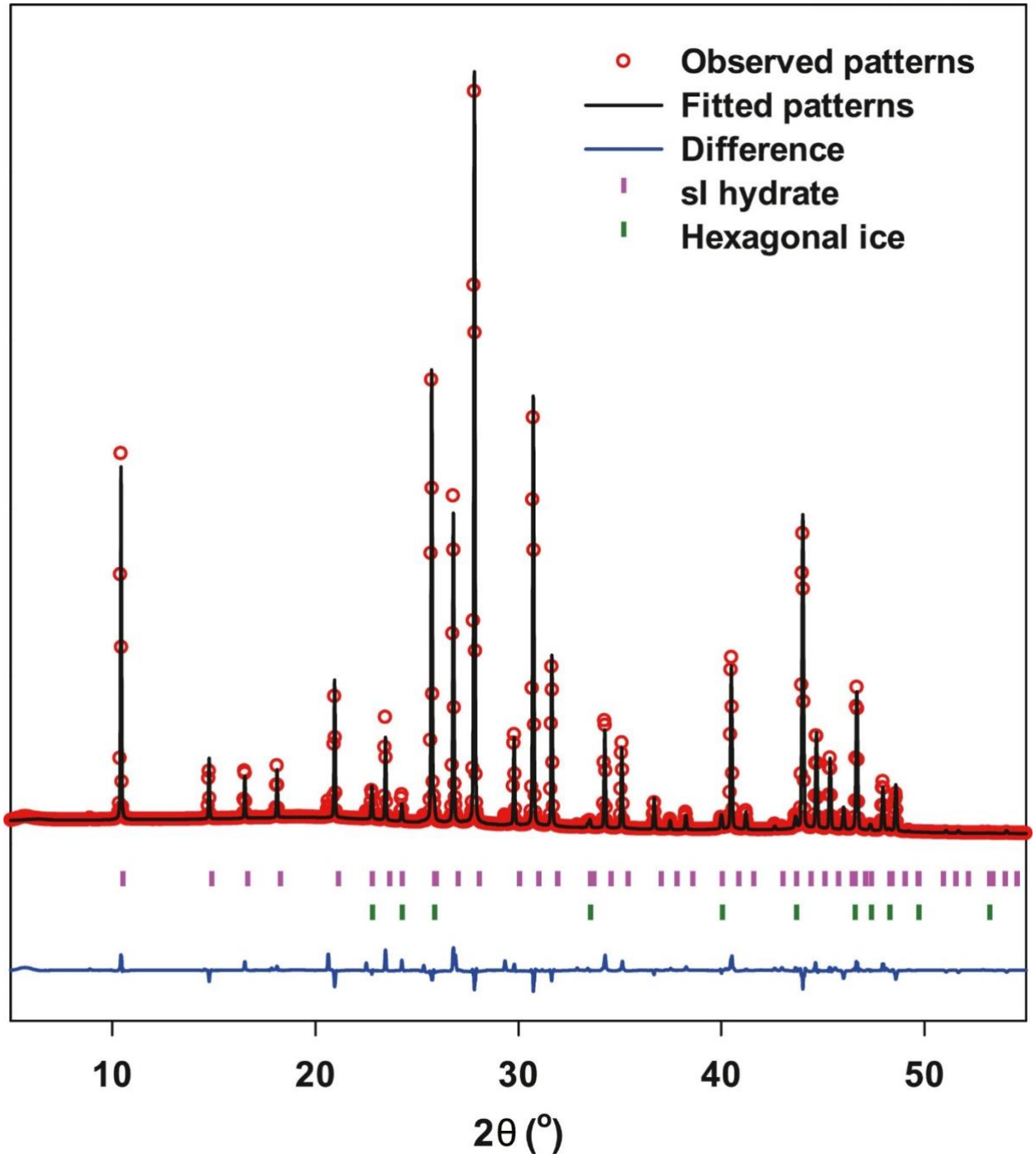


Fig. 4.1.2. PXRD patterns of R152a hydrate.

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Because the continuous HBD process requires the repeated formation and dissociation of gas hydrates, the heat transfer (absorption or liberation) during the phase changes of the gas hydrates is a critical factor in establishing an energy balance of the process. In this study, the heat flows during the dissociation of R152a hydrate formed with pure water and NaCl (3.5 wt%) solution were observed using the HP μ-DSC, as depicted in Fig. 4.1.3 (a) and (b). The R152a + NaCl (3.5 wt%) hydrate showed a much broader endothermic heat-flow curve than the pure R152a hydrate. Because salt is excluded from hydrate crystallization, the NaCl concentration in the residual solution after R152a hydrate formation was expected to be much higher than 3.5 wt%. Therefore, as can be seen in Fig. 3, the temperature at which the R152a hydrate formed with NaCl (3.5 wt%) solution started to dissociate was much lower than that of the pure R152a hydrate, and even lower than the expected dissociation temperature for the NaCl (3.5 wt%) solution. As the dissociation of R152a hydrate progressed, the NaCl concentration in the residual solution gradually decreased due to the dilution caused by the water supply from dissociated hydrates; it finally approached the initial NaCl concentration. Because of the gradual and progressive dissociation of gas hydrates in the salt-containing systems, it was very difficult to measure the accurate onset temperature of R152a + NaCl hydrate using the dynamic DSC method. However, the peak temperatures of the endothermic heat-flow curves in Fig. 4.1.3 (a) and (b) indicated that the presence of NaCl inhibited the hydrate formation and accordingly, the dissociation temperature of the gas hydrate was shifted to the lower one for saline water.

Since the values of cage occupancy and the hydration number of R152a hydrate were quantitatively given by the Rietveld refinement of the PXRD patterns, the dissociation enthalpy (ΔH_d) value of the R152a hydrate was obtained by integrating the endothermic heat-flow curve measured by the HP μ-DSC. The ΔH_d values of the R152a hydrates formed with pure water and NaCl (3.5 wt%) solution were found to be 82.2 ± 0.2 kJ/mol-gas and 81.9 ± 0.5 kJ/mol-gas, respectively, which were consistent with the literature value (85.5 kJ/mol-gas) calculated using the Clausius-Clapeyron equation[109]. A slightly larger error in the ΔH_d value for the NaCl (3.5 wt%) solution was attributable to a slightly larger uncertainty in integrating the broader and more progressive heat-flow curve for the NaCl (3.5 wt%) solution, which might contain R152a hydrate as well as a small amount of solid NaCl and NaCl ·$2H_2O$ at low temperatures[91].

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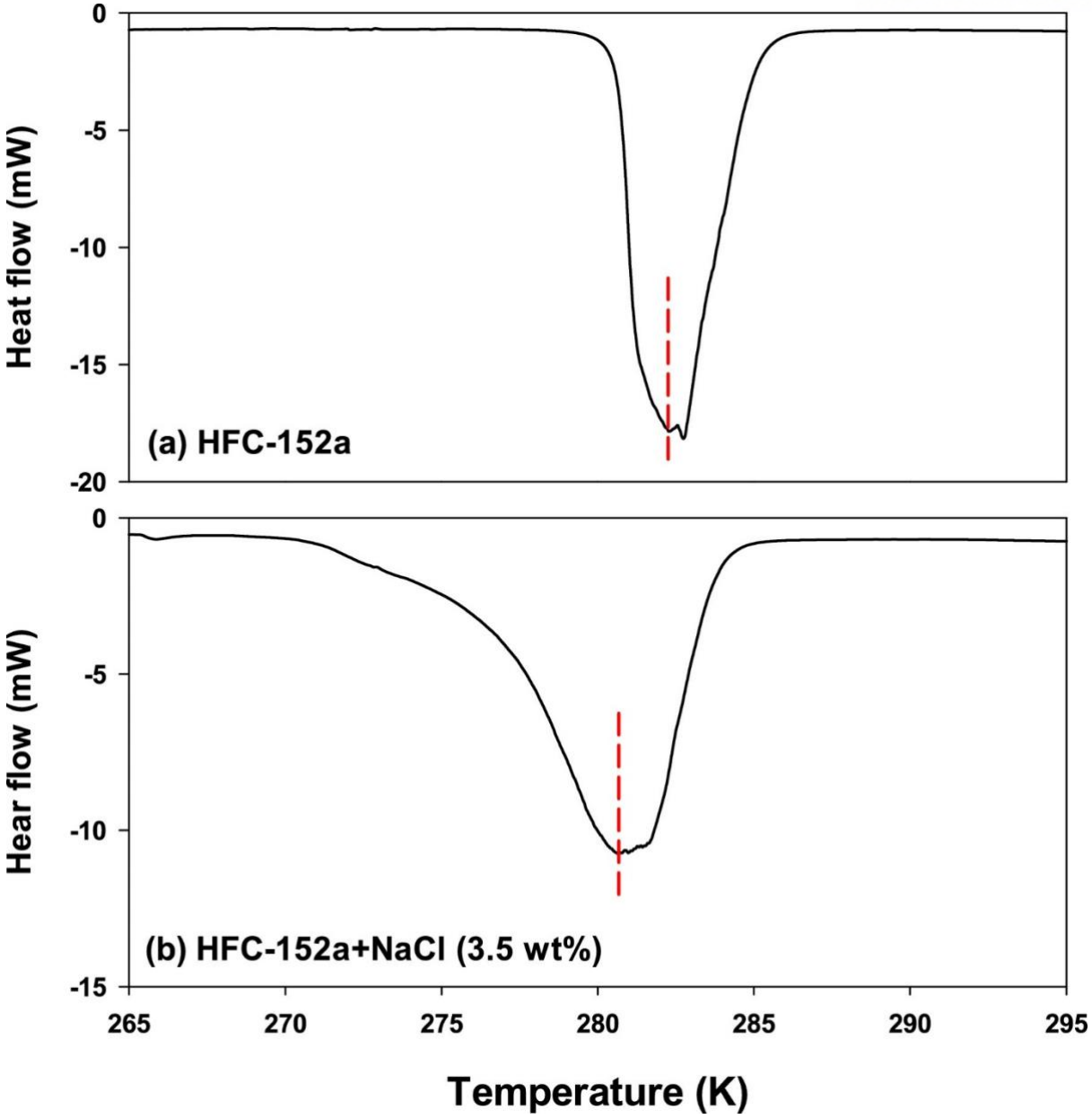


Fig. 4.1.3. Heat flows during dissociation of R152a hydrates formed with pure water and NaCl (3.5 wt%) solution.

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4.1.3. Thermodynamic properties of gaseous hydrate formers in a saline system

An understanding of hydrate phase equilibria should precede the design of the hydrate-based process to determine the appropriate thermodynamic driving force for the process because the reaction conversion and the kinetics of the hydrate formation are directly related to the driving force. Fig. 4.1.4 shows the three-phase [hydrate (H) – liquid water (L_w) – vapor (V)] equilibria of propane, R134a, R22, and R152a hydrates in the pure water system[107,110-112]. Fluorinated gases (F-gases) used as refrigerants have been reported to form thermodynamically very stable gas hydrates. In Fig. 4.1.4, three F-gas hydrates (R134a, R22, and R152a hydrates) exhibited much milder equilibrium conditions (lower pressures and higher temperatures) compared to propane hydrate. For example, the hydrate equilibrium temperatures were 275.5 K for propane, 281.8 K for R134a, 282.3 K for R22, and 285.3 K for R152a at 0.3 MPa, indicating that the equilibrium temperatures of R134a, R22, and R152a hydrates were 6.3 K, 6.8 K, and 9.8 K higher than that of propane hydrate, respectively. The hydrate equilibrium pressures were 0.42 MPa for propane, 0.10 MPa for R134a, 0.15 MPa for R22, and 0.10 MPa for R152a at 277.2 K, indicating that the equilibrium pressures of R134a, R22, and R152a hydrates were 0.32 MPa, 0.27 MPa, and 0.32 MPa lower than that of propane hydrate, respectively. The hydrate phase equilibria demonstrated that the gas hydrate-based process using these refrigerant F-gases could be operated under milder thermodynamic conditions, referring to higher temperatures at any given pressure or lower pressures at any given temperature.

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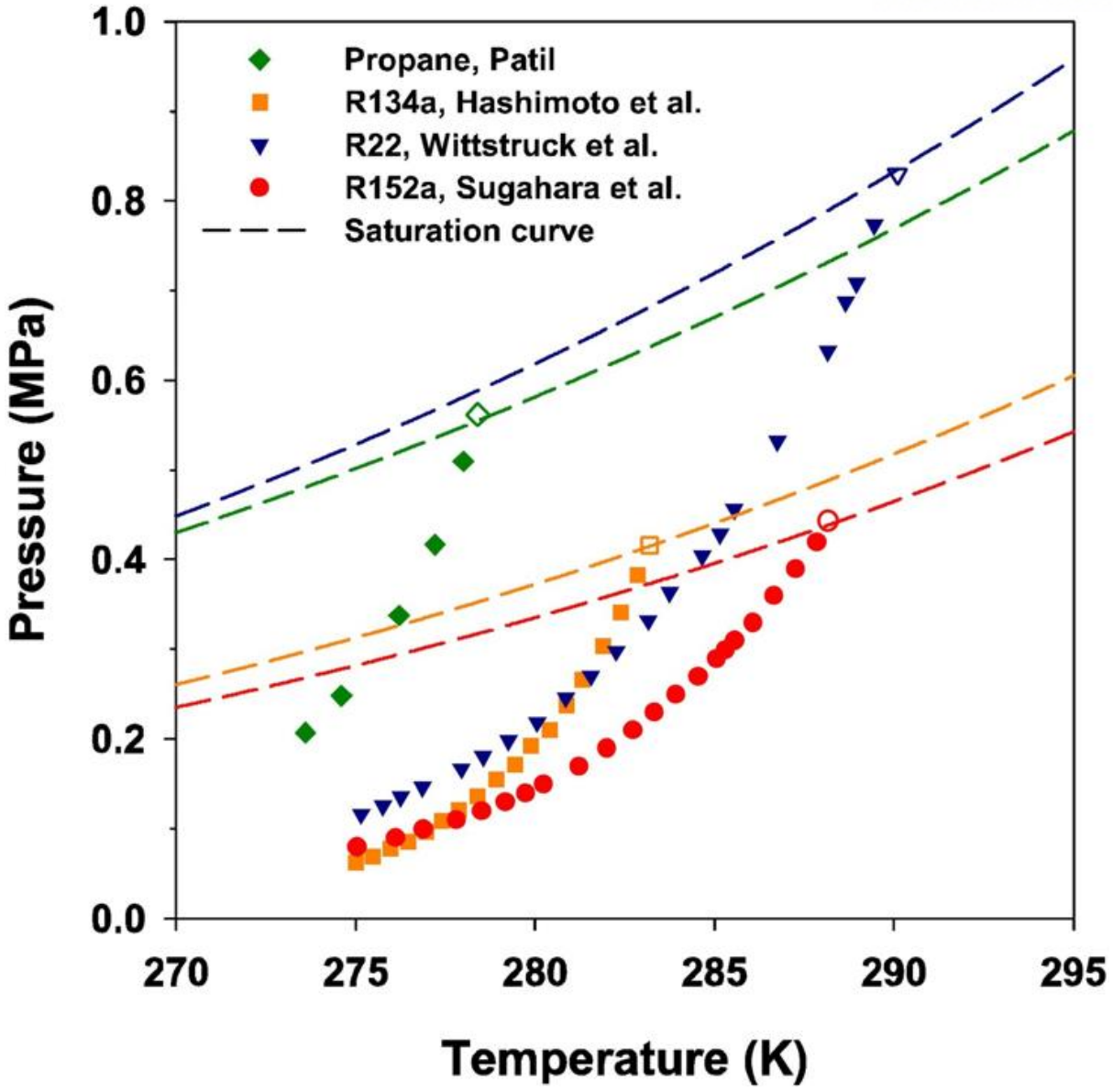


Fig. 4.1.4. Three-phase equilibria of hydrate former (propane, R134a, R22, and R152a) + water systems. The saturation curve of each gas was obtained from NIST data[113]

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Because the slopes of the hydrate phase equilibrium curves vary with enclathrated guest molecules, the equilibrium curves of two different gas hydrates can intersect at certain pressure and temperature condition. From Fig. 4.1.4, the equilibrium curves of R134a and R22 hydrates intersected at 281.2 K and 0.25 MPa, and the equilibrium curves of R134a and R152a hydrates intersected at 277.2 K and 0.10 MPa. This indicates that the order of hydrate equilibrium pressure is R134a > R22 > R152a at T > 281.2 K, R22 > R134a > R152a at 281.2 K > T > 277.2 K, and R22 > R152a > R134a at T < 277.2 K at each specified temperature. The HBD process should be operated at pressure and temperatures lower than the upper quadruple point [Q_2, hydrate (H) – liquid water (L_w) – liquid guest (L_g) – vapor (V)] of each gas hydrate, which is located at an upper termination point of the three-phase (H – L_w – V) equilibrium curve. At Q_2, further heating of the system induces the dissociation of gas hydrates, and further pressurization of the system results in the liquefaction of gaseous hydrate formers (propane, R134a, R22, and R152a). Therefore, the Q_2 is practically considered an upper pressure and temperature limit of gas hydrates with the vapor phase. From Fig. 4.1.4, the Q_2 (shown as open symbols) temperatures for propane, R134a, R22, and R152a hydrates were 278.4 K, 283.2 K, 290.1 K, and 288.2 K, respectively[111,114,115]. This implies that R22 hydrate has the highest operating temperature for the hydrate-based process.

The HBD process inevitably entails the formation and dissociation of gas hydrates, and thus, information about the dissociation enthalpy (ΔH) of each gas hydrate is necessary to establish the energy balance of the process. Table 4.1.2 shows the hydrate structure and dissociation enthalpy of propane, R134a, R22, and R152a hydrates[3,45,116,117]. Both propane and R134a form sII hydrates, whereas both R22 and R152a form sI hydrates. The unit cells of the sI and sII hydrates comprise $6(5^{12}6^2)\cdot 2(5^{12})\cdot 46(H_2O)$ and $8(5^{12}6^4)\cdot 16(5^{12})\cdot 136(H_2O)$, respectively[1]. The gaseous hydrate formers listed in Table 4.1.2 are known to be captured only in the large cages of each hydrate with almost full occupancy due to their larger molecular size. Despite the full occupancy of guest molecules in the large cages, the sII hydrate formers showed a larger hydration number, which refers to the larger mole ratio of water molecules to guest molecules in the unit hydrate lattice, than the sI hydrate formers due to the different cage construction in each hydrate structure. The differential scanning calorimetry measurements during the hydrate dissociation can provide the ΔH (kJ/mol-water) value of each hydrate, and then, the ΔH (kJ/mol-guest) value can be calculated by considering both the hydration number and the ΔH (kJ/mol-water) value. The sII hydrates gave lower ΔH (kJ/mol-water) values than the sI hydrates. The encapsulation enthalpy of the guest molecules also affects the ΔH value, and the value varies with the guest molecules[118]. In each hydrate structure, sI R22 hydrate had a lower ΔH (kJ/mol-water) value than sI R152a hydrate, and the ΔH (kJ/mol-water) value of sII R134a hydrate exceeded that of sII propane hydrate. The lower ΔH (kJ/mol-water) values for gas hydrates imply that the energy consumption for

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the phase transition of hydrate formation/dissociation is smaller regarding the amount of treated water. However, the sI hydrates possessed lower ΔH (kJ/mol-guest) values than the sII hydrates due to their smaller hydration number values, which indicates that the sI hydrates require lower energy consumption for hydrate formation/dissociation regarding the amount of captured gas molecules.

Table. 4.1.2. Structure, hydration number, and dissociation enthalpy of propane, R134a, R22, and R152a hydrates

	Type of structure	**Hydration number**	**ΔH (kJ/mol-water)**	**ΔH (kJ/mol-guest)**	
Propane hydrate	sII	17	7.6 ± 0.1	129.2 ± 0.4	Handa et al.[116]
R134a hydrate	sII	17	8.6 ± 0.1	146.0 ± 0.5	Lee et al.[117]
R22 hydrate	sI	7.7	10.3 ± 0.2	79.2 ± 1.2	Mok et al.[3]
R152a hydrate	sI	7.7	10.6 ± 0.1	82.2 ± 0.2	Mok et al.[45]

Fig. 4.1.5 illustrates the three-phase ($H - L_w - V$) equilibria of propane, R134a, R22, and R152a hydrates in the presence of NaCl[109,119-121]. The higher NaCl concentration induced a larger depression of the hydrate equilibrium temperature at any given pressure. In this study, the hydrate depression temperature from the hydrate equilibrium temperature in pure water (ΔT_{dep}) was calculated using the HLS correlation. From Fig. 4.1.5, the calculated hydrate equilibrium curves (marked by solid lines) were in good agreement with the experimental values from the literature.

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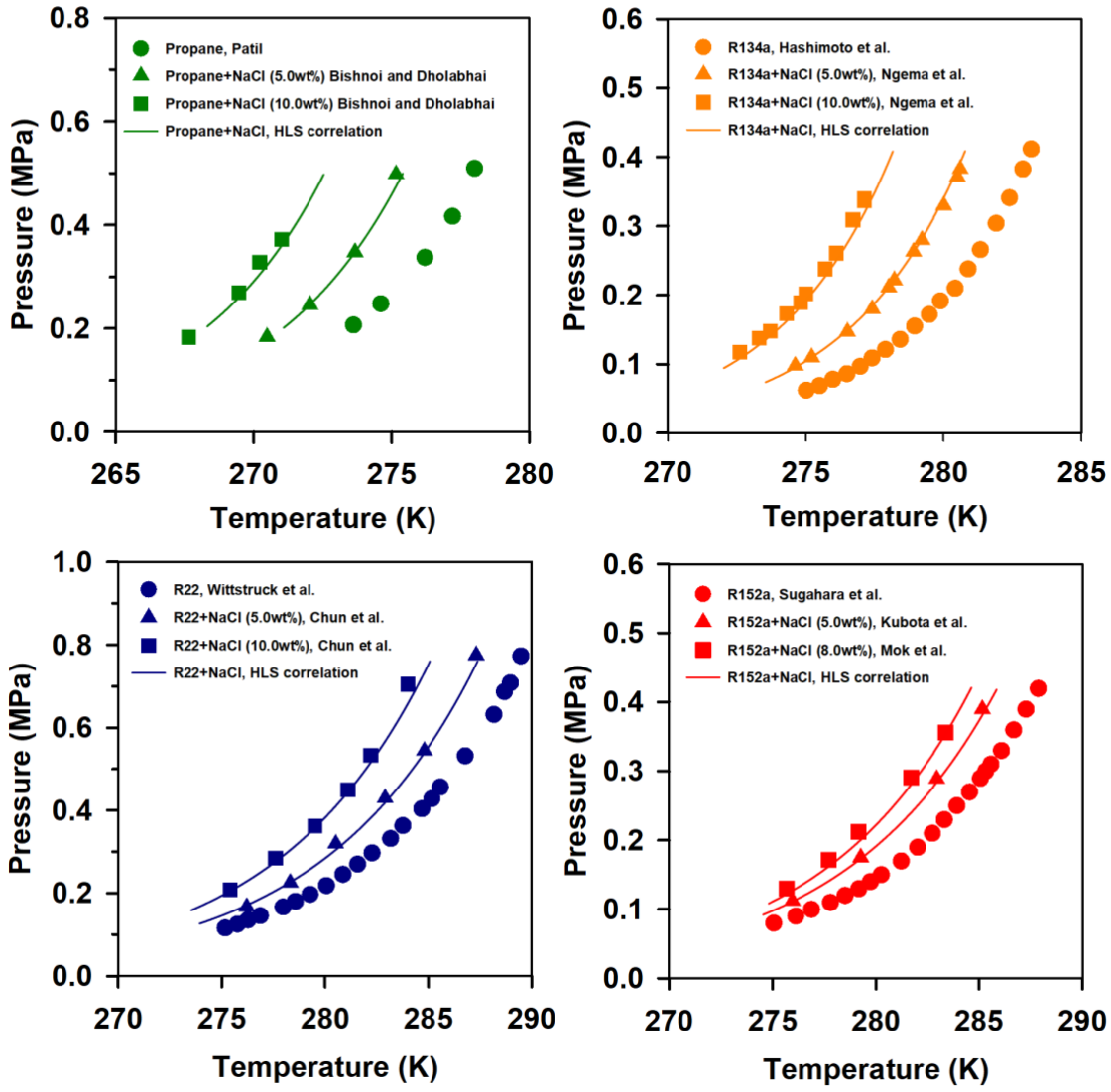


Fig. 4.1.5. Three-phase equilibria of hydrate former (propane, R134a, R22, and R152a) + water + NaCl systems

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Fig. 4.1.6 shows the ΔT_{dep} values of propane, R134a, R22, and R152a hydrates in the presence of NaCl (5 wt% and 10 wt%) at 0.3 MPa. The sII hydrate formers (propane and R134a) showed larger ΔT_{dep} values than the sI hydrate formers (R22 and R152a) at each salinity, and the ΔT_{dep} value of propane hydrate was the largest. In the HLS correlation, the ΔT_{dep} values of gas hydrates depend on both the heat of hydrate formation/dissociation term ($nR/\Delta H_d$) and the water activity term ($ln\, a_w$). At a fixed salinity, the ΔT_{dep} value increases with an increase in the ($nR/\Delta H_d$) term. The values for the ($nR/\Delta H_d$) term calculated using the ΔH (kJ/mol-guest) value and hydration number (n) in Table 4.1.2 were 0.00109 for propane, 0.00097 for R134a, 0.00081 for R22, and 0.00078 for R152a hydrates, indicating that sII hydrates have larger values for the ($nR/\Delta H_d$) term than the sI hydrates. In addition, as the salinity increases, the ΔT_{dep} value for a certain gas hydrate also increases due to the water activity term. Because, at the increased ΔT_{dep}, additional cooling is required to maintain the thermodynamic driving force for the hydrate formation, the sII hydrate formers can be disadvantageous for the HBD process compared to the sI hydrate formers in the aspect of the equilibrium depression behavior, and this disadvantage of the sII hydrate formers will be further intensified with an increase in salinity.

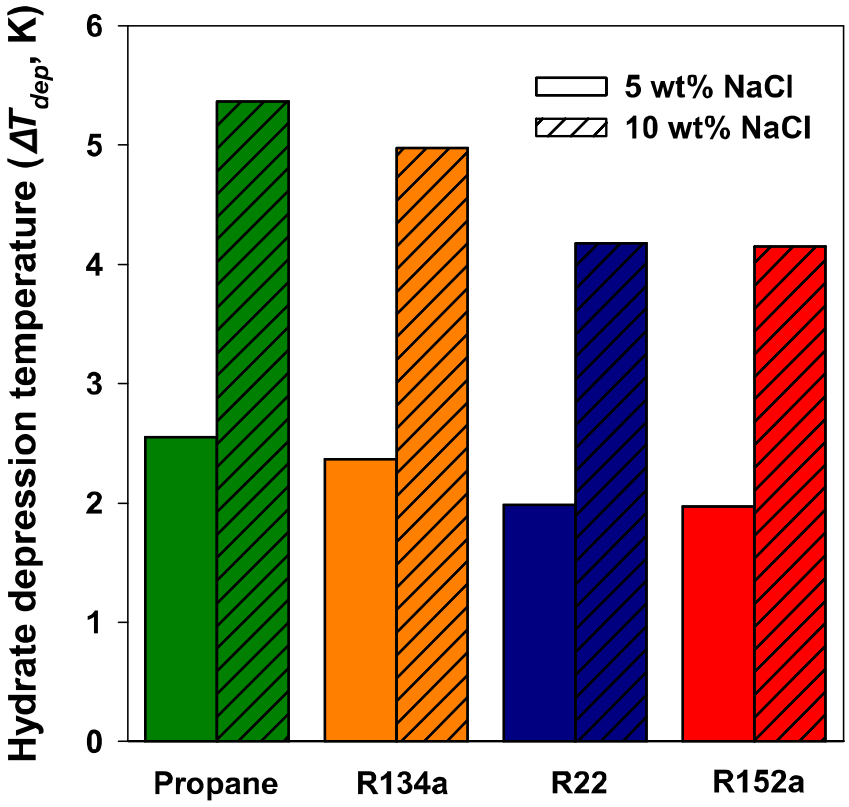


Fig. 4.1.6. Hydrate depression temperatures of propane, R134a, R22, and R152a hydrates in NaCl

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solutions (5 wt% and 10 wt%) at P = 0.3 MPa

4.1.4. Conclusion

The thermodynamic properties of R152a hydrates in saline water were investigated. The three-phase equilibria of R152a hydrates in the presence of NaCl (0 wt%, 3.5 wt%, and 8.0 wt%) were measured. The hydration number and dissociation enthalpy of the R152a hydrate were obtained using the Rietveld refinement of the PXRD patterns and the DSC measurement, respectively. The HLS correlation was employed to predict the equilibrium shift and hydrate depression temperature (ΔT) of both R152a and R134a hydrates in the presence of NaCl. In addition, the thermodynamic (hydrate phase equilibria, hydrate temperature depression, dissociation enthalpy, maximum achievable salinity, and maximum water yield) behaviors of propane, R134a, R22, and R152a hydrates in the presence of NaCl were also investigated closely to determine the energy-efficient gaseous hydrate former for the HBD process. R152a had generally milder equilibrium conditions (lower pressure and higher temperature) than other hydrate formers. The hydrate depression temperatures of sII hydrate formers (propane and R134a) exceeded those of the sI hydrate formers (R22 and R152a) at a given salinity.

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4.2. Estimation of theoretical yield of HBD

This chapter includes the published contents:

Mok, J., Choi, W., & Seo, Y. (2021). Evaluation of kinetic salt-enrichment behavior and separation performance of HFC-152a hydrate-based desalination using an experimental measurement and a thermodynamic correlation. Water Research, 193, 116882.
Mok, J., Choi, W., & Seo, Y. (2022). Theoretically achievable efficiency of hydrate-based desalination and its significance for evaluating kinetic desalination performance of gaseous hydrate formers. Desalination, 524, 115487.
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4.2.1. Abstract

Freshwater can be obtained through gas hydrate formation from saline water, which is called hydrate-based desalination (HBD). In this study, a novel approach that enabled us to quantify the theoretically achievable desalination efficiency for hydrate formers under any given temperature, pressure, and initial salinity conditions was introduced. The theoretically achievable salinity and water yield of HBD at each given thermodynamic condition were calculated using the Hu–Lee–Sum correlation. The theoretical HBD efficiency increased as the initial salinity decreased, the operating pressure decreased, and the initial subcooling temperature increased. At any given thermodynamic conditions, R152a indicated higher desalination efficiency than other candidates (propane, R134a, and R22).

4.2.2. Efficiency evaluation for the R152a and R134a hydrate-based desalination process

HBD is a type of freezing-based desalination and thus, it undoubtedly has a maximum salt-enrichment efficiency which can be achieved under a given pressure and temperature condition because the increased salt concentration in the residual solution causes the hydrate equilibrium depression. In other words, if the temperature, pressure, and initial salinity in the solution are given, the thermodynamically achievable maximum salt concentration in the residual solution can be calculated. Therefore, the quantitative analysis of the maximum salt enrichment under a given thermodynamic condition is vital for the optimal design and operation of the HBD process and its further convergence with other desalination processes. In this study, to quantify the theoretical maximum salt enrichment of the HBD process, a new approach was introduced using the HLS correlation. In the previous section, the shift of the hydrate phase equilibrium caused by the enrichment of NaCl in the residual solution and the changes in the driving force of Δf for the hydrate growth were calculated to examine the kinetic behavior of both

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R152a and R134a hydrates. However, in this section, the maximum achievable salinity in the residual solution by HBD at a given pressure and temperature condition for each hydrate former was predicted from the maximally depressed hydrate equilibrium using the HLS correlation (black-colored dotted line in Fig. 4.2.1).

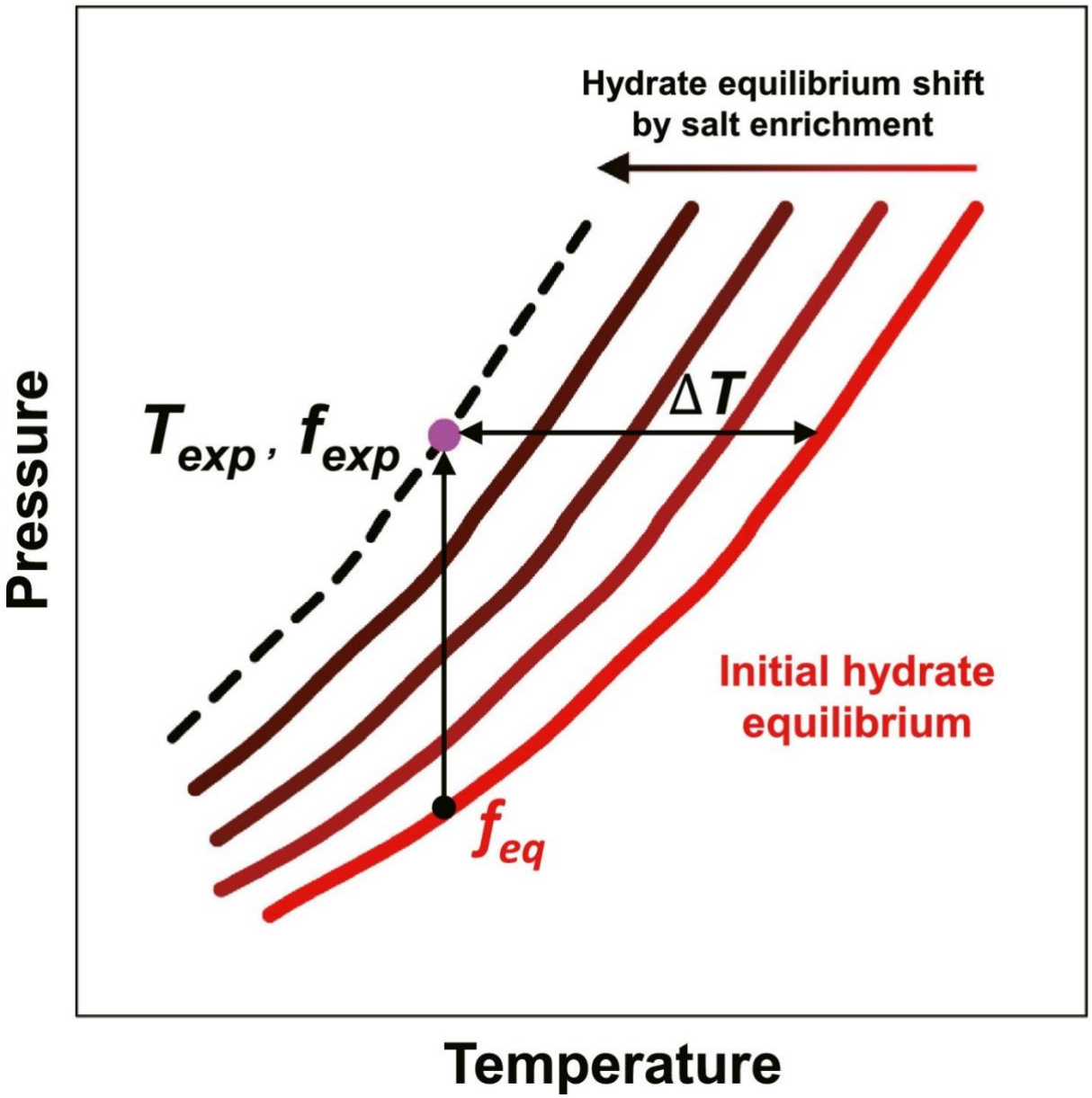


Fig. 4.2.1. Schematic illustration of hydrate equilibrium-shift due to salt enrichment in residual solution.

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Fig. 4.2.2 shows the theoretically achievable maximum salinity of HBD using R152a and R134a for NaCl (3.5 wt% and 8.0 wt%) solutions at P = 0.3 MPa and the subcooling temperature of 3 K. The maximum achievable salinity was 10.6 wt% (R152a) and 9.5 wt% (R134a) for NaCl (3.5 wt%) solution and 13.7 wt% (R152a) and 12.9 wt% (R134a) for NaCl (8.0 wt%) solution, demonstrating that R152a is a potentially superior candidate for HBD. This is because the hydrate equilibrium depression at the same extent of NaCl enrichment in the residual solution was smaller for the R152a hydrate because of the smaller ($\frac{nR}{\Delta H_d}$) term and thus, R152a could achieve a higher conversion of gas hydrates than R134a at the same subcooling temperature of 3 K. In addition, a salinity increment in the residual solution after hydrate formation for the NaCl (8.0 wt%) solution (5.7 wt% for R152a and 4.9 wt% for R134a) was smaller than that of the NaCl (3.5 wt%) solution (7.1 wt% for R152a and 6.0 wt% for R134a) due to the lower conversion of gas hydrates in the NaCl (8.0 wt%) solution.

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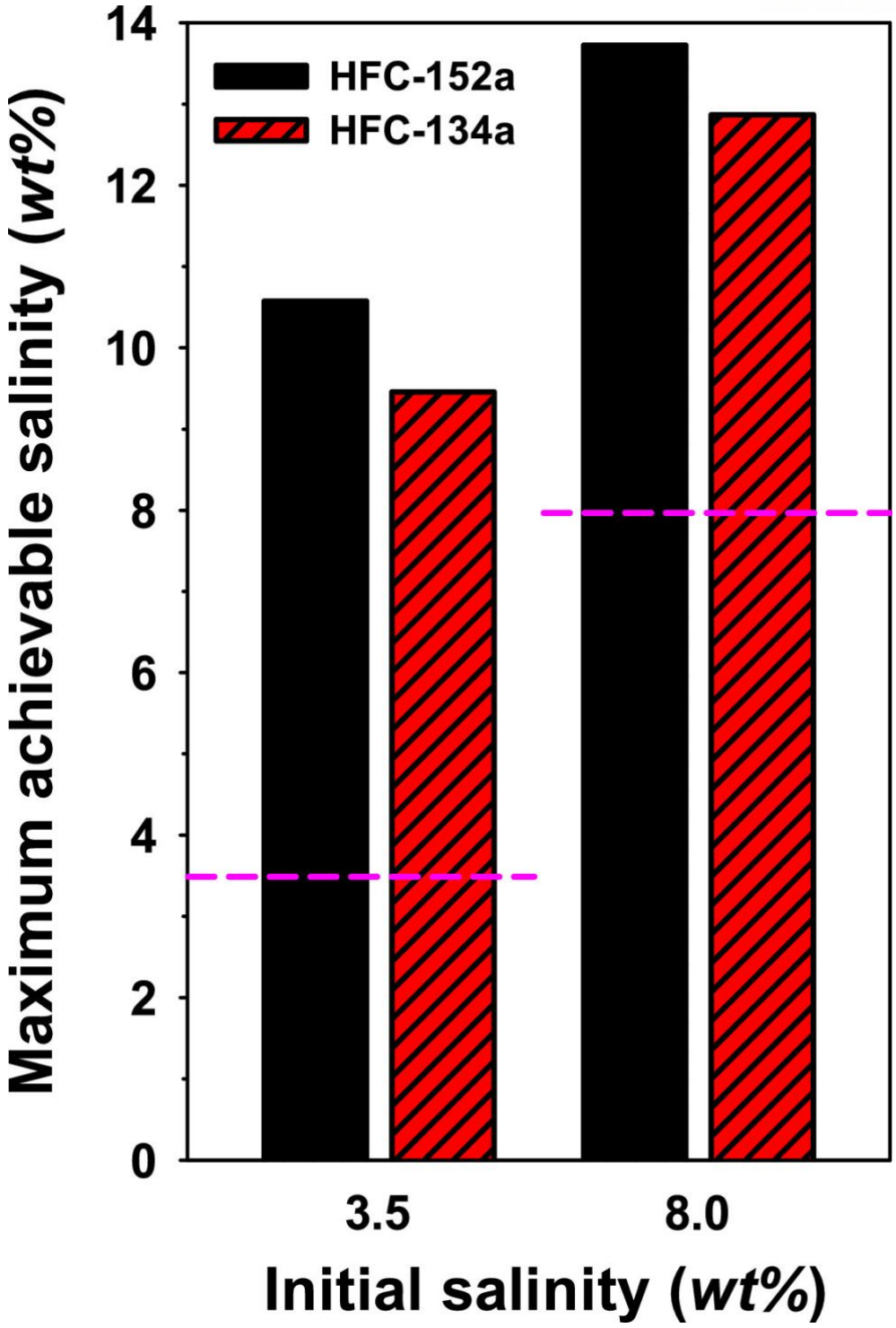


Fig. 4.2.2. Maximum achievable salinity of HBD using R152a and R134a for NaCl (3.5 wt% and 8.0 wt%) solutions at P = 0.3 MPa and the subcooling temperature of 3 K; dotted lines indicate initial salinity.

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To investigate the performance and efficiency of the R152a HBD for saline water with various NaCl concentrations, the maximum achievable salinity and the corresponding maximum water yield, which refers to the ratio of the amount of water included in the R152a hydrate at maximum achievable salinity to the amount of water initially injected, were calculated at P = 0.3 MPa and the subcooling temperature of 3 K using the HLS correlation and the mass balance. Fig. 4.2.3 presents the maximum salinity and the corresponding maximum water yield of R152 HBD at P = 0.3 MPa and the subcooling temperature of 3 K according to the initial salinity. The maximum achievable salinity in the residual solution was 10.6 wt% and 13.7 wt%, and the corresponding maximum water yield was 69.3% and 45.3% for the initial NaCl (3.5 wt%) and NaCl (8.0 wt%) solutions, respectively. Because the hydrate equilibrium depression was larger at higher salinity due to the larger ($ln\, a_w$) term, the capacity for desalination became smaller for the saline solution with a higher initial NaCl concentration at P = 0.3 MPa and the subcooling temperature of 3 K.

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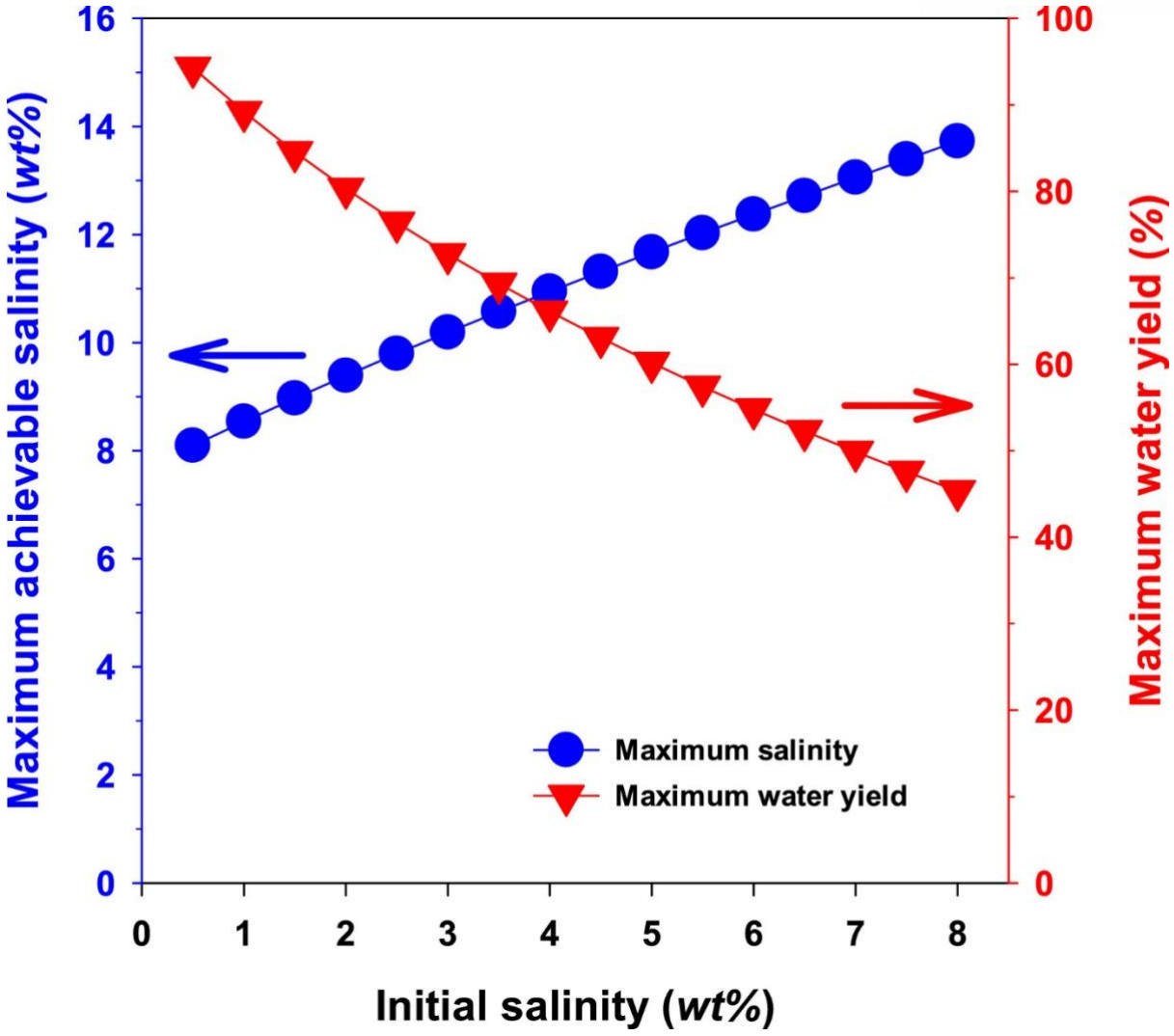


Fig. 4.2.3. Maximum achievable salinity and the corresponding maximum water yield of R152a HBD according to the initial salinity at P = 0.3 MPa and the subcooling temperature of 3 K.

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Because the desalination efficiency is also closely related with the subcooling temperature, the effect of subcooling for hydrate formation on the salt-enrichment efficiency should be examined to determine the optimal subcooling for the desired desalination efficiency. Fig. 4.2.4 demonstrates the maximum water yield of R152a HBD according to various initial levels of salinity in the solution at three different subcooling temperatures of 1 K, 3 K, and 5 K at P = 0.3 MPa. As shown in Fig. 4.2.4, the difference in the maximum water yield between the subcooling temperatures of 3 K and 5 K was significantly smaller than that between the subcooling temperatures of 1 K and 3 K, indicating that the degree of subcooling was not in direct proportion to the salt-enrichment efficiency, even though a better efficiency was observed at a higher subcooling temperature. As seen in Fig. 4.2.4, the changes in the maximum water yield of R152 HBD at P = 0.3 MPa, according to initial salinity, were very susceptible to the degree of subcooling. For instance, the maximum water yield decreased from 84.2% to 23.2% at the subcooling temperature of 1 K, while it decreased from 96.4% to 55.8% at the subcooling temperature of 5 K when the initial salinity changed from 0.5 wt% to 8.0 wt%. A decrease in the maximum water yield with an increase in the initial salinity was much smaller at higher subcooling, and the difference in maximum water yield at lower initial salinity between lower and higher subcooling was not significant.

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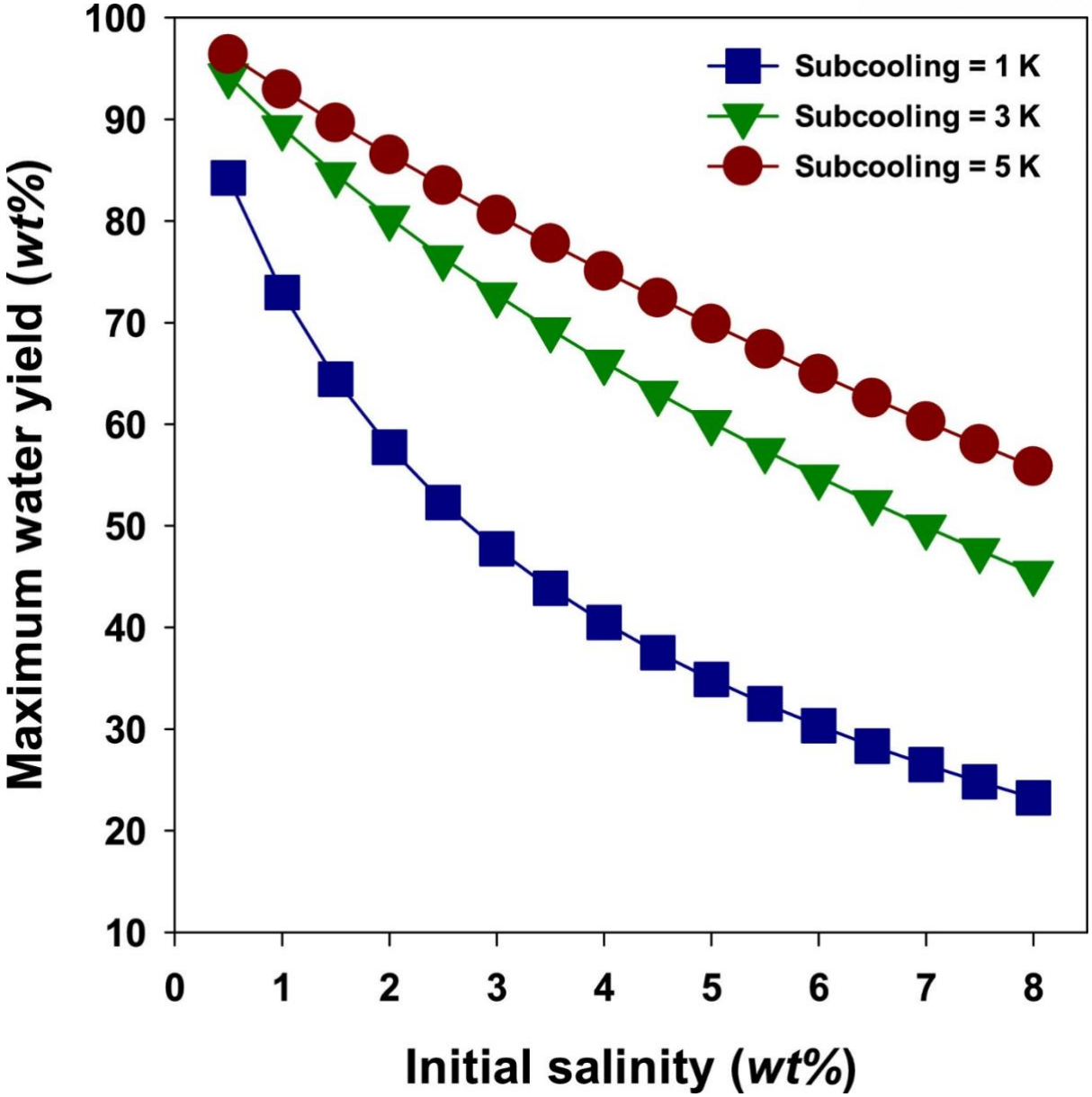


Fig. 4.2.4. Maximum water yield of R152 HBD according to initial salinity at P = 0.3 MPa and three different subcooling temperatures (1 K, 3 K, and 5 K).

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However, higher subcooling for hydrate formation requires more energy for cooling and can also cause difficulties in separating salts from the surface of the hydrate crystals due to the altered crystallization mechanism and the hydrate morphology at higher subcooling[122]. Therefore, the optimal operating conditions in the HBD process should be determined by considering the maximum water yield according to the target initial salinity as well as the trade-off between salt-enrichment efficiency and solid-liquid separation efficiency[123].

4.2.3. Estimation and comparison of theoretically achievable desalination efficiency

Fig. 4.2.5 (a) presents the maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates in the initial salinity range of NaCl 0.5 wt%–5.0 wt% at P = 0.3 MPa and initial ΔT_{sub} = 2 K. The maximum achievable salinity increased with an increase in the initial salinity for all gaseous hydrate formers. The sI hydrate formers (R22 and R152a) exhibited higher maximum achievable salinity than the sII hydrate formers (propane and R134a). R22 and R152a had similar maximum achievable salinity, and propane showed the lowest one in the entire salinity range examined in this study. This is because, for the hydrate former with the smaller temperature depression value in the saline water, more water can be converted into gas hydrates at a fixed initial ΔT_{sub}. From Fig. 4.1.6, the sI hydrate formers offered smaller ΔT_{dep} values at a fixed salinity, which was attributed to their lower $nR/\Delta H_d$ values. The similar $nR/\Delta H_d$ values of R152a and R22 induced similar maximum achievable salinity values at a given salinity, whereas the highest $nR/\Delta H_d$ value of propane induced the lowest maximum achievable salinity. The maximum water yields for all the hydrate formers decreased with an increase in the initial salinity, which resulted from the higher enriched NaCl concentration in the residual solution and the resulting lower hydrate conversion at the higher initial salinity. The sI hydrate formers also exhibited higher maximum water yield values in the whole initial salinity range than the sII hydrate formers due to their higher maximum achievable salinity values.

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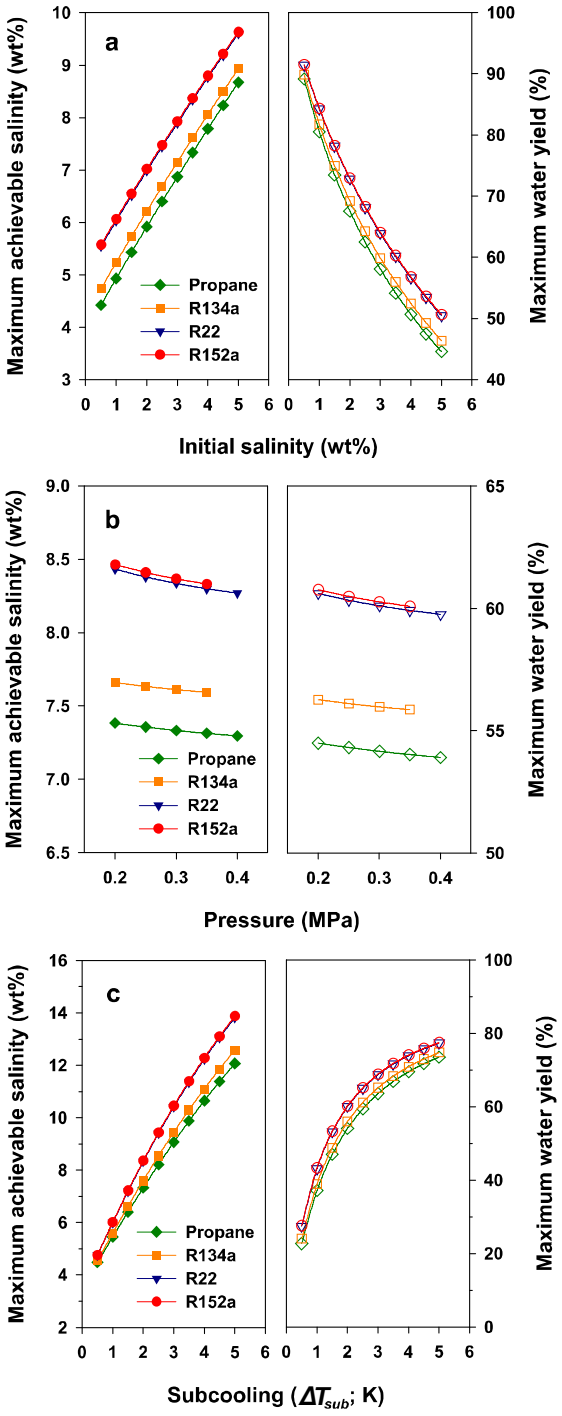


Fig. 4.2.5. Maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates at (a) different initial salinity at P = 0.3 MPa and initial ΔT_{sub} = 2 K, (b) different pressures at initial salinity = NaCl 3.5 wt% and initial ΔT_{sub} = 2 K, and (c) different initial ΔT_{sub} at the initial salinity = NaCl 3.5 wt% and P = 0.3 MPa

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Fig. 4.2.5 (b) shows the maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates in the operating pressure range of 0.2–0.4 MPa at the initial salinity = NaCl 3.5 wt% and the initial ΔT_{sub} = 2 K. For R134a and R152a, the values were calculated up to 0.35 MPa to avoid the liquefaction of the gaseous hydrate formers. R152a had the highest maximum achievable salinity and maximum water yield, while propane had the lowest values. The maximum achievable salinity and maximum water yield values for all hydrate formers decreased slightly with increasing pressure because the ΔT_{dep} value was marginally larger at a higher pressure. This indicates that lowering the operating pressure of HBD can result in reduced energy consumption and improved theoretical HBD efficiency.

The cooling temperature for hydrate formation can also affect the HBD efficiency. Fig. 4.2.5 (c) depicts the maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates in the initial ΔT_{sub} range of 0.5–5.0 K at the initial salinity = NaCl 3.5 wt% and P = 0.3 MPa. Because the hydrate equilibrium temperature can be more significantly depressed at the larger initial ΔT_{sub}, the maximum achievable salinity and maximum water yield increased with an increase in initial ΔT_{sub} (Fig. 4.2.5 (c)). As with the results in Fig. 4.2.5 (a) and (b), R152a had the highest maximum achievable salinity and maximum water yield, while propane had the lowest one in the initial ΔT_{sub} range covered in this study. Therefore, it can be concluded that the higher maximum achievable salinity and maximum water yield for HBD can be achieved with sI hydrate formers because of their smaller hydrate equilibrium depression and that R152a can give the highest theoretical HBD efficiency, whereas propane can offer the lowest one at any given initial salinity, pressure, and initial ΔT_{sub} conditions.

Because CAPEX and OPEX as well as the HBD efficiency are also strongly affected by the operating conditions, considerations for both the thermodynamic nature of guest gases and the theoretical HBD efficiency will be of great help to operators or researchers who are looking for the economically feasible operating conditions of the HBD process. In this study, the HLS correlation was employed as a novel approach to predict the hydrate equilibrium shift and the maximum achievable salinity of the HBD process. However, for scale-up and commercialization of the HBD process, active follow-up studies should be continued. The theoretical desalination efficiency calculated by more complex hydrate phase equilibria considering various ions (Na^+, K^+, Mg^{2+}, Ca^{2+}, B^{3+}, Cl^-, and SO_4^{2-}) should be compared with the HBD efficiency using real seawater. Furthermore, advanced process design considering the kinetic behavior, hydrate desalination capacity, and theoretical desalination efficiency for given hydrate formers should also be conducted to maximize the energy and economic efficiency of the HBD process.

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On the other hand, rather than being used as single process for desalination, the HBD process can be coupled with an existing desalination process like RO. A hybrid process of HBD and RO can generate significant synergy because RO is vulnerable to corrosion or fouling caused by foreign matter or high salinity, whereas HBD is less sensitive to those issues and thus can be used as an effective pretreatment process[124]. Therefore, thermodynamic and kinetic considerations for the R152a HBD process examined in this study will be helpful for determining the optimal operating conditions of the HBD process to increase its energy efficiency and for constituting a hybrid process coupled with other methods.

4.2.4. Conclusion

The maximum achievable salinity and maximum water yield were also calculated using the HLS correlation to quantitatively examine the performance of HBD. The R152a hydrate showed a higher maximum salinity than the R134a hydrate at 0.3 MPa and the subcooling temperature of 3 K. The calculated maximum water yield of the R152a HBD process in the NaCl concentration range of 0.5 wt% to 8.0 wt% demonstrated that the salt-enrichment efficiency decreased with an increase in the initial salinity, and it could be significantly enhanced by increasing the subcooling. The theoretically achievable HBD efficiencies at different thermodynamic conditions (initial salinity, pressure, and initial subcooling) were calculated using the HLS correlation, which was employed as the quantitative standard for estimating the kinetic performance of HBD for each gaseous hydrate former. The theoretical achievable HBD efficiency increased as the initial ΔTsub increased, pressure decreased, and the initial salinity decreased. The sI hydrate formers offered higher maximum achievable salinity and maximum water yield values than the sII hydrate formers at any given initial salinity, pressure, initial ΔTsub condition due to their smaller hydrate temperature depression. The novel approach examined in this study to quantify the performance of HBD using the calculation of hydrate depression temperature will be applicable to any HBD process, regardless of the types of guest molecules, operating conditions, and salt concentrations of the target solution, and thus will be useful for assessing the desalination efficiency of the HBD process.

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4.3. Evaluation of desalination kinetics of HBD candidates

This chapter includes the published contents:

Mok, J., Choi, W., & Seo, Y. (2021). Evaluation of kinetic salt-enrichment behavior and separation performance of HFC-152a hydrate-based desalination using an experimental measurement and a thermodynamic correlation. Water Research, 193, 116882.
Mok, J., Choi, W., & Seo, Y. (2022). Theoretically achievable efficiency of hydrate-based desalination and its significance for evaluating kinetic desalination performance of gaseous hydrate formers. Desalination, 524, 115487.
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4.3.1. Abstract

In this study, the kinetic features of propane, R134a, R22, and R152a hydrates in the presence of NaCl were examined to determine the energy-efficient gaseous hydrate former. At a fixed initial subcooling (2 K), R134a gave faster formation kinetics in the early stage, but R22 eventually offered highest hydrate conversion. At a fixed temperature (272 K), R152a showed fastest formation kinetics and highest HBD efficiency due to its milder hydrate equilibrium conditions. The overall results demonstrated that the thermodynamic stability and the intrinsic formation kinetics of gas hydrates significantly impact HBD efficiency and that the theoretical HBD efficiency can be the quantitative criterion for evaluating the kinetic performance and desalination capability of hydrate formers.

4.3.2. Kinetic performance of gaseous hydrate formers for HBD

The gas uptake measurements were conducted for propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.3 MPa, and initial ΔT_{sub} = 2 K to observe the hydrate formation kinetics. Fig. 4.3.1 (a) presents the accumulated amount of gas consumption during hydrate formation. The sI hydrate formers (R22 and R152a) demonstrated much higher gas uptakes than the sII hydrate formers (propane and R134a). The higher gas uptakes of sII hydrate formers were attributed to their smaller hydration number and the resulting requirement for a larger number of gas molecules per water molecule in the unit cell. Since the amount of hydrate formation is closely related to the HBD efficiency, the time-dependent hydrate conversion for propane, R134a, R22, and R152a was calculated considering both hydration number and time-dependent gas consumption (Fig. 4.3.1 (b)). At the early stage of rapid hydrate growth (for the initial 4 h), R134a exhibited the fastest hydrate conversion. However, at the final stage (~ 12 h), the sI hydrate formers (R22 and R152a) gave the better hydrate conversion, and

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R22 showed the highest final hydrate conversion (45.3% for R134a, 56.2% for R22, and 50.8% for R152a). In our previous study, it was observed that, at a fixed initial ΔT_{sub}, R134a had faster hydrate growth kinetics than R152a at the early stage of the hydrate formation due to the larger initial driving force of fugacity difference (Δf), but the final hydrate conversion of R152a surpassed that of R134a in the saline water because the formation rate of R134a hydrate decreased rapidly as the Δf value decreased rapidly following the salinity enrichment[97]. However, propane showed slow hydrate growth behavior, which resulted in lower hydrate conversion after 12 h compared to other hydrate formers (45.4% for propane). Sulfur hexafluoride (SF_6) was also reported to have slow hydrate growth behavior in saline water, making SF_6 an inferior hydrate former for the HBD process. Therefore, it is reasonable to say that R22 is the best option, and propane is the worst option for HBD regarding the hydrate formation kinetics and desalination efficiency at a fixed initial ΔT_{sub}.

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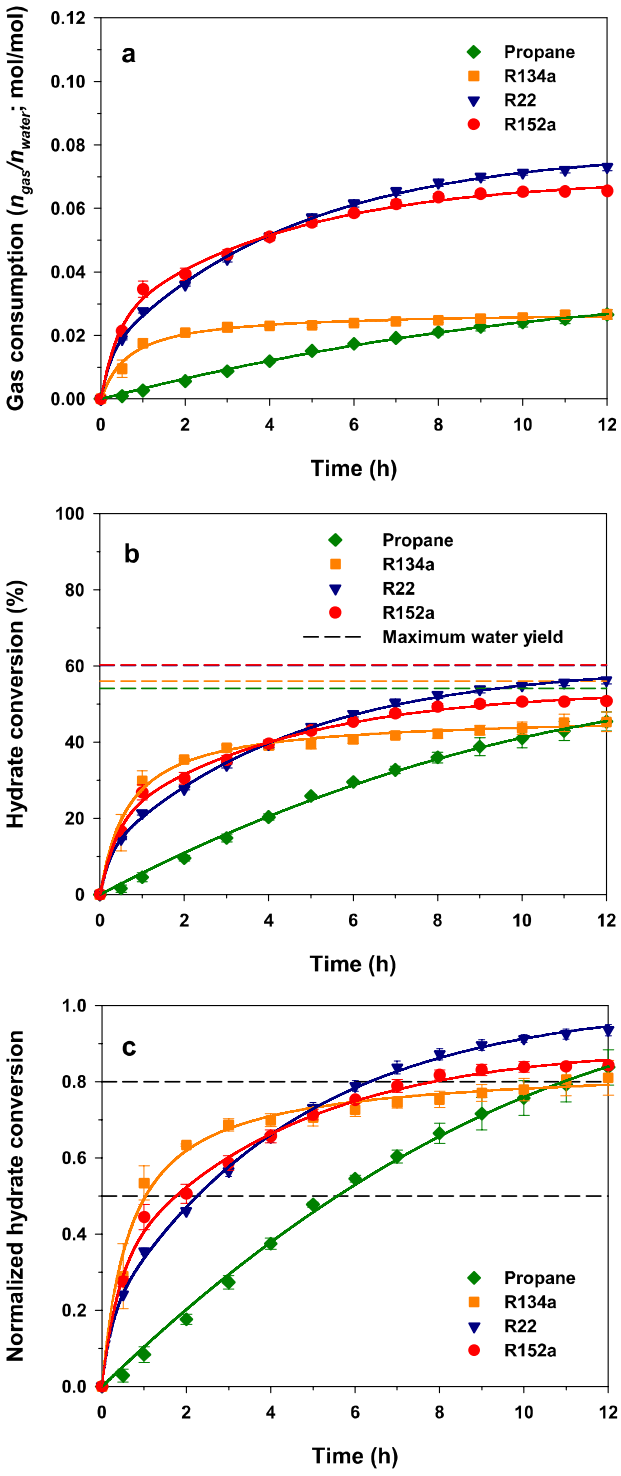


Fig. 4.3.1. Time-dependent (a) gas consumption, (b) hydrate conversion, and (c) normalized hydrate conversion of propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.3 MPa, and initial ΔT_{sub} = 2 K

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It should be noted that the theoretically achievable HBD efficiency varies with the types of hydrate formers, even at the same initial NaCl concentration, pressure, and initial ΔT_{sub}. The maximum water yield of each hydrate former at the initial salinity = NaCl 3.5 wt%, P = 0.3 MPa, and initial ΔT_{sub} = 2 K is depicted as a dotted line in Fig. 4.3.1 (b) (54.2% for propane, 56.0% for R134a, 60.1% for R22, and 60.3% for R152a). The maximum water yield, i.e., the theoretically achievable HBD efficiency at the given thermodynamic condition, can be considered a quantitative criterion for evaluating the kinetic performance and desalination capability of the hydrate formers. The time-dependent normalized hydrate conversion, which was calculated by dividing the time-dependent hydrate conversion ($H_{c,t}$) by the maximum water yield at the given thermodynamic condition, is illustrated in Fig. 4.3.1 (c). The time-dependent normalized hydrate conversion enabled us to evaluate the hydrate formation kinetics and to quantify the amount of gas hydrates formed relative to its theoretical maximum desalination capability. Fig. 4.3.1 (c) shows the time-dependent normalized hydrate conversion of propane, R134a, R22, and R152a at the initial salinity = NaCl 3.5 wt%, P = 0.3 MPa, and initial ΔT_{sub} = 2 K. R22 had the highest normalized hydrate conversion after 12 h (0.84 for propane, 0.81 for R134a, 0.94 for R22, and 0.84 for R152a). Also, the times taken to accomplish 50% and 80% of the maximum HBD capacity, i.e., the normalized hydrate conversion (t_{50} and t_{80}), were obtained from Fig. 4.3.1 (c) and are listed in Table 4.3.1. R134a had the smallest t_{50} value, but its t_{80} value was relatively larger due to its slower hydrate formation kinetics in the latter part, indicating that R134a might be a suitable hydrate former for the fast-cycle HBD process. The sI hydrate formers gave smaller t_{80} values, which demonstrates that both R22 and R152a can be good options to obtain higher desalination efficiency with one cycle of the HBD process. Propane offered the largest t_{50} and t_{80} values due to its slowest hydrate growth rate.

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Table. 4.3.1. T_{exp}, t_{50}, and t_{80} values of propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.3 MPa, and initial ΔT_{sub} = 2 K

	Propane	**R134a**	**R22**	**R152a**
T_{exp} **(K)**	271.7	278.2	279.0	282.0
ΔT_{sub} **(K)**	2.0	2.0	2.0	2.0
t_{50} **(h)**	5.4	0.9	2.4	2.0
t_{80} **(h)**	11.2	11.1	6.9	7.3

In this study, the gas uptake experiments were conducted at a fixed initial ΔT_{sub} (2 K), i.e., at different T_{exp} values as shown in Table 4.3.1, to set the same driving force for gas hydrate formation of different hydrate formers because the hydrate equilibrium temperatures vary with hydrate formers at the given initial salinity and pressure. To compare the HBD efficiency of each hydrate former for the real desalination process, the gas uptake and conversion of each hydrate former should be observed at a fixed T_{exp}. Fig. 4.3.2 demonstrates the maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.25 MPa, and T_{exp} = 272 K. R152a gave the highest maximum achievable salinity (5.1 wt% for propane, 15.6 wt% for R134a, 17.2 wt% for R22, and 20.1 wt% for R152a) and maximum water yield (32.7% for propane, 80.3% for R134a, 82.6% for R22, and 85.6 % for R152a). Propane showed the poorest performance for HBD because the initial ΔT_{sub} for propane hydrate formation was the smallest (only 0.8 K) at the given condition.

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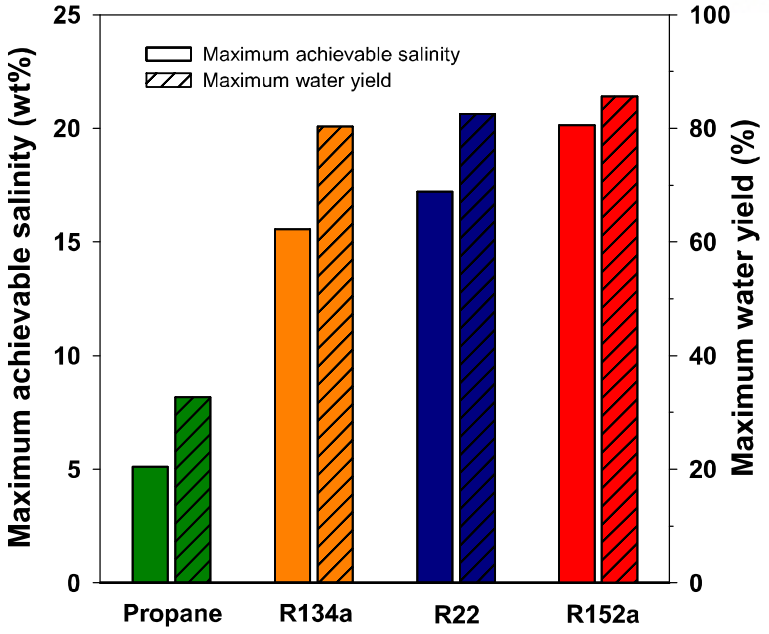


Fig. 4.3.2. Maximum achievable salinity and maximum water yield of propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.25 MPa, and T_{exp} = 272 K

To examine the HBD kinetics of each hydrate former at a fixed temperature, gas uptake measurements were also conducted at the initial salinity = NaCl 3.5 wt%, P = 0.25 MPa, and T_{exp} = 272 K (Fig. 4.3.3). Compared to the results in Fig. 4.3.1, the accumulated amount of gas consumption and the time-dependent hydrate conversion for R134a, R22, and R152a increased drastically because the initial ΔT_{sub} for the hydrate formation increased from 2 K to 7.4 K (R134a), 7.6 K (R22), and 10.5 K (R152a), whereas those for propane decreased due to the decrease in the initial ΔT_{sub} from 2 K to 0.8 K at T_{exp} = 272 K (Table. 4.3.2). After 12 h, the hydrate conversions were 23.1%, 74.1%, 77.9%, and 82.3% for propane, R134a, R22, and R152 hydrates, respectively. The maximum water yield values for each hydrate former at T_{exp} = 272 K, the initial salinity = NaCl 3.5 wt%, and P = 0.25 MPa are also marked as dotted lines in Fig. 4.3.3 (b). The time-dependent normalized hydrate conversion for each hydrate former at a fixed temperature (T_{exp} = 272 K) is depicted in Fig. 4.3.3 (c). The high normalized hydrate conversion values for R134a, R22, and R152a (0.92, 0.94, and 0.96, respectively) at T_{exp} = 272 K after 12 h indicate that more than 90% of the HBD capability was achievable using these hydrate formers. Compared to the results at the initial ΔT_{sub} = 2 K, the normalized hydrate conversion for R134a, R22, and R152a was improved, whereas that for propane was reduced (0.84 → 0.71). In addition, the hydrate conversion kinetics for R134a, R22, and R152a at T_{exp} = 272 K were also accelerated due to the increase in the initial ΔT_{sub} value. The t_{50} and t_{80} values for R134a, R22, and R152a at T_{exp} = 272 K (Table 4.3.2) were significantly shortened compared to those at the initial ΔT_{sub} = 2 K. However, the HBD kinetics

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of propane was decelerated at the initial ΔT_{sub} = 2 K; t_{50} increased and t_{80} was not even measurable within the experimental time (12 h). This indicates that the thermodynamic driving force for hydrate formation is a critical factor for HBD kinetics and that an increase in the thermodynamic driving force can improve the throughput of the HBD process.

Table. 4.3.2. Initial ΔT_{sub}, t_{50}, and t_{80} values of propane, R134a, R22, and R152a hydrates at the initial salinity = NaCl 3.5 wt%, P = 0.25 MPa, and T_{exp} = 272 K

	Propane	**R134a**	**R22**	**R152a**
T_{exp} **(K)**	272.0	272.0	272.0	272.0
ΔT_{sub} **(K)**	0.8	7.4	7.6	10.5
t_{50} **(h)**	8.1	0.6	0.6	0.3
t_{80} **(h)**	-	4.8	1.2	0.6

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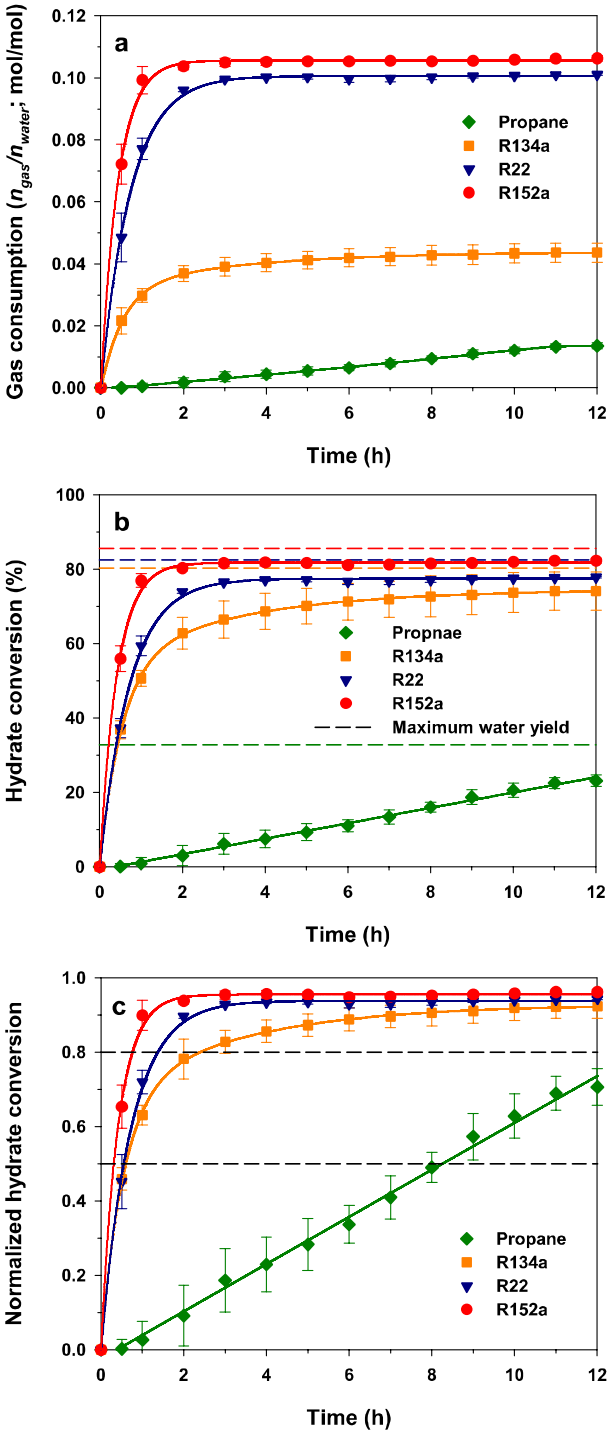


Fig. 4.3.3. Time-dependent (a) gas consumption, (b) hydrate conversion, and (c) normalized hydrate conversion of propane, R134a, R22, and R152a hydrates at initial salinity = NaCl 3.5 wt%, P = 0.25 MPa, and T_{exp} = 272 K

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R22 gave the highest hydrate conversion and the fastest hydrate growth kinetics at a fixed initial ΔT_{sub} (2 K) (Fig. 4.3.1), but at a fixed T_{exp} (272 K), R152a showed a better conversion and kinetic performance than R22 (Fig. 4.3.3) because the initial ΔT_{sub} for hydrate formation was larger for R152a due to its milder hydrate equilibrium conditions (e.g., higher equilibrium temperature at any given pressure). The results signify that the thermodynamic hydrate equilibrium conditions and the hydrate growth kinetics of the hydrate formers significantly impact the HBD efficiency of the real process. Therefore, the selection of thermodynamically stable hydrate formers, considering the actual operating conditions of the HBD process, would be indispensable for improving the efficiency of the HBD process.

Recently, various studies on the design and economic evaluation of the HBD process have been reported to increase its efficiency[60]. A cogeneration model of the propane-HBD process utilizing waste cold energy of LNG showed that it could significantly lower the production cost of fresh water by adding a power generation unit (an expander) to the process and that it could even produce energy[125]. The experimental results obtained in this study can provide the kinetic data and achievable HBD efficiency values for various hydrate formers at different conditions, which will be useful for the process assessment and also will be helpful for the selection of energy-efficient hydrate formers in the thermodynamic aspect. In addition, the maximum water yield for each hydrate former is expected to act as a quantitative standard for evaluating its water separation capability in the HBD process. Therefore, the experimental results and concept presented in this study will enable us to make a more integrative approach to the HBD process, and will provide valuable insights into the design of the HBD process.

4.3.3. Conclusions

In this work, kinetic (time-dependent gas uptake, hydrate conversion, and normalized hydrate conversion) behaviors of propane, R134a, R22, and R152a hydrates in the presence of NaCl were investigated closely to determine the energy-efficient gaseous hydrate former for the HBD process. For the hydrate formation in NaCl 3.5 wt% solution at a fixed initial ΔT_{sub} (2 K), R134a hydrate showed faster formation behavior at the early stage, but R22 achieved the highest hydrate conversion at the end of the experiment (12 h). Propane gave slow formation kinetics, which would be disadvantageous for HBD. For hydrate formation in NaCl 3.5 wt% solution at a fixed T_{exp} (272 K), R152a gave the fastest formation kinetics and the highest HBD efficiency (hydrate conversion: 82.3% and normalized hydrate conversion: 0.96) because the larger driving force was given to the thermodynamically milder hydrate former at a fixed T_{exp}. The overall results suggested that the thermodynamic properties and the kinetic properties of the hydrate formers are critical factors in determining the HBD efficiency. They also indicated that theoretically achievable HBD efficiency can be the reasonable standard for evaluating

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the kinetic performance of HBD, and thus, it can be a powerful tool for selecting the energy-efficient hydrate former for the HBD process.

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4.4. Liquid phase-hydrate former for HBD at ambient pressure

4.4.1. Abstract

The hydrate-based desalination process is a promising technology for the production of freshwater from seawater. This study presents a study on the thermodynamic properties of CP hydrates in NaCl solution, and provides a theoretical framework for estimating the efficiency of cyclopentane (CP) hydrate-based desalination. A high-pressure micro-differential scanning calorimeter was employed to measure the CP hydrate dissociation enthalpy. The equilibrium dissociation temperature of CP hydrate at atmospheric pressure was determined by visual observation. The theoretical CP hydrate-based desalination efficiency was analyzed by the HLS correlation. The operating condition-dependent cooling demand for the process was simply examined from a thermodynamic point of view. The study provides a valuable theoretical foundation for the further development of gas hydrate-based desalination technology.

4.4.2. Thermodynamic properties of CP hydrate for hydrate-based desalination

Accurate thermodynamic properties of hydrates are indispensable factor for establishing the efficiency of the hydrate-based process. During the HBD process the continuous repetition of hydrate formation and dissociation will be conducted, and thus, heat flow during the phase change of hydrates is critical information for the process. Several studies reported the dissociation enthalpy value of CP hydrate but there was some disparity between the reported values[126-129]. The reason can be conjectured that for CP dispersion method using surfactant which was selected for the previous studies could be caused the inadequate conversion of CP to hydrate, and thus, the dissociation enthalpy of CP hydrate may be under-evaluated. As mentioned in experimental part, in this study, to measure accurate dissociation enthalpy of pure CP hydrate, the hydrate formation was conducted in CP-excess circumstance in first time and the heat cycle was repeated until full conversion of water or ice to CP hydrate. As shown in the heat flow during the CP hydrate dissociation in Fig. 4.4.1, only endothermic peak of CP hydrate dissociation was observed which indicates that the added water was fully converted to CP hydrate. The measured dissociation enthalpy of CP hydrate was 6.7 kJ/mol-water, and by additionally considering the hydration number of CP hydrate, 17, 113.7 $\pm$ 1.7 kJ/mol-guest. The experimental result is well matched with the literature which achieved full water-hydrate conversion using a heat-flow calorimeter with a stirred cell (115.4 $\pm$ 7.6 kJ/mol-guest)[129]. Compared to well-studied HBD candidates such as propane and R134a which have identical crystal structure as sII hydrate with CP hydrate, the dissociation enthalpy of CP hydrate was 12% and 22% lower than that of propane and R134a hydrate, respectively[116,117]. Therefore, CP-HBD not only can be operated at the atmospheric pressure but energy-efficient than previous HBD

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candidates of sII hydrates for same extent of water treatment in thermodynamic point of view.

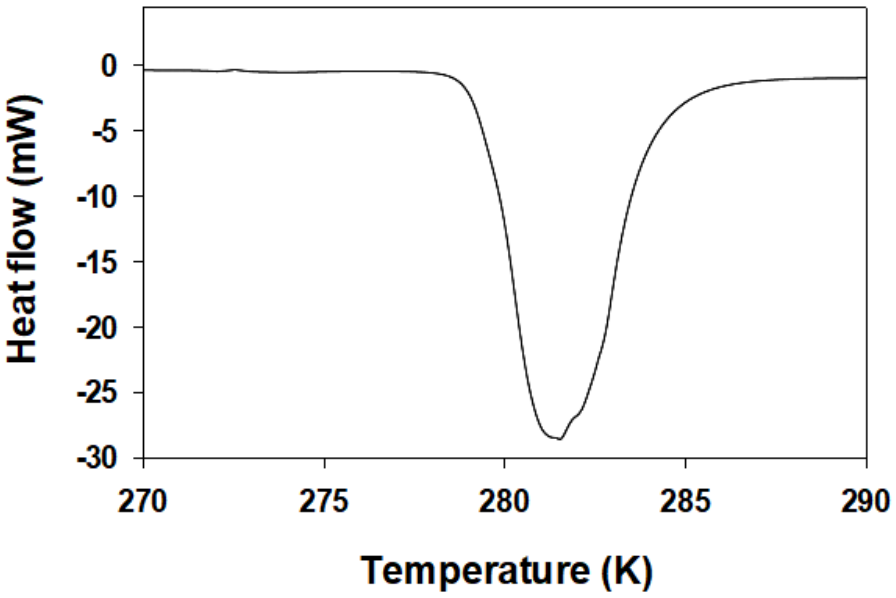


Fig. 4.4.1. Heat flow during dissociation of CP hydrate formed with pure water

The thermodynamic equilibrium of hydrates are also important for hydrate-based process because the kinetics and extent of hydrate formation for the HBD are directly affected by the thermodynamic driving force for the hydrate formation which is estimated by hydrate equilibrium and process operating conditions. The phase diagram of CP hydrate in NaCl solution with various concentration conditions (0-20 wt%) was confirmed in Fig. 4.4.2 with literature values. It should be noted that in several literature, during the measurement of the phase diagram of CP hydrate in the saline solution, the equilibration time for each temperature step was expected that too short for achieve equilibrium state. In the saline system, required time for achieving equilibrium state of hydrates are usually longer than that of the pure water system, and the tendency is enhanced at higher salinity. Therefore, in this study, the plenty time, 24 h, for achieving the equilibrium state in each temperature rising step was selected which was longest time in any reported literature of CP hydrate phase diagram. In addition, the equilibrium conditions of CP hydrates in NaCl solution were estimated using HLS correlation using our experimental data of the dissociation enthalpy of CP hydrate, which is also plotted in Fig. 4.4.2[129-133]. The calculated equilibrium conditions by HLS correlation are well-matched with experimental data (AAD = 1.42 K and 0.13 K for all reference and only our experimental data obtained in this study, respectively), indicating that HLS correlation can also well estimate the phase equilibria of CP hydrates with NaCl conditions.

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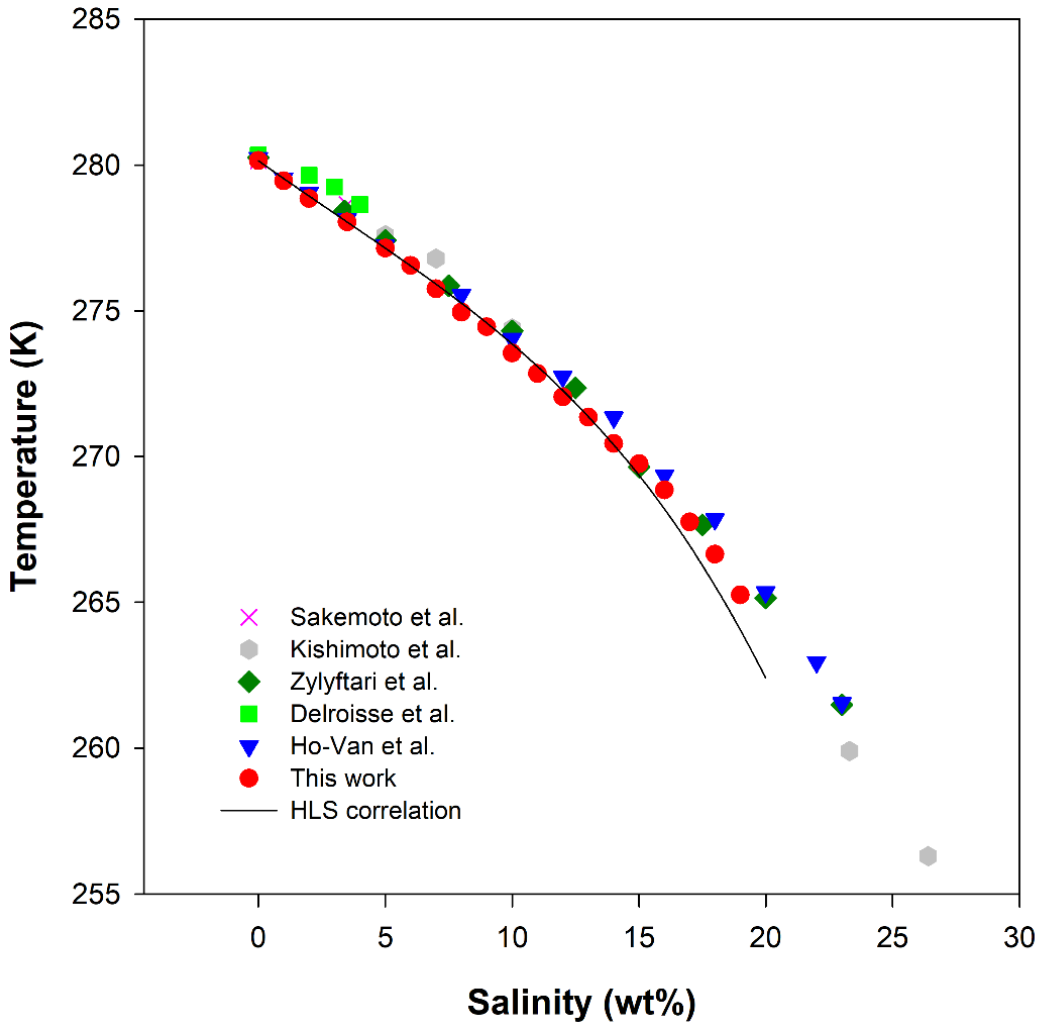


Fig. 4.4.2. Phase diagram of CP hydrate in NaCl solution

4.4.3. Estimation of theoretically achievable desalination efficiency of CP hydrate

The subcooling temperature (ΔT_{sub}) (the difference between the actual operating temperature (T_{exp}) and hydrate equilibrium temperature (T_{eq}) at a given pressure: $\Delta T_{sub} = T_{eq} - T_{exp}$) is generally employed as the thermodynamic driving force for hydrate formation. As hydrate formation using saline water progresses in the closed system, the hydrate equilibrium curve will shift gradually to the harsher region (the lower-temperature and higher-pressure regions) because the salt concentration in the residual solution increases. Theoretically, the shift in the hydrate equilibrium curves can continue until the thermodynamic driving force of subcooling is exhausted. That is, the hydrate equilibrium temperature can continue to be theoretically depressed until the subcooling temperature becomes zero ($\Delta T_{sub} = 0$),

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and the extent of hydrate conversion is maximized at that point[134].

First of all, a theoretically achievable maximally enriched salinity in the residual solution after the hydrate formation, i.e. maximum achievable salinity, of CP hydrate in NaCl solution was estimated using the HLS correlation with operating temperature between 268-280K and initial salinity up to 8 wt% (Fig. 4.4.3). It should be highlighted that as shown in Fig. 4.4.3, the maximum achievable salinity is identical for same hydrate formation temperature conditions regardless of the initial salinity of the target solution, and the maximum achievable salinity at a certain hydrate formation temperature is identical with the NaCl concentration of the hydrate equilibrium of CP + NaCl + water system at the temperature condition. This is because whatever the initial salinity of the target solution, the salinity of the residual solution will be gradually concentrated during the hydrate formation and theoretically, the hydrate formation will be terminated when the salinity of the residual solution becomes identical to the NaCl concentration of the CP hydrate equilibrium conditions at a given hydrate formation temperature. In this regard, the boundary between the calculatable region of the maximum achievable salinity (rainbow-colored region in Fig. 4.4.3) and purple region is identical with the hydrate phase diagram line of CP + NaCl + water system, and because the hydrate cannot be formed outer region of the hydrate equilibrium conditions, the maximum achievable salinity of the purple region which is the outer region of the boundary cannot be calculated and featured as 0 wt%. Therefore, the maximum achievable salinity is increased as the hydrate formation temperature decreased.

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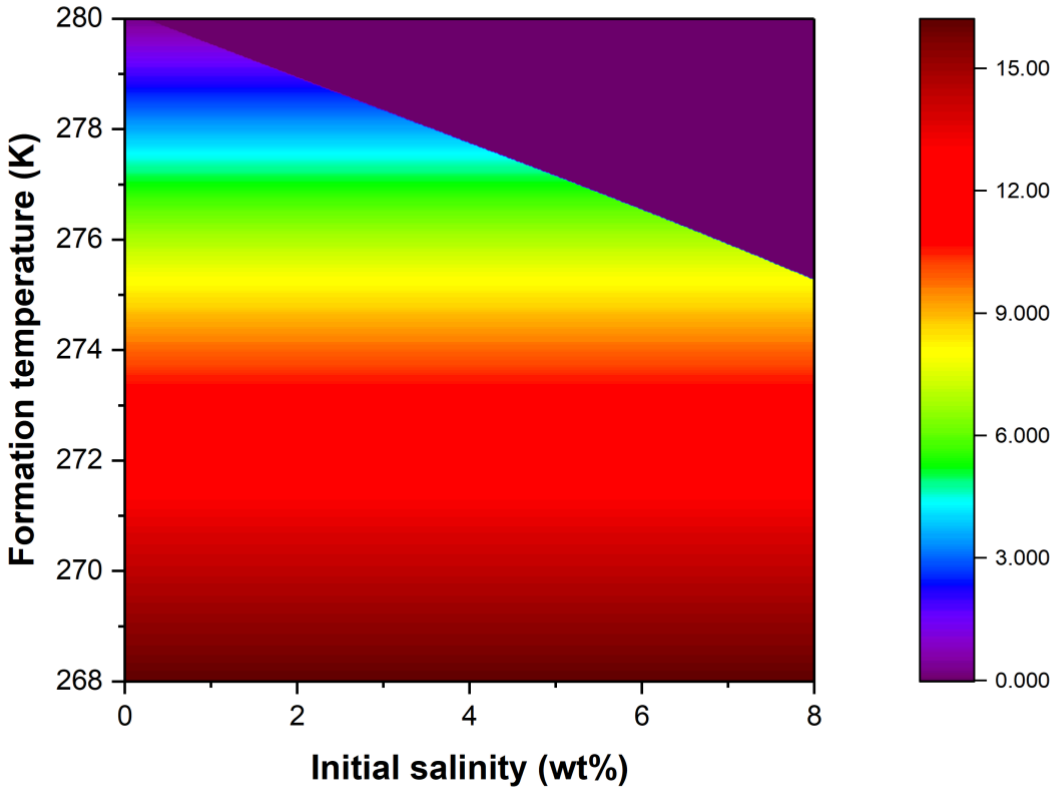


Fig. 4.4.3. Maximum achievable salinity CP hydrate at different initial salinity (up to 8 wt%) and temperature (268~280 K)

Then, based on a mass balance equation with the maximum achievable salinity, the ratio of water converted into gas hydrates to water initially injected i.e. maximum water yield, of CP hydrate in NaCl solution was quantitatively calculated with operating temperature between 268-280K and initial salinity up to 8 wt% (Fig. 4.4.4). It is assumed that in the range of calculation conditions, ice was not formed and the consumed water contents was all converted to CP hydrate. Firstly, in fixed hydrate formation temperature cases, the maximum water yield is decreased as the initial salinity increased. This is because in case of high initial salinity, the salinity in the residual solution is quickly reached to the maximum achievable salinity with relatively small amount of hydrate formation compared to the case of low initial salinity. In case of fixed initial salinity cases, the maximum water yield is increased as the hydrate formation temperature decreased. This is because the maximum achievable salinity is increased as the hydrate formation temperature decreased, more amount of hydrate can be formed until the salinity

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reaching the maximum achievable salinity at lower hydrate formation temperature in fixed initial salinity cases. These results indicates that theoretically achievable maximum water yield of CP hydrates can be determined for any given hydrate formation temperature and initial salinity of the treating solution.

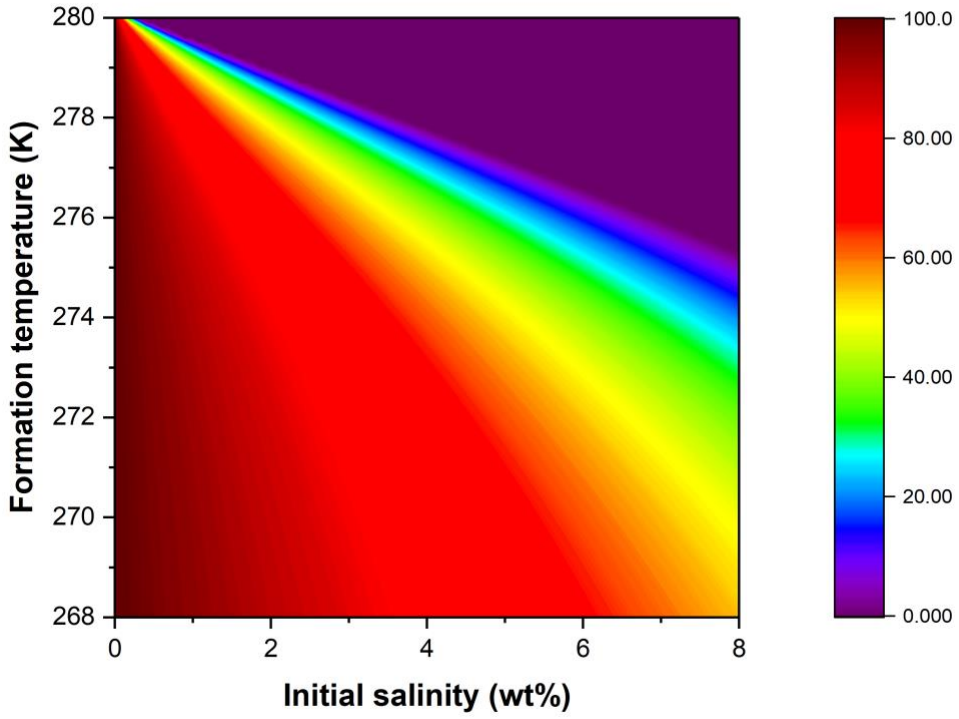


Fig. 4.4.4. Maximum water yield of CP hydrate at different initial salinity (up to 8 wt%) and temperature (268~280 K)

4.4.4. Estimation of energy demand of cooling for CP hydrate-based desalination

In previous section, theoretically achievable maximum water yield of CP hydrate-based desalination was estimated in various hydrate formation temperature and initial salinity conditions, and it was confirmed that the maximum water yield is increased as the hydrate formation temperature and initial salinity decreased. However, these results do not directly represent the higher desalination efficiency at lower temperature and initial salinity conditions because an energy-demand for cooling the system will be increased as an operating temperature decreased, and also sensible heat changes as salinity changed.

In this study, therefore, the operating condition-dependent cooling demand for the process was simply

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examined in thermodynamic point of view. It was assumed that CP and NaCl solution is cooled to the operating temperature for the hydrate formation in the closed system. Then, the theoretically formable maximum amount of CP hydrate was formed at the condition. For the estimation, several assumptions should be build; 1. The system is insulated. 2. The heat flow is only considered for cooling (sensible heat) and hydrate formation (latent heat). 3. The mechanical procedure and kinetics of the hydrate formation was not considered. Through these procedures, the main two energy demand for the CP-HBD, cooling and hydrate formation, could be compared.

For the NaCl solution, the empirical equation based on experimentally measured heat capacity data of real seawater ($c_{p,sal}$ (J/g)) was simply employed as followed[135]:

$$c_{p,sal} = 4.184 \times [1.007 - 0.01488S + (2.8618 \times 10^{-4}S^2) + \{(0.42997 + 0.15192S) \times 10^{-4}T\}]$$

The second-order of the heat capacity equation of the liquid CP ($c_{p,CP}$ (J/g)) is established based on the experimentally measured heat capacity data in the literatures and employed as shown in below[136,139].

$$c_{p,CP} = (111.4777 - 0.2511T + 0.001T^2)/70.1$$

It is assumed that the reactor is filled with NaCl solution and stoichiometric amount of liquid CP to water (CP 19 wt% for CP hydrate) in the saline solution. The Initial temperature of the system is estimated as average seawater temperature in the world (287 K). Then, the sensible heat of the solutions for the cooling to the hydrate formation temperature, (H_s), is calculated as shown in below:

$$H_s = \int_{T^*}^{287} \{c_{p,sal} + 0.19(1 - S)c_{p,CP}\}dT$$

Secondly, for the heat transfer for the hydrate formation (H_L), the theoretically achievable maximum water yield was assumed in each thermodynamic condition and calculated by following equation:

$$H_L = (1 - S) \times MWE \times \Delta H$$

MWE is maximum water yield at the given conditions. From these calculation results, heat ratio between H_L and H_S (H_L/H_S) could be calculated. The heat ratio value was calculated for operating temperature between 268-280K and initial salinity up to 8 wt% (Fig. 4.4.5), and the results indicates that the extent of heat for the hydrate formation is several times larger than that for cooling in our calculation conditions. This suggests that when cooling for the operation of the HBD process, cooling for the hydrate phase change is dominant among the total heat flow. This is remarkably confirmed under

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the same temperature conditions. The heat ratio rapidly decreases when the initial salinity increases under the same temperature conditions due to the decrease of maximum water yield, which is directly involved in H_L. In the case of the same initial salinity conditions, it was confirmed that although maximum water yield increased as the formation temperature decreased, the heat ratio also decreases as the formation temperature decreases, which indicates that the rise in energy for cooling is enormous, and thus, it suggests that unconditional low-temperature conditions will not be efficient. One interesting thing here is that in same initial salinity conditions, there is a specific temperature condition in which the heat ratio is maximum. It indicates that there is an optimum point where energy consumed by phase change relative to input energy is maximize. Therefore, if the economic evaluation is conducted under more practical conditions for the operation of the HBD process in the future, the process operation conditions with optimal economic feasibility according to the salinity of target solution will be able to specify.

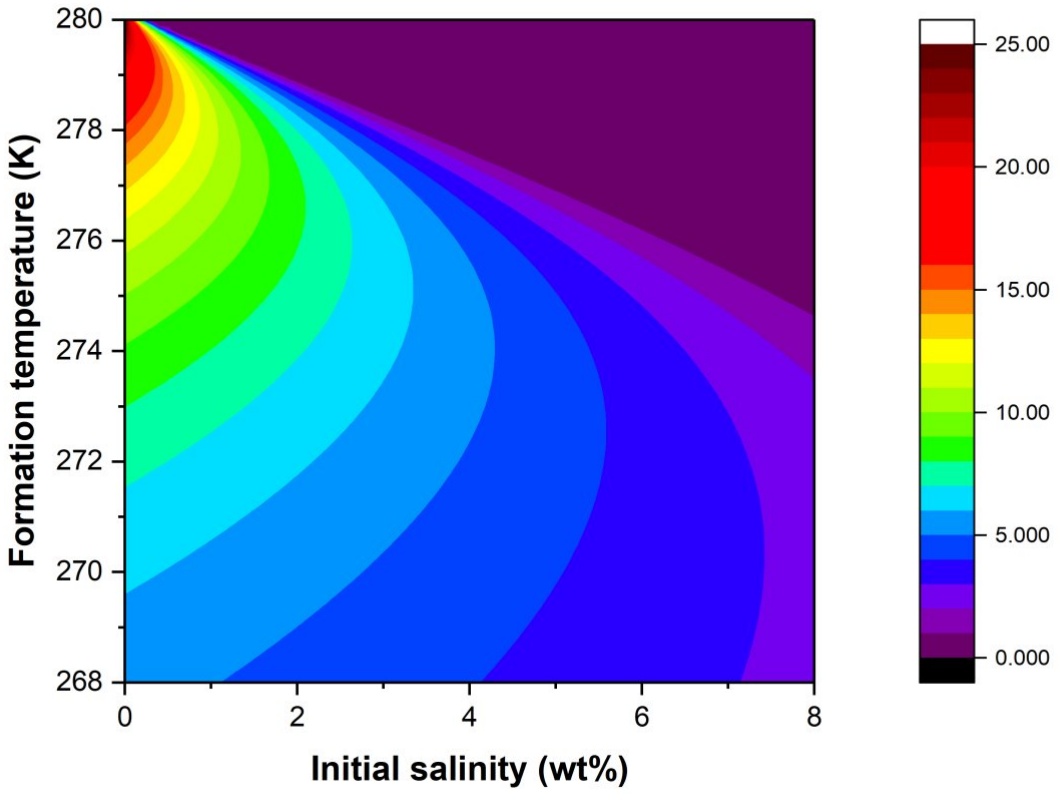


Fig. 4.4.5. Heat ratio of CP hydrate at different initial salinity (up to 8 wt%) and temperature (268~280 K)

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4. Conclusions

In this study, the thermodynamic properties of CP hydrate were experimentally determined, and the desalination efficiency of the CP hydrate-based desalination process was theoretically analyzed. The dissociation enthalpy of CP hydrate was measured to be 113.7 ± 1.7 kJ/mol-guest, which is lower than that of the well-studied HBD candidates, propane and R134a, by 12% and 22%, respectively. The phase diagram of CP hydrate in NaCl solution was observed. The HLS correlation was employed to predict the hydrate equilibrium temperature of CP hydrate at a given pressure and a specific NaCl concentration. The maximum achievable salinity and maximum water yield of CP hydrate in NaCl solution was quantitatively calculated with operating temperature between 268-280K and initial salinity up to 8 wt%. The operating condition-dependent cooling demand for the process was simply examined in thermodynamic point of view. The results suggests that when cooling for the operation of the HBD process, cooling for the hydrate phase change is dominant among the total heat flow. This is remarkably confirmed under the same temperature conditions. The heat ratio rapidly decreases when the initial salinity increases under the same temperature conditions due to the decrease of maximum water yield. In same initial salinity conditions, there is a specific temperature condition in which the heat ratio is maximum, which suggested that there could be an optimum point where energy consumed by phase change relative to input energy is maximize. Therefore, if the economic evaluation is conducted under more practical conditions for the operation of the HBD process in the future, the process operation conditions with optimal economic feasibility according to the salinity of target solution will be able to specify.

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Chapter 5. Future works

5.1. One-dimensional reactor for N_2 - assisted CH_4 - CO_2 replacement

During the flue gas injection into hydrate-bearing sediment in an actual environment, the injected gas will react as it spreads by replacement or hydrate formatioin/disssociaiton, and accordingly, various reaction patterns will appear through changes in the gas composition (Fig. 5.1.1). Therefore, one-dimensional reactor will be employed to observe gradational reaction during CO_2 + N_2 gas injection into CH_4 hydrate. In addition, the representative disadvantage of guest replacement technology is "slow" reaction kinetics. In this study, the dual functional roles of N_2 gas was observed. Therefore, in future works, the possibility of novel hybrid technology which is combination of inhibitor injection (N_2 preflush) and replacement. Therefore, we observe the dissociation of hydrate according to nitrogen injection conditions and the substitution behavior according to subsequent carbon dioxide injection, and try to optimize the infusion conditions that can increase the productivity of substitution techniques.

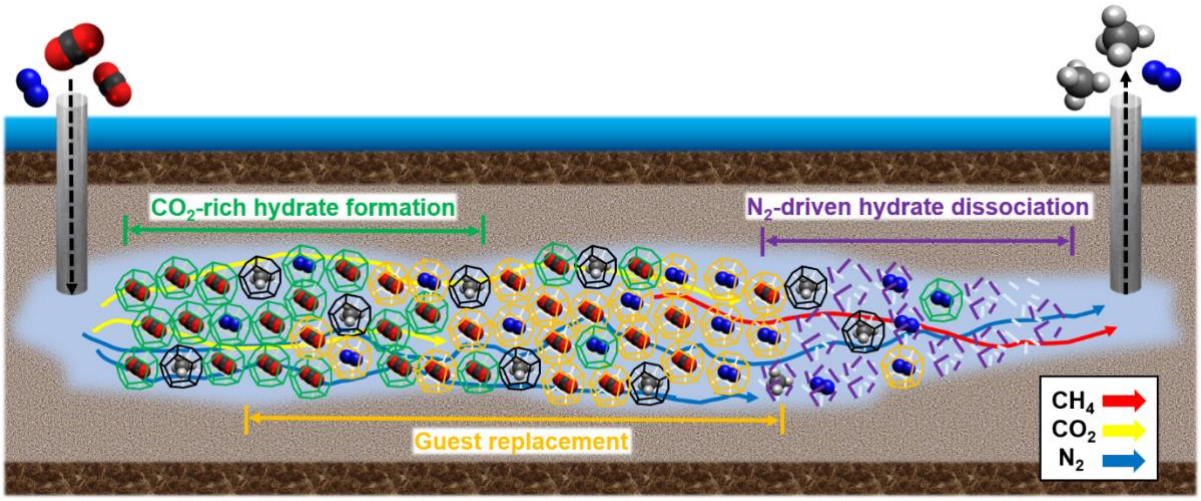


Fig. 5.1.1. Schematic illustration of expected complex phenomenon during CH_4 – CO_2 + N_2 replacement

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5.2. Efficiency evaluation of HBD using liquid-phase hydrate former

To verify the economically optimal operating conditions of the HBD process, the kinetics of the HBD reaction is essential. In future work, therefore, the hydrate formation kinetics according to thermodynamic conditions will be measured with the standard of theoretical HBD efficiency. Furthermore, if the economic evaluation of the process is carried out with the data, the optimal process operating conditions according to the target liquid and hydrate guests can be confirmed (Fig. 5.2.1). Applying this approach to liquid-phase hydrate formers, process operating conditions can be optimized to maximize the operation efficiency of the HBD technology.

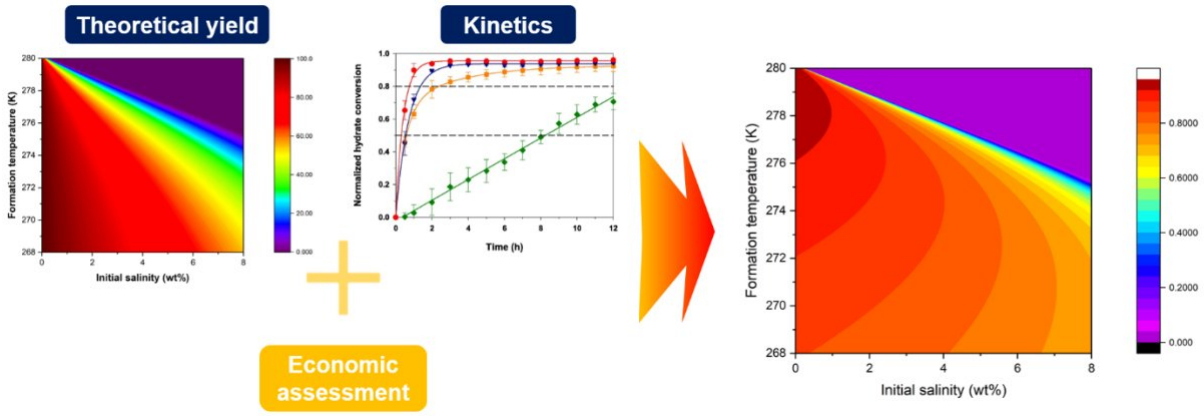


Fig. 5.1.2. Schematic illustration of optimization of guest and condition-dependent HBD efficiency

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우선은 대학원 과정 동안 제 지도 교수님이셨던 스승님, 서용원 교수님께 감사의 말씀드리고 싶습니다. 지도 교수라는 말에 걸맞게, 아니 그 이상으로 너무나도 많은 것들을 지도해 주셔서 하루하루가 배움의 연속이었습니다. 연구적인 지식과 통찰력은 물론이려니와 겸손한 자세와 사람을 대하는 태도까지, 제가 앞으로 살아가는 데 있어서 필요할 많은 지혜와 영감을 주셨습니다. 앞으로 제 삶에 바쁜 나날들이 이어지겠지만, 자주 인사드리고 찾아뵙겠습니다.

또 같은 연구실에서 지낸 모든 분들께도 감사의 말씀드리고 싶습니다. 서툴고 부족했던 저를 항상 도와주셨던 선배님들. 특히 사수로서 오랜 시간을 함께하며 연구적으로 도움을 많이 주셨던 원중이 형. 선배님들이 일궈놓고 가신 밑바탕이 있었기에 제 연구가 꽃피울 수 있었습니다. 끼도 많고 능력도 출중한 후배님들. ‘내가 저 때는 이렇게 잘했었나?’ 하는 생각이 항상 들 정도로 다들 잘하고 열정이 넘쳐서 항상 많은 자극을 받을 수 있었습니다. 그리고 대학원 생활을 가장 오래 함께한 하나뿐인 동기, 우진이. 대학원 과정 중에 많이 의지했었고, 연구뿐만 아니라 삶에 있어서도 많은 걸 배워갈 수 있어서 너무 고마웠다.

무엇보다 부모님께도 감사의 말씀드리고 싶습니다. 제가 하고 싶은 일을 할 수 있도록 믿고 기다려주신 점, 그리고 제게 주신 모든 것들 항상 당연하게 생각하지 않고 감사히 여기며 살고 있습니다. 앞으로 잘 살아가는 모습 보여드리겠습니다. 건강히 오래오래 지켜봐 주세요.

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목정훈.

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